

PROJECT REPORT

Predicting Traffic Volumes from Satellite Imagery for Urban Planning

A case study of Melbourne

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Cover image: Nearmap satellite imagery around Alexandra Parade and Wellington Street, Melbourne (2020). Its right side shows an illustrative SAM 3 segmentation of the roofs, roads, vegetation and parking or paved ground, corrected by hand in two places.

Executive summary

Problem

Urban development changes travel demand and, in turn, the traffic volumes on the road network. Planning approval requires this effect to be assessed before a development is built, but conventional traffic forecasting relies on recent measured volumes and cannot be applied to a development that does not yet exist. Estimates based on land use mostly give annual or daily totals and cannot show how traffic varies through the day. Planning assessment therefore needs a method that relies only on information available at the planning stage, predicts traffic volumes throughout the day and presents development proposals visually.

Scope and deliverables

The study area is central Melbourne. The observed data are traffic volumes recorded by the Sydney Coordinated Adaptive Traffic System (SCATS) at 81 signalised intersections from 2014 to 2024, excluding 2020 and 2021. They were aligned by location and year with Nearmap satellite imagery, the Census of Land Use and Employment (CLUE) and the OpenStreetMap road network, with 2024 as the test year. The generalisation of the method was tested on 371 signal sites across the City of Melbourne. The deliverables are this report, the traffic volume prediction model and its evaluation, the results of the land-use sensitivity analysis, the development scenario imagery tool with its interactive web interface, and the organised project code.

Outcomes

1. The project developed a planning-oriented traffic volume prediction model. SAM 3 and DINOv3 turn satellite imagery into region features, and a large language model embedding (LLM embedding) encodes each intersection's position relative to its neighbouring signalised intersections. Combined with land-use attributes and calendar features, these inputs feed a temporal convolutional network that predicts traffic volumes in all 96 fifteen-minute intervals of the day. The model uses only information available at the planning stage, and no traffic information from the test year enters it in any form. (Chapter 4)
2. The project analysed how sensitive predicted traffic is to land use. With the trained models held fixed, the analysis changes the scale and mix of office, residential apartment and retail uses in the test year, and quantifies how predicted traffic volumes respond and how the response differs between times of day and road classes. (Chapter 5)
3. The project built a development scenario imagery tool. From building footprints, heights and uses, the tool generates satellite imagery of new building, alteration and demolition scenarios. It constrains generation by gpt-image-2 with 3D massing, an editable region and prompts, and runs in an interactive web interface. The generated imagery was also tested as input to the prediction model on 17 real developments completed in 2023–2025. (Chapter 6)

Key results

1. In the test year, the configuration that uses imagery, the road network and land use together is the most accurate. Its mean absolute error (MAE) is 59.18 vehicles per 15 minutes (veh/15 min), its median GEH statistic is 3.84 and 59.9% of intervals have a GEH below 5. Its MAE is 4.9% lower than that of the time-only baseline. The gains are concentrated in the weekday AM peak, where MAE falls by 9.4, and on main roads, where it falls by 4.4 on average.
2. Among the inputs, satellite imagery makes the most consistent contribution. It reduces MAE by 1.67 veh/15 min (p = 0.002), with improvements in all 5 random seeds. Land use reduces the MAE of the configuration that adds imagery only (Route 1) by 1.04 (p = 0.028), and the contribution of the road network features is not statistically significant.
3. Across 371 signal sites in the City of Melbourne, the MAE of the four configurations with imagery differs by no more than 0.49. At the 78 intersections shared with the study area that have test data, the MAE of Route 2 + land use is 58.02, close to the 57.15 of the study-area model, so the method extends to a wider area. At this scale, land use brings no improvement to Route 1 but gives the configuration with both imagery and the road network (Route 2) a small reduction in MAE of 0.48, with lower MAE in 4 of the 5 seeds.
4. Predicted responses to land-use change follow the direction of the change and grow roughly linearly with its size, and office is the most sensitive use. When the office share doubles while the total number of floors stays the same, the all-day responses of the two configurations with land use are +2.59 and +2.99 veh/15 min in the model output, about 0.7%–0.8% of all-day volume, and +1.55 and +1.80 veh/15 min in the final predictions, which blend the model output with the historical profile. The response arises mainly on secondary and local roads. In the weekday AM peak, the response on trunk roads is only about 0.01% of volume, against 0.7%–1.8% on secondary and tertiary roads.
5. In the scenario imagery tool, the position, footprint and height of a generated building follow the input geometry, and edits stay within the building massing and its shadow. In all 18 saved examples, no pixel outside the editable region changed.
6. When a generated building is added to satellite imagery taken before construction and used as input to the prediction model, the median GEH at the affected intersections is 3.50–4.06 and the MAE changes by no more than 0.15 veh/15 min, so accuracy is essentially unchanged. The responses to the full scenario, +0.49 and +0.19 veh/15 min for the developments completed in 2023–2024 and in 2025, are positive but not statistically significant.

Limits of application

The results are intended for screening and comparing development proposals at the planning stage. Impacts are best measured as the difference between the same model's predictions before and after a proposal, and the results cannot replace a formal traffic impact assessment, traffic simulation or field surveys. The model can only predict for observation sites it was trained on and has been evaluated on a single test year, 2024, apart from a check on 2025 at the intersections around the test developments of Section 6.4. The land-use features rely on the CLUE census, which covers only the City of

Melbourne, so applying the method to new sites or other regions requires equivalent data, retraining and validation.

The sensitivity results reflect statistical associations between land use and traffic volume that the model learned from the training data (2014–2019 and 2022). They are not causal effects and do not include changes that a development may bring to the road network, signal timing or travel modes, and land-use changes beyond the range of the training data are extrapolations. The model has no capacity constraints or route choice, so it cannot represent how traffic redistributes across the network. Building appearance in the scenario imagery is synthesised by an image model and serves only to show the location and bulk of a proposal, not its architectural design. Used as input to the prediction model, generated imagery leaves accuracy essentially unchanged, but the predicted change from a single development is small and not statistically significant (Section 6.4), so the imagery is not on its own a basis for traffic assessment.

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1 Introduction

Urban development changes travel demand and, through it, the traffic volumes on the road network. Assessing this effect during planning approval requires traffic volumes to be predicted before a development is built, but conventional traffic forecasting relies on recent measured volumes and is difficult to apply to developments that do not yet exist.

This project proposes a planning-oriented approach to traffic volume prediction whose core is the use of satellite imagery to describe the built environment. Satellite imagery directly records the physical form of buildings, roads, vegetation and car parks, covers large areas and can be updated every year. The project uses large pretrained vision foundation models and a large language model embedding (LLM embedding) to turn imagery and road network structure into features a model can use, without manual labelling for this project. The observed data are traffic volumes recorded by the Sydney Coordinated Adaptive Traffic System (SCATS) at 81 signalised intersections in central Melbourne from 2014 to 2024. The project combines imagery features with the land-use census, road network structure and calendar information to predict traffic volumes in all 96 fifteen-minute intervals of the day. The model uses only information available at the planning stage, and the method was tested on 371 signal sites across the City of Melbourne.

Building on the prediction model, the project analysed how changes in the scale and mix of office, residential and retail land use affect predicted traffic volumes, and how the effects differ between times of day and road classes. The project also built a development scenario imagery tool with an interactive web interface. From a building's footprint, height and use, the tool generates satellite imagery of the site after the development is built, so that a proposal can be presented as imagery of the same kind as the model's input.

1.1 Background

1.1.1 Traffic impact assessment of developments

Urban development affects travel demand by changing the scale and use of land. New office, residential and commercial floor space generates and attracts trips, which enter the road network and change its traffic volumes (Ortúzar & Willumsen, 2011). Planning approval therefore needs an estimate of a development's traffic impact before it is built, to judge whether the surrounding roads and transport facilities need to be adjusted.

Current traffic impact assessment usually estimates new trips from trip generation rates for the type and size of development, then distributes and assigns them to the surrounding network to assess how the roads will perform. This process has two shortcomings. First, trip generation rates come from surveys of similar sites elsewhere, while trip generation depends strongly on the built environment around a site. In urban areas with high population and employment densities and good public transport and walking conditions, applying these rates directly often overestimates vehicle trips (Schneider et al., 2015), and the rates need to be adjusted for the surrounding built environment (Clifton et al., 2015). Second, each assessment is made project by project with manually set parameters, which makes it hard to compare alternative proposals

quickly and consistently. Meanwhile, the volume data that traffic signal systems have accumulated over many years have not yet been fully used in development impact assessment.

1.1.2 Describing the built environment with remotely sensed imagery

Travel demand is largely shaped by the built environment. Empirical studies show that travel distances and mode choices are associated with the density, land-use mix, street design and destination accessibility of the built environment (Ewing & Cervero, 2010). A data-driven model that is to learn the relationship between the built environment and traffic volumes first needs a consistent and comparable description of the built environment.

Traditionally, such descriptions have relied on land-use surveys, planning zones and population and employment statistics. These data are compiled specifically by local governments, their definitions vary from place to place, and their update frequency and spatial detail are limited by survey conditions, which makes them hard to use consistently across regions and years.

Satellite imagery and other remotely sensed data offer an alternative. Remotely sensed imagery directly records buildings, roads, vegetation, car parks and other features on the ground, covers large areas and can be acquired repeatedly, so that changes at a location can be traced back over many years. Studies have shown that imagery combined with machine learning can estimate the socio-economic conditions of an area (Jean et al., 2016) and predict road traffic volumes (Ganji et al., 2022). For planning, imagery has a further advantage: planned developments can also be presented as imagery, so existing conditions and planning proposals can be described within the same representation.

1.1.3 New models and data

Conventional methods of extracting information from imagery require large numbers of manually labelled samples and a dedicated model for each task. Their cost has limited the use of imagery in traffic prediction. Vision foundation models have changed this. Pretrained on large-scale data, they can be applied directly to new tasks without retraining. SAM 3 segments the features in an image from text prompts (Carion et al., 2025), and DINOv3 extracts general-purpose visual features for any region of an image (Siméoni et al., 2025). A large language model embedding can encode structured information, such as a location in the road network and its connections, as numerical vectors (Zhang et al., 2025). Generative image models, meanwhile, can now make local edits to an image from text instructions, which makes it possible to generate imagery of a planned development as built.

On the data side, Melbourne's traffic signals are controlled by SCATS (Sims & Dobinson, 1980). Its detectors continuously count passing vehicles, and the state government publishes the counts aggregated to 15-minute intervals. The City of Melbourne's Census of Land Use and Employment (CLUE) records the number of floors and the predominant use of every building in the municipality each year, and satellite imagery is available year by year. Long, continuous volume records combined with annual imagery and land-use data make it possible to learn the relationship between the built environment and traffic volumes from data.

1.2 Planning-oriented traffic prediction

Most research and practice in traffic prediction serves operational management. It predicts traffic conditions minutes to hours ahead from recently measured volumes, for signal control, traffic guidance and incident management (Vlahogianni et al., 2014). Deep learning methods perform strongly in these tasks (Li et al., 2018), and their accuracy depends largely on the short-term continuity in recent measurements.

Planning assessment deals with developments that are not yet built. Post-development volumes do not yet exist, and measured data for the forecast year are not available. The information available is limited to the calendar and public holidays, the imagery, land use and road network that describe the built environment, and the planning proposal itself. Volumes measured in earlier years can serve as a reference, but they reflect conditions before the development. Estimating traffic volumes from land use and other static information is not new. Zhao and Chung (2001) used regression models to relate annual average daily traffic to the surrounding land use and road network characteristics, but such methods mostly give annual or daily totals and cannot reflect variation within the day.

Table 1.1 compares the inputs available in the two settings. This project adopts the planning setting: the model uses only information available at the planning stage to predict traffic volumes in each of the 96 fifteen-minute intervals of a day, and no traffic information from the test year enters the model in any form (Section 4.9).

Table 1.1: Inputs available in the operational and planning settings

Input	Operational setting (short-term prediction)	Planning setting (this project)
Recent measured volumes (previous few intervals)	Available	Not available
Lagged volumes (previous day, same day of the previous week)	Available	Not available
Measured statistics for the forecast year	Available	Not available
Calendar and public holidays	Available	Available
Historical profile (average daily pattern measured in earlier years)	Available	Available (existing data)
Satellite imagery	Available	Available (current or scenario imagery)
Land use	Available	Available (from the planning proposal)
Road network structure	Available	Available

1.3 Objectives

The main objectives of the project are to:

1. develop a 15-minute traffic volume prediction model, built around satellite imagery and combining land-use and road network information, using only information available at the planning stage
2. quantify how land-use changes of different uses and scales affect predicted traffic volumes, and compare the effects across times of day and road classes

3. build a development scenario imagery tool with an interactive web interface, which generates satellite imagery of a development as built from information on the development.

1.4 Work undertaken

The project addressed these objectives through the following work:

- **Multi-source data integration**: SCATS volumes recorded at the 81 signalised intersections of the study area from 2014 to 2024 were aligned by location and year with Nearmap satellite imagery, CLUE land use and the OpenStreetMap road network, and the patterns in the data that affect modelling were analysed (Chapter 3).
- **Imagery-based prediction modelling**: pretrained models for image segmentation, visual representation and LLM embedding turn satellite imagery and road network structure into features. These are combined with land-use and calendar features in a temporal convolutional network that predicts volumes for all 96 intervals of the day (Chapter 4).
- **Contribution of each input**: six input configurations were compared to test the role of imagery, the road network and land use, and the method was tested on 371 signal sites to check whether it extends to a wider area (Chapter 4).
- **Land-use sensitivity analysis**: with the trained models held fixed, the scale and share of office, residential apartment and retail uses were changed to analyse the response of predicted volumes and how it differs between times of day and road classes (Chapter 5).
- **Development scenario imagery tool**: satellite imagery of new building, alteration and demolition scenarios is generated from building footprints, heights and uses, in an interactive web interface, and real developments completed in 2023–2025 were used to test generated imagery as input to the prediction model (Chapter 6).

Chapter 2 describes the overall framework of the project and the structure of the report.

2 Project scope and report outline

The project's core task is traffic volume prediction at the planning stage. Its research framework has four parts: data integration, traffic volume prediction, land-use sensitivity analysis and development scenario imagery. The framework uses satellite imagery as the core data describing the built environment and combines it with the land-use census, the road network and observed traffic volumes, forming a complete process from data to prediction and on to planning applications.

The parts of the framework are linked by data and models. The integrated multi-source data are used to train and evaluate the traffic volume prediction model. The trained model is then used to analyse how land-use change affects traffic volumes, and the development scenario imagery tool presents planning proposals as satellite imagery consistent with the model's input.

2.1 Project framework

Figure 2.1 shows the project framework. Five types of data feed the traffic volume prediction model, the trained models are used in the land-use sensitivity analysis, and the development scenario imagery tool generates imagery in the same form as the model's input. The three parts lead to three types of output.

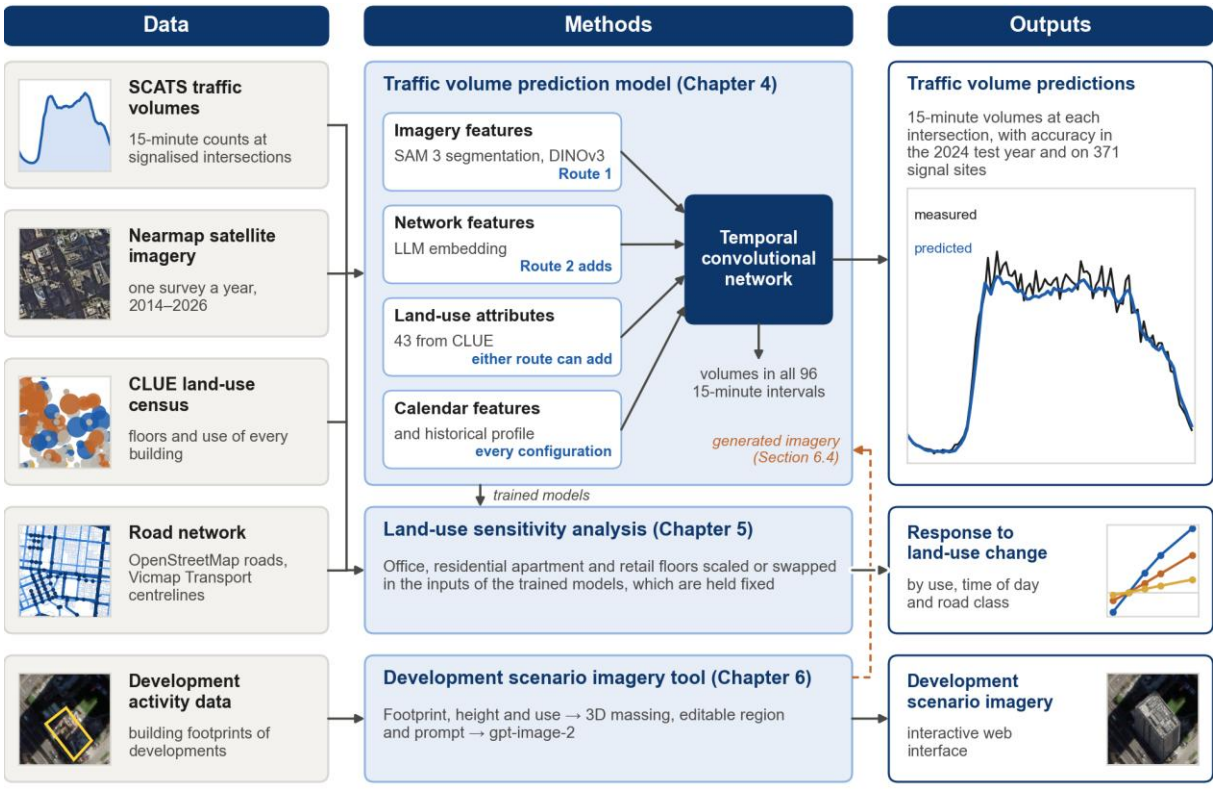


Figure 2.1: Project framework. The left column shows the five types of data (Section 2.1.1), the middle column the traffic volume prediction model with its two routes, the land-use sensitivity analysis and the development scenario imagery tool (Section 2.1.2), and the right column the three types of output (Section 2.1.3). The dashed line shows the use of generated imagery as model input (Section 6.4).

2.1.1 Data

The project uses five types of data. Nearmap satellite imagery, one survey a year covering every traffic observation site, is the core data describing the built environment. The Census of Land Use and Employment (CLUE) records the number of floors and the predominant use of every building each year, and these records are aggregated into land-use attributes. The road network data comprise OpenStreetMap roads and Vicmap Transport road centrelines. The former are used to group results by road class, and the latter serve as prior information on roads in image segmentation.

Traffic volumes from the Sydney Coordinated Adaptive Traffic System (SCATS) give vehicle counts for every 15 minutes at each signal site. They are the observations used to train and evaluate the model and to compute the historical profiles, and the locations of the signal sites are used to build the adjacency graph that represents position in the road network. The City of Melbourne's development activity data provide the building footprints of developments, which form the development library of the scenario imagery tool. (Chapter 3)

2.1.2 Methods

The traffic volume prediction model corresponds to the two-route multimodal modelling in Figure 2.1. Route 1 is the visual-semantic route, which uses pretrained vision foundation models such as SAM 3 and DINOv3 to extract built-environment features from satellite imagery. Route 2 adds network features to Route 1: a large language model embedding (LLM embedding) encodes each observation site's position relative to its neighbouring signalised intersections. Both routes can also incorporate land-use attributes and, together with calendar and public holiday information, predict traffic volumes in all 96 fifteen-minute intervals of the day.

The land-use sensitivity analysis changes the scale and mix of office, residential apartment and retail uses in the inputs of the trained models, and compares the response of predicted traffic volumes and how it differs between times of day and road classes.

The development scenario imagery tool generates satellite imagery of a development as built from its building footprints, heights and uses, and supports new building, alteration and demolition scenarios. (Chapters 4–6)

2.1.3 Outputs

The project produces three types of output. The first is 15-minute traffic volume prediction and its accuracy assessment, including the evaluation in the 2024 test year and the generalisation test on 371 signal sites. The second is the response of traffic volumes to land-use change. The third is development scenario imagery with its interactive web interface. (Chapters 4–6)

2.2 Project scope

Within a study area in central Melbourne, the project completed the full process from data integration, feature construction, and model training and evaluation to land-use sensitivity analysis and development scenario imagery. The study area covers the eastern half of the Hoddle Grid, Carlton, East Melbourne, Southbank and the Hoddle

Street–Punt Road corridor. It extends about 2.3 km east–west and 3.4 km north–south and includes traffic volume observations at 81 signalised intersections. To test how well the method generalises, the project also trained and evaluated the model on 371 signal sites across the City of Melbourne, of which 305 are intersection signals and 66 are pedestrian crossing signals.

The project uses data from 2014–2019 and 2022–2024 at a resolution of 15 minutes. Traffic volumes in 2020 and 2021 fell sharply under COVID-19 lockdowns and are not used for modelling. The data are split by year: 2014–2019 and 2022 form the training set, 2023 the validation set and 2024 the test set. The development scenario imagery tool can use satellite base images from 2014 to 2026, each centred on the development, 400 m across and at a resolution of about 0.39 m per pixel.

2.3 Report outline

The remainder of the report is organised as follows:

- **Chapter 3 Study area and exploratory analysis of datasets** introduces the study area and each dataset, analyses the patterns in the data that shape the model design, and describes the data split and the conditions of use.
- **Chapter 4 Traffic volume prediction model** describes how the four types of features (imagery, land use, road network and time) are built, the model architecture and the training settings. It compares the accuracy of six input configurations and tests how the method generalises on 371 signal sites.
- **Chapter 5 Land-use sensitivity analysis** uses the trained models to analyse how changes in the scale and mix of office, residential apartment and retail uses affect predicted traffic volumes, and compares the effects across times of day and road classes.
- **Chapter 6 Development scenario imagery tool** explains how image generation is constrained by 3D massing, the editable region, prompts and the generative model, presents generated results for new building, alteration and demolition scenarios, and tests generated imagery as input to the prediction model.
- **Chapter 7 Conclusion** summarises the main findings and limitations of the project and makes recommendations for application and future work.

Appendices A to E give, in turn, the data dictionary, the definitions of the land-use attributes, the full experimental results, the prompts used by the development scenario imagery tool and the test cases of the generated imagery.

3 Study area and exploratory analysis of datasets

Building a planning-oriented traffic volume prediction model requires observed traffic volumes and the multi-source data describing the built environment to be matched in space and time. The project uses traffic volumes recorded by the Sydney Coordinated Adaptive Traffic System (SCATS) as the observed data, and Nearmap satellite imagery, the Census of Land Use and Employment (CLUE) and the OpenStreetMap road network to describe the built environment around each observation site. The four datasets are aligned by location and year and form the data basis of the prediction model (Chapter 4) and the land-use sensitivity analysis (Chapter 5). In addition, Vicmap Transport road data are used in image segmentation (Section 4.2), and the City of Melbourne's development activity data provide building footprints for the development scenario imagery tool (Chapter 6). The daily patterns in the data and the differences in volume level and between road classes also determine the time windows and groupings used in the later analyses.

3.1 Study area

3.1.1 Defining the study area

The study area comprises 81 signalised intersections in central Melbourne, covering the eastern half of the Hoddle Grid (the central city street grid), Carlton, East Melbourne, Southbank and the Hoddle Street–Punt Road corridor (Figure 3.1). The intersections span 2.3 km east–west and 3.4 km north–south and are closely spaced: the median distance from an intersection to its nearest neighbour is 126 m.

The intersections were selected from the 143 SCATS intersections of the original study area in two steps:

1. **Complete volume data**: 139 intersections appear in the volume data in every year used by the model (2014–2019 and 2022–2024), although a few site-years are missing entirely (Section 3.2.2).
2. **Land-use data coverage**: of these, 81 intersections have at least one CLUE building within the 500 m square centred on the intersection. CLUE covers only the City of Melbourne, so the land-use attributes of the remaining 58 intersections are all zero and cannot be used to study how land use relates to traffic volume.

The final study area contains all 72 City of Melbourne intersections of the original set and 9 City of Yarra intersections on the municipal boundary. The 500 m squares of the latter extend into the City of Melbourne, so they also have land-use data (Table 3.1).



Figure 3.1: The study area. Filled circles are the 81 study intersections, and open circles are the other SCATS intersections of the original study area, all of which lie outside the coverage of the CLUE census (yellow dots are CLUE buildings). Grey lines are the OpenStreetMap road network.

Table 3.1: Overview of the study area

Item	Value
Signalised intersections	81 (72 City of Melbourne, 9 City of Yarra)
Extent	2.3 km east–west × 3.4 km north–south
Median spacing to the nearest intersection	126 m
CLUE small areas covered	6
CLUE buildings within the intersection squares (2024)	3,594
Years of traffic data used	2014–2019, 2022–2024

3.1.2 Extended validation area

To test whether the modelling approach also holds beyond the study area, the project applied the same method to a larger set of SCATS signal sites (Section 4.13). There are 446 signal sites within the CLUE census area, of which 371 appear in the volume data in every year from 2014 to 2026 and have CLUE buildings within their square in every census year. As in Section 3.1.1, a site appears in a year's data even when none of its counts in that year are valid, as at intersection 4889 from 2024. These 371 signal sites spread across the City of Melbourne, about 7.2 km east–west and 8.4 km north–south. Of them, 305 are intersection signals and 66 are pedestrian crossing signals, and they

include 79 of the 81 study intersections (Figure 3.2). The extended area is used only for this validation, and all other results in the report refer to the 81 study intersections.



Figure 3.2: The extended validation area. Dark grey circles are the 305 intersection signals and triangles the 66 pedestrian crossing signals. Blue circles are the 81 study intersections, 79 of which also belong to the extended set. Yellow dots are CLUE buildings in 2024, marking the census coverage (the City of Melbourne). Grey lines are main roads.

3.2 SCATS traffic volume data

3.2.1 Data description

Traffic volumes come from the Sydney Coordinated Adaptive Traffic System (SCATS), which controls all traffic signals in Melbourne (Sims & Dobinson, 1980). The Victorian Department of Transport and Planning publishes the volume data as "Traffic Signal Volume Data" on the state government's open data platform, one file per month (Department of Transport and Planning, 2026a). Each record gives the number of vehicles that passed one inductive loop detector in each 15-minute interval of a day, 96 intervals per day.

The project sums the counts of all detectors at each intersection, giving one record per intersection per day, each holding the volumes of 96 intervals. The prediction target of the model is the traffic volume in each 15-minute interval at each observation site.

3.2.2 Data preparation

The monthly files for 2014–2026 were merged into one table, which holds 263,083 site-days in the years used by the model. Data quality issues were handled as follows:

- **Missing intervals**: negative values are not vehicle counts and are treated as missing, together with empty values. Missing intervals make up 0.48% of all intervals and are excluded from training and evaluation. About 67% of them come from days with no record at all: 840 site-days (0.32%), mainly at intersection 4603 in 2018 (33% of days) and intersection 4889 in 2024 (100% of days). Intersection 4889 therefore has no data in the test year (2024), and the test results in Chapter 4 cover the other 80 intersections.
- **Days recorded as zero**: days with records but a daily total of zero are kept as they are. There are 3,351 such site-days (1.27%), 97% of them at 4 intersections. One is the access road intersection of the Melbourne Cricket Ground (MCG), which carries traffic mainly on event days and records zero on 59–79% of days in each year. The others are intersections 4586 (2022), 4588 (2023 and 2024) and 4603 (2022), where zeros run on for weeks to months, most likely because of detector faults. The affected site-years are listed in Appendix A.

Table 3.2 summarises the data by year. Apart from December 2018, when no intersection has any record, the data are essentially complete.

Table 3.2: SCATS data for the study intersections by year

Year	Intersections	Site-days	Missing intervals (%)	Days with no record (%)	Days recorded as zero (%)	Use
2014	81	29,559	0.26	0.1	0.9	Train
2015	81	29,422	0.63	0.3	0.8	Train
2016	81	29,646	0.26	0.1	0.8	Train
2017	81	29,484	0.35	0.1	0.9	Train
2018	81	27,054	0.85	0.7	0.8	Train
2019	81	29,484	0.12	0.0	1.0	Train
2020	81	29,403	0.05	0.0	1.6	Not used
2021	81	29,565	0.05	0.0	2.9	Not used
2022	81	29,403	0.06	0.0	2.5	Train
2023	81	29,466	0.34	0.3	1.8	Validation
2024	81	29,565	1.43	1.3	1.8	Test

3.2.3 Daily and weekly patterns

Weekdays and weekends follow clearly different daily patterns (Figure 3.3).

- **Weekdays**: the morning peak is at 08:30 (596 vehicles per 15 minutes per intersection on average) and the evening peak at 17:15 (622 vehicles), slightly higher than the morning peak. Between the two peaks, volumes stay at about 500 vehicles per 15 minutes. The overnight minimum occurs at 03:30 (62 vehicles).

- **Weekends**: there is only one broad midday peak, at about 12:30 (524 vehicles), and late-night volumes are higher than on weekdays.

Within the week, daily volume per intersection rises from 34,057 vehicles on Monday to 39,597 on Friday and is lowest on Sunday (31,538 vehicles).

The sensitivity analysis in Chapter 5 reports responses in three time windows centred on these peaks. Each window covers the peak interval and the two intervals on either side:

- Weekday AM peak: 08:00–09:15
- Weekday PM peak: 16:45–18:00
- Weekend midday: 12:00–13:15

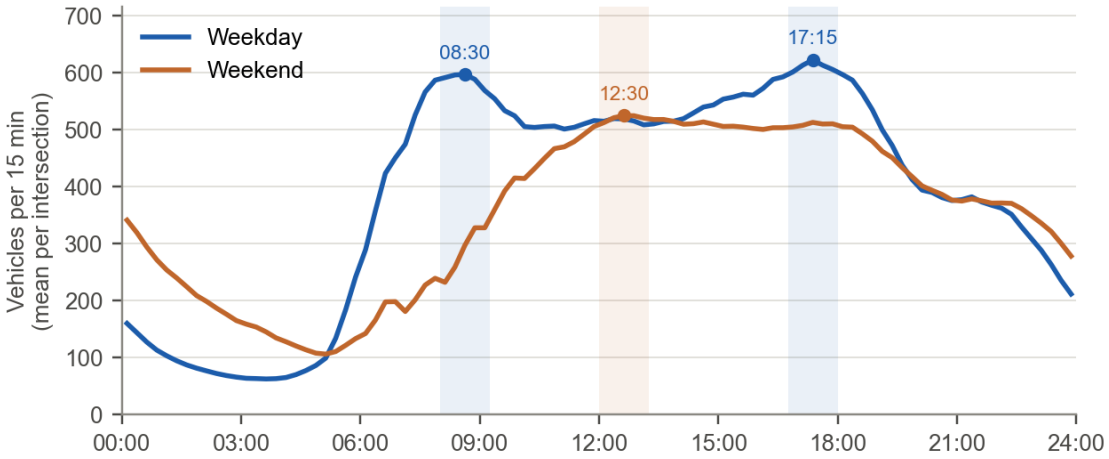


Figure 3.3: Mean volume per intersection in each 15-minute interval on weekdays and weekends (2014–2019 and 2022–2024). Shading marks the three analysis windows used in Chapter 5.

3.2.4 Volume levels across intersections

Volumes differ widely between intersections (Figure 3.4). On an average weekday, the median intersection carries 29,531 vehicles a day, and the busiest, where Hoddle Street meets Albert Street and Elizabeth Street, carries 127,592. The intersection at the 90th percentile carries about 19 times the volume of the one at the 10th percentile.

For the error analysis in Section 4.12, the intersections are divided into three equal groups by average weekday daily volume:

- Low volume: below about 20,000 vehicles
- Medium volume: about 20,000–45,000 vehicles
- High volume: above about 45,000 vehicles

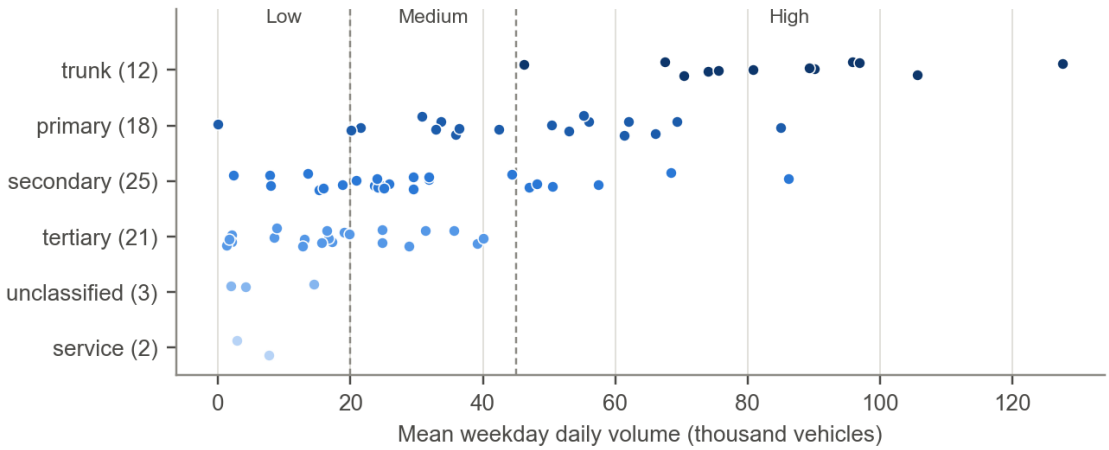


Figure 3.4: Average weekday daily volume at each study intersection, in rows by OpenStreetMap road class (see Section 3.5). Dashed lines divide the intersections into three equal volume groups.

3.3 Nearmap satellite imagery

3.3.1 Data description

Satellite imagery comes from Nearmap through the University of Melbourne's subscription. For each intersection and year, a square image patch centred on the intersection is downloaded from Nearmap's vertical imagery tiles (Table 3.3). A resolution of 0.49 m per pixel is enough to distinguish individual buildings, roofs, trees, car parks and lane markings.

Table 3.3: Nearmap imagery specifications

Item	Value
Provider	Nearmap, vertical imagery (University of Melbourne subscription)
Image patch	One per intersection per year, centred on the intersection
Ground coverage	500 m × 500 m
Image size	1024 × 1024 pixels, RGB
Ground resolution	0.49 m per pixel (tiles at zoom 18, resampled)
Years	2014–2026, one survey per year (2025 and 2026 not used in Chapters 4 and 5)
Images used	81 intersections × 9 years = 729

3.3.2 Survey selection

Nearmap captures Melbourne several times a year: between 2014 and 2024, 4–11 surveys a year cover the study area. To keep season, sun angle and vegetation comparable across years, the survey closest to 1 February is used in each year. The survey selected in each year covers every study intersection, and no capture date is more than 31 days from 1 February (Table 3.4).

Table 3.4: Nearmap surveys used in each year

Year	Capture date	Days from 1 February	Surveys available	Use
2014	09 Jan 2014	−23	7	Train
2015	05 Feb 2015	+4	5	Train
2016	05 Feb 2016	+4	4	Train
2017	07 Feb 2017	+6	6	Train
2018	17 Feb 2018	+16	5	Train
2019	11 Jan 2019	−21	5	Train
2020	17 Feb 2020	+16	6	Not used
2021	22 Jan 2021	−10	11	Not used
2022	12 Feb 2022	+11	7	Train
2023	01 Jan 2023	−31	6	Validation
2024	03 Feb 2024	+2	8	Test

3.3.3 Example imagery

Figure 3.5 shows the imagery of the Exhibition / Little Lonsdale intersection in four years. Among all study intersections, the total number of floors within this intersection's 500 m square grew the most between 2014 and 2024. The imagery clearly records the new high-rise buildings and the change in their shadows, which is exactly what the imagery features in Section 4.2 are meant to describe.



Figure 3.5: Nearmap imagery of intersection 2911 (Exhibition / Little Lonsdale). (a) 2014. (b) 2018. (c) 2022. (d) 2024. Each image covers 500 m × 500 m, and the cross marks the intersection.

3.4 CLUE land-use data

3.4.1 Data description

The Census of Land Use and Employment (CLUE) is conducted by the City of Melbourne, and its data are published on the City of Melbourne Open Data platform. Its building information dataset (City of Melbourne, 2026a) lists every building in the municipality each year, 14,094 in 2024. The project uses the following fields from every census year from 2014 to 2024:

- location
- number of floors above ground
- predominant use

- construction year
- number of bicycle spaces

The predominant uses are grouped into 13 classes: 12 common uses, chosen by their frequency across the whole municipality (Appendix B.3), plus “other”.

Table 3.5 shows a sample of the raw data. The 500 m square of the Exhibition / Little Lonsdale intersection (Figure 3.5) contained 193 buildings in 2024, and the table lists the tallest building of each use, with fields and values as recorded in the original dataset. Each record also has longitude and latitude, which are omitted here. The uses in the table are, in order, Office, Residential Apartment, Commercial Accommodation, Student Accommodation, Educational/Research, Retail - Shop, Entertainment/Recreation - Indoor and House/Townhouse.

Table 3.5: Sample of raw CLUE building data (2024, within the square of the Exhibition / Little Lonsdale intersection, with the address suffix MELBOURNE VIC 3000 omitted)

Building name	Street address	Construction year	Floors above ground	Predominant space use	Bicycle spaces
Telstra Centre	242-284 Exhibition Street	1992	47	Office	350
Adobe 318 Apartments	302-324 Russell Street	2014	58	Residential Apartment	146
Oaks on Lonsdale	23-33 Lonsdale Street	2006	30	Commercial Accommodation	0
–	42-50 La Trobe Street	2018	43	Student Accommodation	10
The Greek Centre	166-170 Lonsdale Street	2014	13	Educational/Research	0
–	300 Russell Street	1925	3	Retail - Shop	0
Harwood House	37-41 Little Bourke Street	1920	4	Entertainment/Recreation - Indoor	0
–	15-17 Punch Lane	1989	4	House/Townhouse	–

3.4.2 Aggregation by intersection

Building records are aggregated by intersection and year within the same 500 m square as the image patch, so that imagery and land-use attributes describe the same area. The median number of buildings within a study intersection’s square is 140. Aggregation gives 43 attributes per intersection per year:

- **Building stock (17 attributes)**: the number of buildings, the mean, maximum, total and 90th percentile of the number of floors, the numbers of buildings with 10 or more and 30 or more floors, bicycle parking spaces, the number of buildings completed in the last five years and the median building age. Distance is expressed in three ways: the same statistics within the central 250 m square, totals weighted by the inverse of the distance to the intersection, and the distances to the nearest buildings with 10 or more and 30 or more floors.

- **Land-use mix (26 attributes)**: each of the 13 use classes has two attributes, its share by number of buildings and its share by number of floors (that is, weighted by building height).

The full definition of each attribute is given in Appendix B. The sensitivity analysis in Chapter 5 changes the floor-weighted shares of office, residential apartment and retail use and the total number of floors.

3.4.3 Land-use mix in the study area

Office and residential uses coexist in the study area, but their proportions vary widely between intersections (Figure 3.6). In 2024:

- **Office**: the median share of floors within an intersection's square is 14%, reaching 46% in the Hoddle Grid, and office is the largest use at 19 intersections.
- **Residential** (residential apartments, houses and townhouses, and student accommodation): the median share is 46%, and residential is the largest use at 46 intersections, mainly in Carlton and East Melbourne.
- **Retail**: few buildings have retail as their predominant use. The median share is 1% and the maximum 12%. Street-level shops in mixed-use buildings are recorded under the predominant use of the whole building and are not counted as retail.

The building stock also changes from year to year: between 2014 and 2024, the median growth in the total number of floors within an intersection's square was 9%.

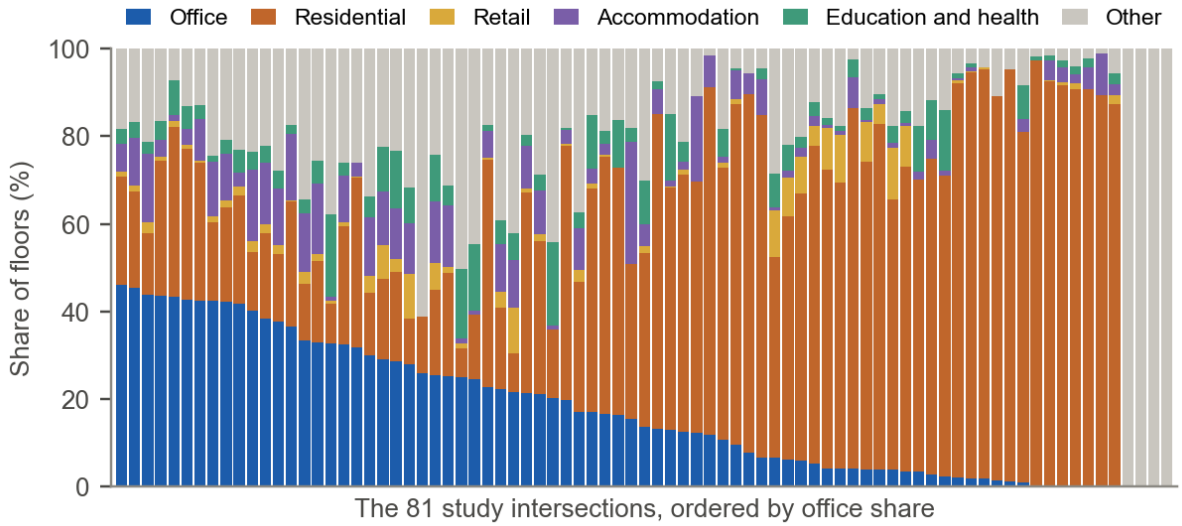


Figure 3.6: Land-use mix at the 81 study intersections in 2024, as the share of floors of each use within each intersection's 500 m square. The 13 CLUE use classes are grouped into six for display.

3.5 Road network data

3.5.1 OpenStreetMap data extraction

Road geometry and attributes for the study area and its surroundings were downloaded from OpenStreetMap (OpenStreetMap contributors, 2026) through the Overpass API,

10,755 road segments (ways) in all, with classes from motorway to service. The road class, number of lanes, speed limit and name of each segment are used.

3.5.2 Intersection matching and road-class groups

Each intersection takes the highest OpenStreetMap road class (OSM class) among the roads that pass within 40 m of it. All 81 intersections were matched: 12 to trunk roads, 18 to primary, 25 to secondary, 21 to tertiary, 3 to unclassified and 2 to service roads.

In the sensitivity analysis, the 55 trunk, primary and secondary intersections form the **main-road group**, and the 26 intersections of tertiary and lower classes form the **local-road group**. The intersections of each class are located as follows:

- **trunk**: on Hoddle Street, Punt Road, Alexandra Parade, Princes Street and City Road.
- **primary**: mainly on Victoria Parade, Wellington Parade and Nicholson Street.
- **secondary**: mainly on Flinders Street, Russell Street, Spring Street, Rathdowne Street and Elgin Street.
- **Local-road group**: lower-volume streets in the Hoddle Grid (such as Exhibition Street, Collins Street, Bourke Street and the pedestrian-priority section of Swanston Street) and residential streets in Carlton and East Melbourne.

Figure 3.7 maps the road classes, and Table 3.6 compares the traffic characteristics of each class.



Figure 3.7: OpenStreetMap road classes in the study area. Intersections are coloured by their class, and their size shows the weekday AM peak volume.

3.5.3 Traffic patterns by road class

Volumes fall step by step with road class (Table 3.6). In the weekday AM peak window, trunk intersections carry 1,257 vehicles per 15 minutes (veh/15 min) on average, 4.2 times as many as tertiary intersections (299 vehicles).

The shape of the daily profile also differs between classes (Figure 3.8). Trunk intersections rise first in the morning and then stay near their peak for most of the working day: on an average weekday they spend 11 hours above 90% of their peak. Primary, secondary and tertiary intersections have distinct morning and evening peaks and spend only 3–5.25 hours near their peak. This flat, sustained profile shows that trunk roads operate under heavy load for most of the day. Chapter 5 returns to this pattern when explaining how roads of different classes respond to land-use change.

Table 3.6: Road classes of the study intersections and their traffic characteristics (weekdays, 2014–2019 and 2022–2024)

OSM class	Intersections	Mean lanes	AM window volume (veh/15 min)	Hours within 90% of peak	Peak / daily mean
trunk	12	3.6	1,257	11	1.47
primary	18	2.9	740	3.5	1.73
secondary	25	1.9	491	5.25	1.58
tertiary	21	1.4	299	3	1.70
unclassified	3	1.5	128	2.5	1.86
service	2	–	87	2.25	1.66

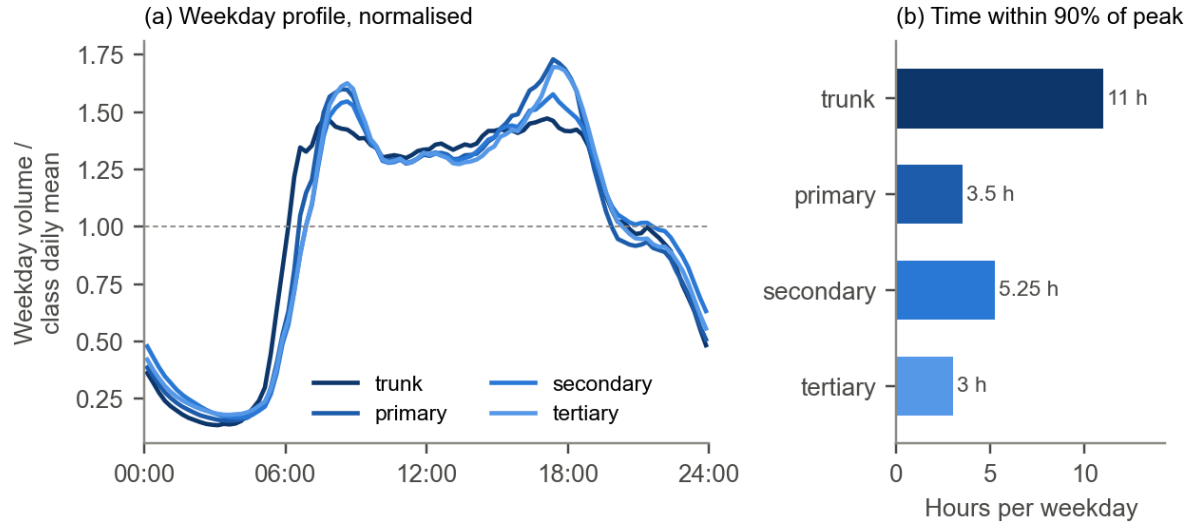


Figure 3.8: Weekday traffic characteristics by road class. (a) Mean volume in each interval divided by the all-day mean volume of the class. (b) Number of hours in which the mean volume of the class is above 90% of its peak.

3.5.4 Vicmap Transport road data

The Vicmap Transport road centreline data are published by the Victorian Department of Transport and Planning (Department of Transport and Planning, 2026b) and are the official road centrelines for the whole state. They are not an input to the prediction model and are used only in image segmentation (Section 4.2.3), to fill in road surfaces that the segmentation model misses.

3.6 City of Melbourne development activity data

3.6.1 Data description

The City of Melbourne publishes two datasets on developments in the municipality:

- **Development Activity Monitor**: records each development's planning application number, status, completion year, height, number of dwellings, and commercial and industrial floor area (City of Melbourne, 2026c).
- **Development Activity Model Footprints**: gives the building footprint of each development as a polygon (City of Melbourne, 2026b).

Linked by development, 355 developments have both a record and a footprint, with completion years from 2012 to 2026.

3.6.2 Development library of the tool

The development scenario imagery tool (Chapter 6) includes 20 of these developments, which can be placed on any base image and cover a range of heights, footprints and uses (Table 3.7). Users can also draw their own building footprints, so the library is only a starting point and does not limit the tool's use.

Table 3.7: Developments with building footprints and the development library of the tool

	All developments with footprints	Development library
Developments	355	20
Completion years	2012–2026	2018–2026
Height range (m)	4.0–318.7	11.8–211.2
Median height (m)	44.0	43.2
Median footprint (m^2)	887	813
Developments with dwellings	225	16
Developments with commercial floor area	202	12

3.7 Data split

The data are split by year rather than at random, because the model's task is to predict future years from past years (Table 3.8):

- **Training set**: 2014–2019 and 2022.
- **Validation set**: 2023, used to decide when to stop training and to select the post-processing weights (Section 4.7).
- **Test set**: 2024, used only for the results reported in Chapter 4.

The historical profile that the model's predictions refer to (Section 4.7.1) is computed from the most recent training year, 2022.

The years 2020 and 2021 are not used. Melbourne's COVID-19 lockdowns caused large and uneven falls in traffic volumes in those years (Figure 3.9): the median daily volume per study intersection fell from 27,862 vehicles in 2019 to 15,368 in 2020, a fall of 45% (34% across all 143 intersections of the original study area). It was 20,026 in 2021 and recovered to 22,925 in 2022. Volumes in these two years reflect travel restrictions rather than the relationship between land use, the built environment and traffic volumes that the model is meant to learn.

Table 3.8: Training, validation and test sets (the test set includes intersection 4889, which has no data in 2024, so the test results cover 80 intersections)

Set	Years	Intersections	Site-days	15-minute intervals
Training	2014–2019, 2022	81	204,052	19,588,992
Validation	2023	81	29,466	2,828,736
Test	2024	81	29,565	2,838,240

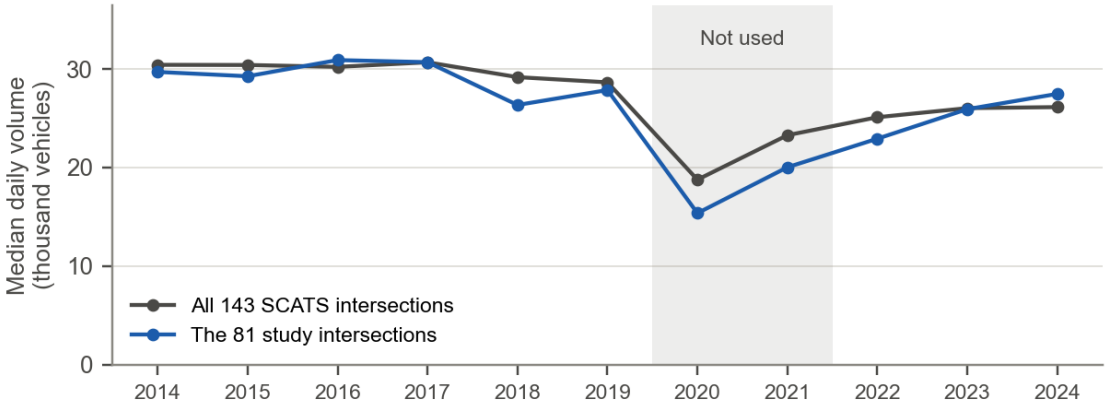


Figure 3.9: Median daily volume per intersection in each year, for all SCATS intersections of the original study area and for the study intersections. Shaded years are not used in the model.

3.8 Data licensing and conditions of use

Apart from the Nearmap imagery, all data are open data that can be reused with attribution (Table 3.9). The Nearmap imagery was obtained through the University of Melbourne's subscription and cannot be redistributed. The project's delivery package therefore provides the data tables processed from open data, but for the imagery only the download scripts and manifests, so that users with access can obtain it again.

Table 3.9: Data sources, licences and redistribution

Dataset	Publisher	Licence	Included in the delivery package
SCATS traffic signal volumes	Department of Transport and Planning, Victoria	CC BY 4.0	Scripts and processed tables
CLUE building information	City of Melbourne	CC BY 4.0	Scripts and processed tables
Development Activity Monitor and footprints	City of Melbourne	CC BY 4.0	Scripts and processed tables
OpenStreetMap roads	OpenStreetMap contributors	ODbL 1.0	Scripts and derived road classes
Vicmap Transport road lines	Department of Transport and Planning, Victoria	CC BY 4.0	Download scripts
Nearmap satellite imagery	Nearmap	Commercial subscription (University of Melbourne)	Download scripts and manifests only

3.9 Key findings

1. The study area comprises 81 signalised intersections in central Melbourne: all of the original 143 intersections that appear in the volume data in every year used and are covered by CLUE land-use data.
2. The weekday AM and PM peaks are at 08:30 and 17:15, while weekends have a single midday peak around 12:30. These peaks determine the three analysis windows used in Chapter 5.

3. Volumes vary widely between intersections, with the 90th percentile about 19 times the 10th. Volumes fall with road class, and the AM peak volume at trunk intersections is 4.2 times that at tertiary intersections.
4. On weekdays, trunk intersections spend 11 hours above 90% of their peak, against only 3–5.25 hours for primary, secondary and tertiary intersections, which shows that trunk roads carry heavy loads for most of the day.
5. The land-use mix varies markedly across the study area: the Hoddle Grid is dominated by office use (up to 46% of floors), and Carlton and East Melbourne by residential use.
6. Traffic volumes in 2020 and 2021 were markedly lower under the COVID-19 lockdowns (the median at the study intersections fell by 45% in 2020), so these years are not used. The training set is 2014–2019 and 2022, the validation set 2023 and the test set 2024.

4 Traffic volume prediction model

Traffic volume predictions for planning assessment must be made before a development is built, so they can draw only on information available at the planning stage. The project therefore built a traffic volume prediction model centred on satellite imagery. For each Sydney Coordinated Adaptive Traffic System (SCATS) observation site and each date, the model predicts the traffic volumes in the 96 fifteen-minute intervals of that day from satellite imagery, land use, the intersection's network position and calendar information, without using recent measured volumes. Large pretrained vision foundation models turn the satellite imagery into built-environment features, and a large language model embedding (LLM embedding) encodes each intersection's position relative to its neighbouring signalised intersections. Both are fed, together with land-use attributes and calendar features, into a temporal convolutional network. A comparison of six input configurations shows that satellite imagery makes the most consistent contribution and that the configuration using imagery, the road network and land use together is the most accurate. The method also extends to 371 signal sites across the City of Melbourne.

4.1 Model overview

Each sample of the model is one site-day, and the output is the volume in each of that day's 96 intervals. The inputs fall into five groups:

- **Imagery features**: built-environment features extracted from the 500 m satellite image centred on the intersection (Section 4.2).
- **Land-use features**: Census of Land Use and Employment (CLUE) attributes within the intersection's 500 m square (Section 4.3).
- **Network features**: the intersection's position relative to its neighbouring signalised intersections (Section 4.4).
- **Temporal features**: calendar and public holidays (Section 4.5).
- **Historical profile**: the intersection's mean volume in the same interval on the same day of the week in 2022, used as the reference for the prediction (Section 4.7).

The project keeps the names of its two modelling routes. **Route 1** uses the imagery features, and **Route 2** adds the network features to Route 1. Either route can also take the land-use features. The first three groups describe the built environment around each observation site, with one set per intersection per year that stays fixed within the year, whereas the temporal features change from day to day and from interval to interval. Figure 4.1 shows the model framework.

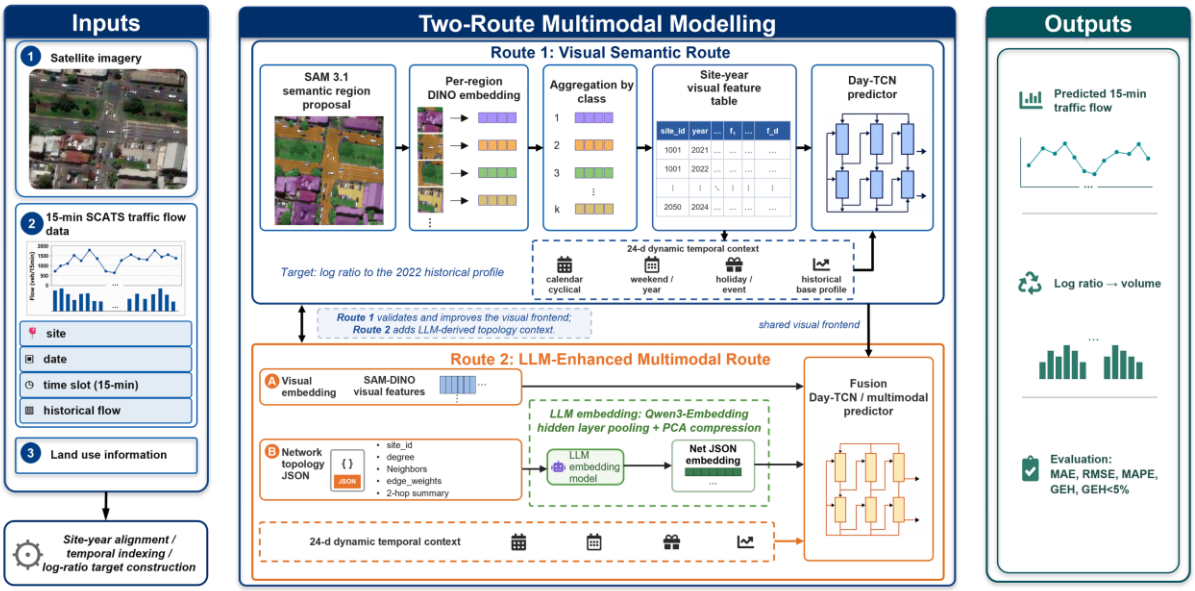


Figure 4.1: Model framework. The left column shows the input data, including the land-use information. The middle column shows the two modelling routes, in which Route 1 describes the satellite imagery with SAM 3 segmentation and DINOv3 region features, and Route 2 adds each intersection's position relative to its neighbouring signalised intersections, encoded by a large language model embedding. The right column shows the prediction outputs and evaluation metrics.

4.2 Imagery features

The imagery features turn each intersection's satellite image for each year into a 135-dimensional vector in four steps (Figure 4.2):

1. **Semantic segmentation**: SAM 3 segments the image into four classes of region, namely road, roof, vegetation, and parking and paved ground.
2. **Region feature extraction**: DINOv3 extracts a 768-dimensional feature vector for each region.
3. **Dimensionality reduction**: principal component analysis (PCA) compresses the 768 dimensions to 16.
4. **Aggregation by intersection**: the features of all regions in an image are aggregated by class, together with geometric measures such as roof area and distances.

SAM 3 and DINOv3 are publicly released pretrained models. The project uses them as released, without retraining or fine-tuning them on local data.

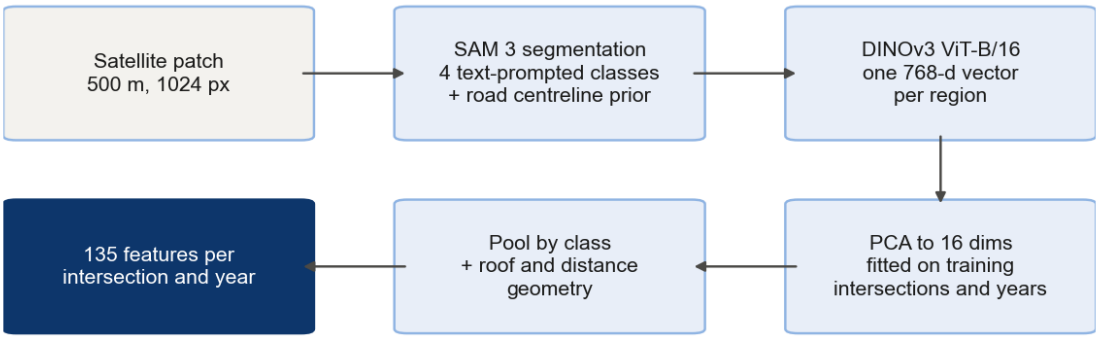


Figure 4.2: Extraction of the imagery features.

4.2.1 Input imagery

The input is the Nearmap image patch of Section 3.3, centred on the intersection and covering 500 m × 500 m in 1024 × 1024 pixels, about 0.49 m per pixel. Segmentation was run on all 1,859 images of the original study area (143 intersections × 13 years), of which the 729 images of the 81 study intersections in the 9 years used enter the model.

4.2.2 The SAM 3 segmentation model

SAM 3 (Segment Anything Model 3) is a foundation model for image and video segmentation released by Meta in 2025 (Carion et al., 2025). It supports "promptable concept segmentation": given a noun phrase such as "building roofs", the model returns masks for all instances of that concept in the image, each with a confidence score. Earlier versions could segment only single objects from points or boxes. The ability to find every instance of a concept at once suits the extraction of roads, roofs and other features that occur many times in satellite imagery.

The upper part of Figure 4.3 shows the structure of SAM 3:

- **Image and text encoders**: encode the image and the prompt into features.
- **Detector**: fuses the image and prompt features, decodes candidate instances and their masks from a set of object queries, and uses a "presence token" to judge whether the prompted concept appears in the image at all, which reduces false detections.
- **Tracker and memory bank**: used for tracking across video frames, and not involved when a single image is processed.

The project uses the SAM 3.1 weights. Images are resized to 1008 × 1008 pixels inside the model, and instances with a confidence below 0.35 are discarded.

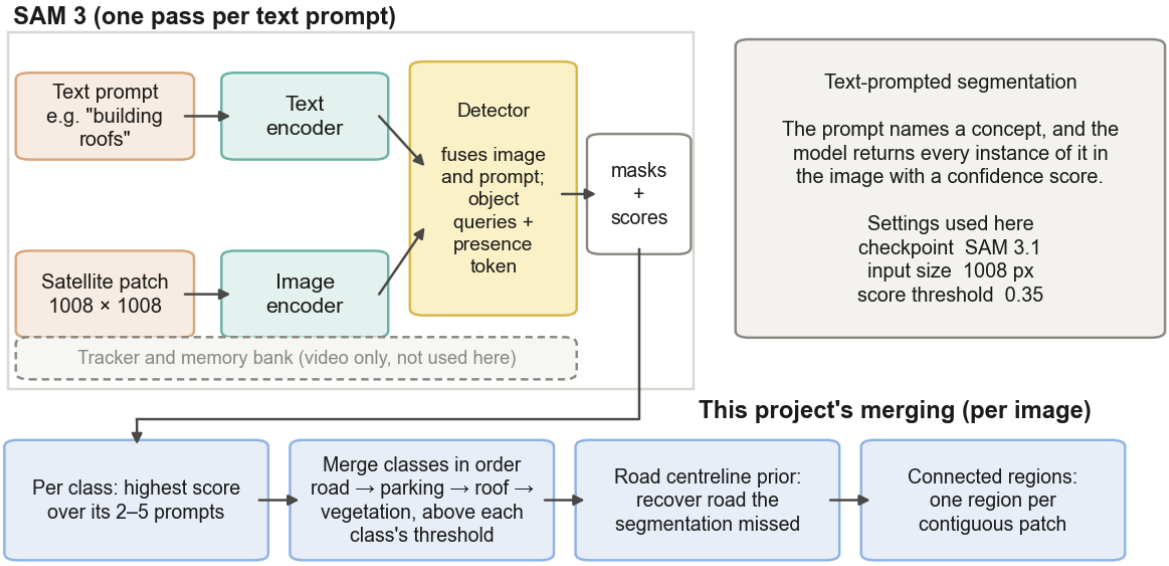


Figure 4.3: Structure of SAM 3 (top) and the project's segmentation and merging process (bottom).

4.2.3 Segmentation process

Each image is segmented in five steps (lower part of Figure 4.3):

1. **Segmentation by prompt**: each of the four classes has 2–5 English prompts (Table 4.1). SAM 3 is run once for each prompt, giving a set of instance masks with confidence scores, and very small fragments are removed.
2. **Merging within a class**: the results of the prompts of one class are merged into one score map, in which each pixel takes the highest confidence among the class's prompts.
3. **Arbitration between classes**: the classes are processed in the order road, parking and paved ground, roof and vegetation. A pixel is reassigned to a class when its score reaches the class's minimum merge score (Table 4.1) and, with the score bonus added, exceeds the score of the class already assigned. Parking and paved ground and roofs can override road when their scores are close to that of road. Pixels assigned to no class are recorded as "other".
4. **Road completion**: roads can be missed where tree canopies or shadows cover them. Vicmap Transport road centrelines (Section 3.5.4) are given widths by road class and rasterised into a road prior. Within the prior, connected areas previously recorded as "other" and not vegetation are reassigned to road.
5. **Region generation**: each connected area of each class becomes a region, and its class, share of the image area, centre position and bounding box are recorded.

Table 4.1: Segmentation prompts and merging parameters

Class	Prompts	Minimum merge score	Score bonus
Road	"asphalt road carriageways and intersections", "street lanes and roadway surfaces seen from above", "urban traffic lanes between kerbs in satellite view"	0.12	+0.00
Parking and paved ground	"surface parking lot with parked cars", "parking bays with white markings", "parking area beside buildings with parked cars", "open parking lot next to buildings"	0.40	+0.00
Building roof	"building roofs", "residential house roofs in overhead satellite view", "commercial and industrial building rooftops", "dark roofs and light roofs of urban buildings", "light colored roof surfaces and roof blocks"	0.29	+0.05
Vegetation	"trees and vegetation", "tree canopy and green vegetation"	0.42	+0.00

The 729 images of the 81 study intersections were segmented into 86,703 regions (Table 4.2). On average, roofs cover 25.8% of the image area, roads 11.6% and vegetation 5.3%.

Table 4.2: Segmentation results for the 81 study intersections (9 years used)

Class	Regions	Mean share of image (%)	Site-years present
Road	43,793	11.6	729 / 729
Building roof	18,855	25.8	673 / 729
Vegetation	24,001	5.3	669 / 729
Parking and paved ground	54	0.1	35 / 729

Segmentation quality varies between areas (Figure 4.4). In the central city, roads, roofs and vegetation are segmented fairly completely. In low-rise residential areas such as Carlton, roads are segmented completely, but in the example image the small residential roofs and the street trees are not detected at all. Figure 4.5 shows how the share of each class varies between images.



Figure 4.4: Segmentation results for the 2014 images of two intersections. (a)(b) Intersection 2024, Spring Street / Flinders Lane, in the central city. (c)(d) Intersection 1034, Faraday Street / Drummond Street, in the Carlton residential area.

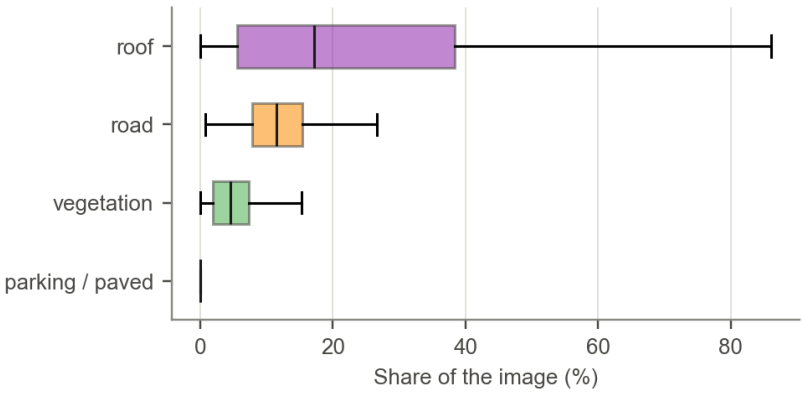


Figure 4.5: Share of the image area covered by each class of region (81 study intersections, 9 years used, one sample per image).

4.2.4 DINOv3 region features

DINOv3 is a self-supervised vision foundation model released by Meta in 2025 (Siméoni et al., 2025), trained on about 1.7 billion unlabelled images (the LVD-1689M dataset). In training, a "student" network learns to reproduce the output of a "teacher" network on different crops of the same image, without any manual labels. Once trained, the model produces features that distinguish differences in appearance such as material, form and texture, and these can be used directly in downstream tasks.

The project uses its ViT-B/16 version, in the process shown in Figure 4.6:

1. **Cropping**: each region is cropped from the image by its bounding box. The surrounding pixels within the box are kept, so the feature reflects both the region and its immediate surroundings.
2. **Preprocessing**: the crop is resized to 224 × 224 pixels and normalised with the mean and standard deviation used in training the model.
3. **Patching and embedding**: the crop is cut into 14 × 14 = 196 patches of 16 × 16 pixels, each linearly mapped to a 768-dimensional vector, and one [CLS] token and four register tokens are added.
4. **Transformer encoding**: the tokens pass through 12 Transformer encoder layers, each with 12-head self-attention (with rotary position embeddings) and a feed-forward network.
5. **Output**: the output of the [CLS] token is the region's 768-dimensional feature vector.

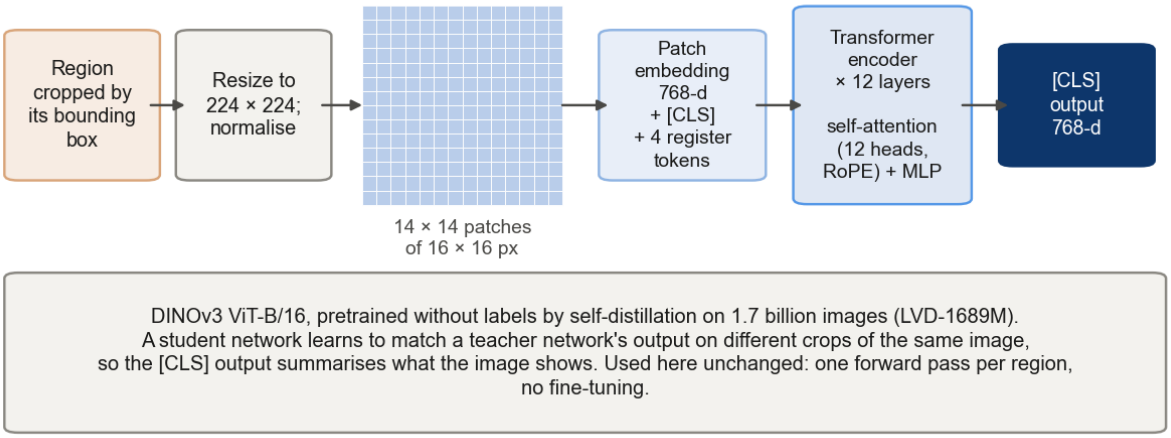


Figure 4.6: Region feature extraction with DINOv3 (ViT-B/16).

4.2.5 Dimensionality reduction

The 768-dimensional features are first standardised dimension by dimension and then reduced to 16 dimensions with PCA. The standardisation and the PCA are fitted only on the regions of the 81 study intersections in the training years (68,630 regions in all). Regions from the validation and test years are only transformed and play no part in the fitting.

Figure 4.7a shows the variance explained by each principal component. The first component explains 16.9% and the first 16 together 48.0%, and the first 32, 64 and 100

together explain 57.9%, 68.7% and 75.7%. Figure 4.7b projects the regions onto the first two components, where vegetation, roads and roofs fall in roughly different places. DINOv3 knows nothing of the segmentation classes, which shows that its features alone distinguish the appearance of different land covers.

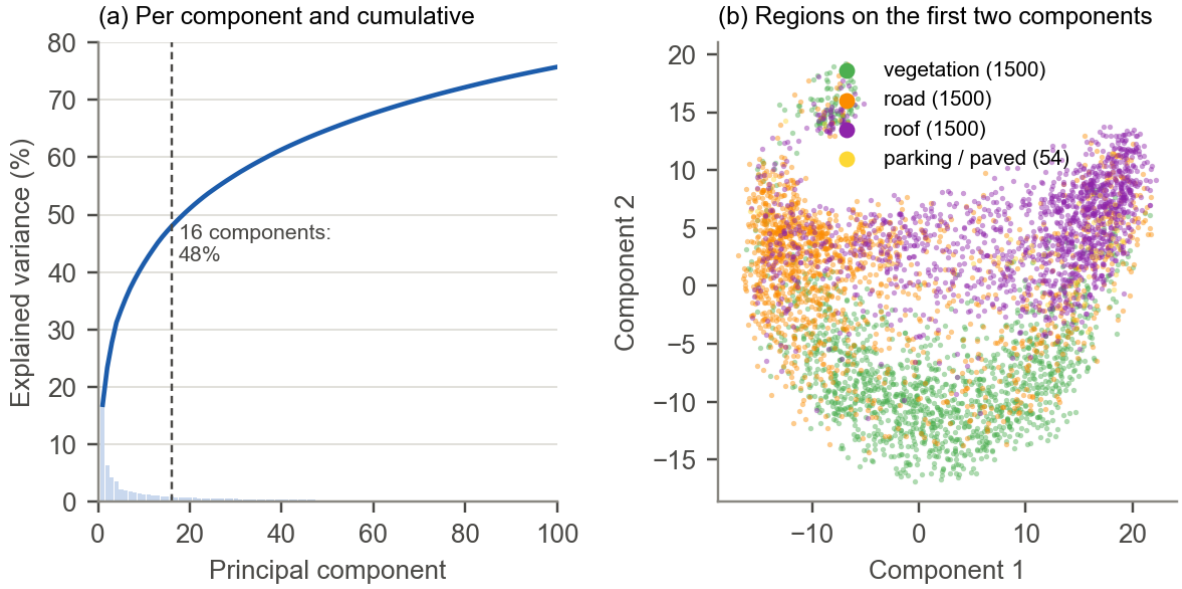


Figure 4.7: Principal component analysis of the region features. (a) Variance explained by each principal component (bars) and cumulative explained variance (line), with the dashed line at the 16 components kept. (b) Distribution of each class of region on the first two components (1,500 regions sampled at random per class, all regions shown for parking and paved ground).

4.2.6 Aggregation by intersection

All regions of an image are aggregated into 135 dimensions as set out in Table 4.3. If an image has K regions, and region k has the 16-dimensional feature $\mathbf{z}_k$ and covers a share a_k of the image, then:

- **Mean of all regions**: $\frac{1}{K}\sum_k \mathbf{z}_k$.
- **Area-weighted mean**: $\sum_k a_k \mathbf{z}_k \,/\, \sum_k a_k$.
- **Class means**: the mean of $\mathbf{z}_k$ over all regions of a class, or 0 if the class has no region.

Geometric measures are computed from the distance between each region's centre and the image centre (the intersection), converted to metres at 0.488 m per pixel. Aggregation is done by class, so the result does not depend on the number or order of the regions. The four classes of the current segmentation include no water class, and regions of the "other" class are not aggregated separately. These two groups, 32 columns in all, are therefore always 0 in this study area and are kept only to preserve the table format.

Table 4.3: Composition of the imagery features

Component	Dimensions	Definition
Mean of all regions	16	Arithmetic mean of the 16-dimensional vectors of all regions in the site-year
Area-weighted mean	16	Mean weighted by each region's share of the image area
Road	16	Mean of the road region vectors (0 if the class has no region)
Building roof	16	Mean of the roof region vectors
Vegetation	16	Mean of the vegetation region vectors
Parking and paved ground	16	Mean of the parking and paved ground region vectors
Water, other	32	Reserved columns, always 0 in this study area
Roof area	4	Roof share of the image area, in three rings by distance from region centre to intersection (0–62, 62–124 and 124–248 m) and in total
Nearest roof distance	1	Distance from the intersection to the nearest roof region centre (m)
Region distance	2	Mean distance from all region centres to the intersection, plain and area-weighted (m)
Total	135	

4.3 Land-use features

The land-use features come directly from the building records of the CLUE census (Section 3.4) and describe the number, height, age and use of the buildings within the square. Compared with the imagery features, they provide information that imagery cannot capture directly, such as the number of floors and the use of a building.

4.3.1 Aggregation method

Each intersection's attributes in each year are aggregated from the CLUE buildings within the 500 m square centred on the intersection. The square is the same as the image patch, so the two kinds of features describe the same area. Aggregation follows these rules:

- Buildings with a missing number of floors count as one floor, and missing bicycle spaces count as zero.
- The inner square is the 250 m square centred on the intersection.
- Distance weighting uses 1/distance as the weight, with a minimum distance of 20 m so that buildings right next to the intersection do not receive excessive weight.
- When the square has no building of 10 (or 30) or more floors, the distance to the nearest building of 10 (30) or more floors is set to 500 m.

4.3.2 Attribute definitions

Aggregation gives 43 attributes (Table 4.4), of which 17 describe the building stock and 26 the land-use mix. Uses fall into 13 classes, namely house or townhouse, residential apartment, office, retail shop, indoor entertainment and recreation, education and research, storage, commercial accommodation, hospital or clinic, student accommodation, workshop or studio, unoccupied and other, and each class has two shares, by number of buildings and by number of floors. The floor-weighted share takes

building height into account: a 40-storey office tower weighs 20 times as much as a two-storey house, which better reflects the floor space actually in use.

Table 4.4: Definitions of the land-use attributes

Attribute	Name	Definition
`buildings`	Buildings	Number of buildings in the square
`floors_mean`	Mean floors	Mean number of floors above ground of the buildings in the square
`floors_max`	Maximum floors	Number of floors of the tallest building in the square
`floors_sum`	Total floors	Sum of the floors of all buildings in the square
`floors_p90`	90th percentile of floors	90th percentile of the number of floors in the square
`tall_over_10`	Buildings of 10+ floors	Number of buildings with 10 or more floors above ground
`tall_over_30`	Buildings of 30+ floors	Number of buildings with 30 or more floors above ground
`bicycle_spaces`	Bicycle spaces	Total bicycle parking spaces recorded for the buildings in the square
`built_last_5y`	Built in the last 5 years	Number of buildings completed within five years before the census year
`median_age`	Median building age	Median of census year minus construction year (years)
`buildings_inner`	Inner buildings	Number of buildings in the central 250 m square
`floors_sum_inner`	Inner total floors	Total floors in the central 250 m square
`floors_max_inner`	Inner maximum floors	Maximum floors in the central 250 m square
`floors_sum_weighted`	Distance-weighted floors	Σ floors ÷ distance to the intersection (minimum distance 20 m)
`buildings_weighted`	Distance-weighted buildings	Σ 1 ÷ distance to the intersection
`nearest_over_10_m`	Distance to nearest 10+ floor building	Distance to the nearest building with 10 or more floors (m, 500 if none)
`nearest_over_30_m`	Distance to nearest 30+ floor building	Distance to the nearest building with 30 or more floors (m, 500 if none)
`share_*` (13)	Share of buildings by use	Buildings of the use ÷ buildings in the square
`floorshare_*` (13)	Floor-weighted share by use	Floors of the buildings of the use ÷ floors in the square

4.3.3 Example

Table 4.5 lists some attributes of the example intersection of Section 3.3.3 (Exhibition / Little Lonsdale) in 2014 and 2024. Over the decade, the total number of floors within its square rose from 1,119 to 1,594, the number of buildings of 30 or more floors from 7 to 16, and the floor-weighted share of student accommodation from 2.0% to 6.0%. These changes correspond to the new towers visible in the imagery in Figure 3.5.

Table 4.5: Land-use attributes of the example intersection (2014 and 2024)

Attribute	2014	2024
Buildings	195	193
Total floors	1,119	1,594
Mean floors	5.7	8.3
Maximum floors	47	64
Buildings of 30+ floors	7	16
Built in the last 5 years	4	6
Inner total floors	328	480
Distance to nearest 30+ floor building (m)	63	63
Office floor share (%)	25.3	21.4
Residential apartment floor share (%)	29.8	37.0
Student accommodation floor share (%)	2.0	6.0
Commercial accommodation floor share (%)	14.6	9.7

4.3.4 Distribution and change

Figure 4.8 compares the main attributes of the 81 intersections in 2014 and 2024. Over the decade, the total number of floors increased at 75 intersections, and its median rose from 613 to 679. The land-use mix also changed: the median floor-weighted share of office fell from 20% to 14%, and that of residential apartments rose from 16% to 19%. The median number of buildings completed in the previous five years fell from 3 to 1.

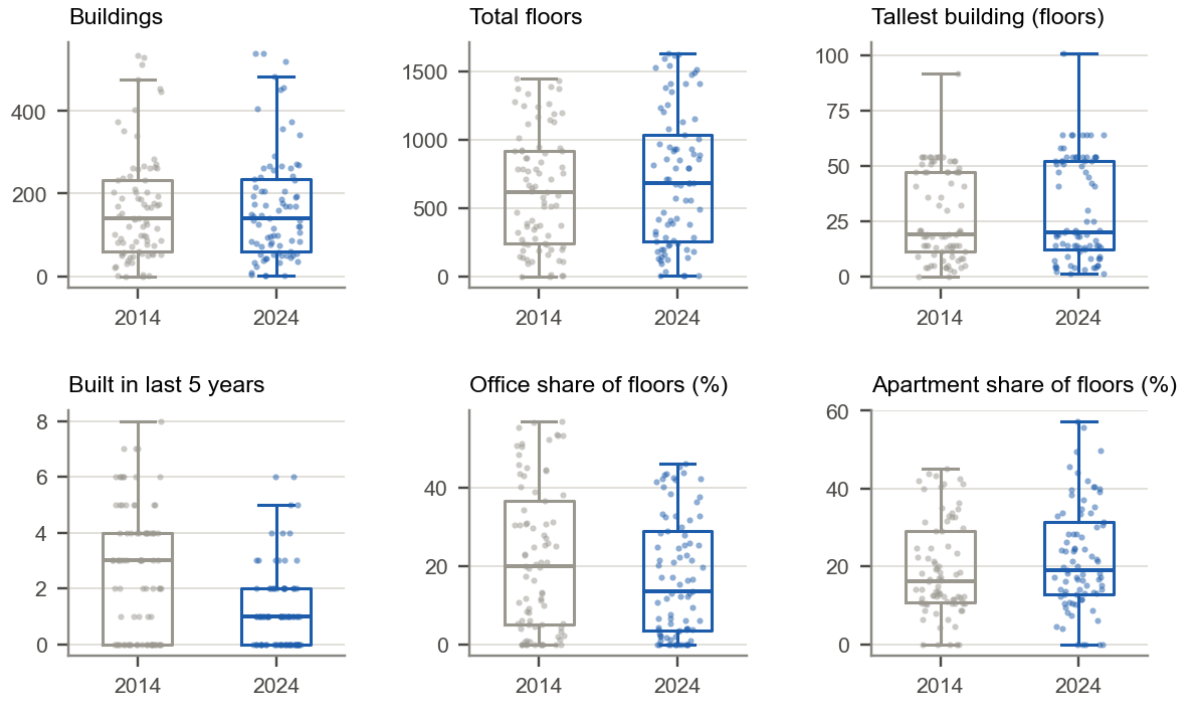


Figure 4.8: Distribution of the main land-use attributes at the 81 study intersections (2014 and 2024). Each dot is an intersection, and the boxes show the interquartile range.

4.3.5 Standardisation

Before entering the model, each attribute is standardised to zero mean and unit variance with the mean and standard deviation of the training years. The imagery and network

features are treated in the same way. The sensitivity analysis in Chapter 5 changes the use shares and the total number of floors in these attributes and examines how the predicted volumes change.

4.4 Network features

The network features describe each intersection's position and connections in the network of signalised intersections. They are built in four steps: an adjacency graph of the intersections is constructed, a structured text is generated for each intersection, a large language model embedding encodes the text as a vector, and the dimensionality is reduced.

4.4.1 Intersection adjacency graph

The adjacency graph is built on all 143 intersections of the original study area, so a study intersection's neighbours can lie outside the study area:

1. For each intersection, the four nearest intersections within 1200 m are found, and each link is weighted by the inverse distance $w_{ij} = 1/d_{ij}$ (d_{ij} in metres).
2. Links are two-way: two intersections are linked when either selects the other as a nearest neighbour, so an intersection can have more than four neighbours.
3. Each intersection is given a self-link, and the rows are normalised so that each row's weights sum to 1.

The resulting graph has 350 links (Figure 4.9). Each intersection has 3–9 neighbours, 4.9 on average, and the median link length is 302 m, with a maximum of 1,166 m. The same graph is used for spatial smoothing in Section 4.7.3.

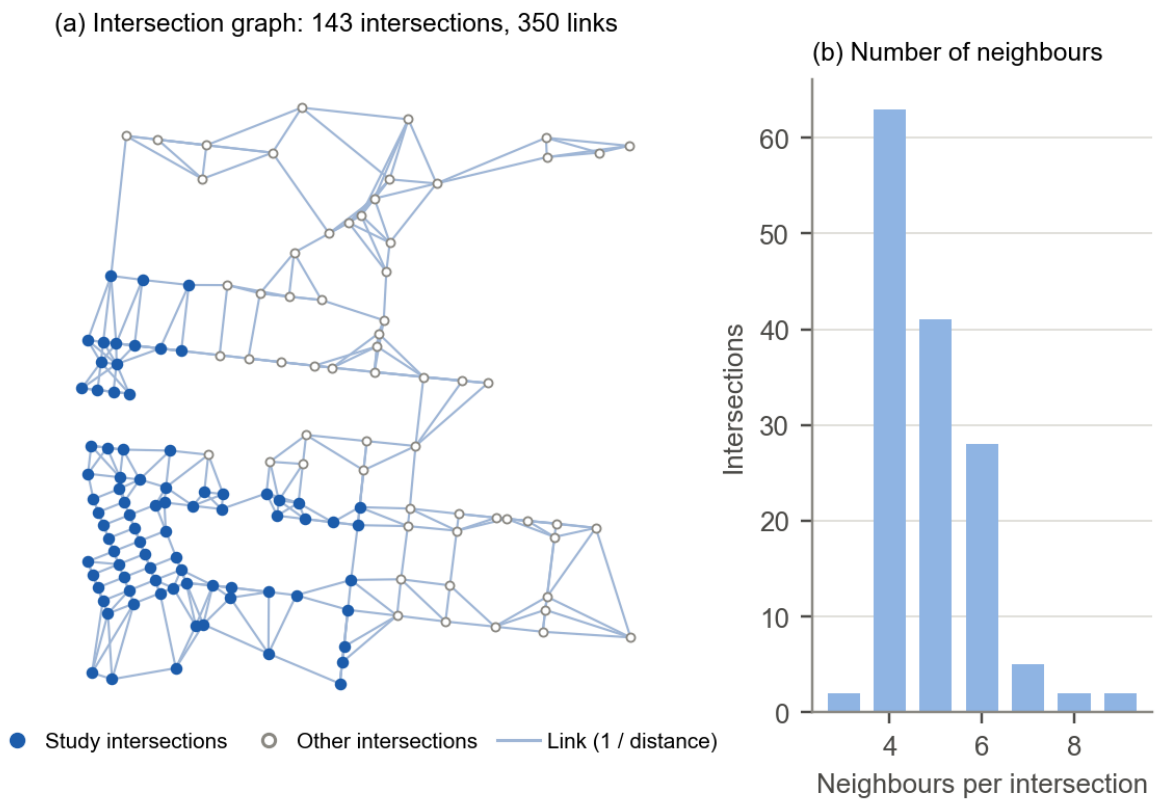


Figure 4.9: Intersection adjacency graph. (a) The 143 intersections and their links, with the study intersections in blue. (b) Distribution of the number of neighbours per intersection.

4.4.2 Structured text

A text in JSON format is generated for each intersection, with the fields listed in Table 4.6 and an example in Figure 4.10. The text contains the intersection's coordinates, its number of neighbours, up to eight neighbours (sorted by weight, each with number, weight and coordinates) and up to six second-order neighbours. The texts for different years are identical apart from the year field and contain no volume information.

Table 4.6: Fields of the network text

Field	Meaning	Example (intersection 2911, 2024)
`sid`	Intersection number	2911
`yr`	Year	2024
`lat`, `lon`	Intersection coordinates	−37.80914, 144.9692
`deg`	Number of neighbours	5
`nbrs`	Neighbours sorted by weight, each with rank `r`, number `id`, weight `w` (1/distance) and coordinates	2919 (0.008), 2902 (0.008), 4405, 2918, 2912
`hop2`	Numbers of second-order neighbours (up to 6)	2920, 4603, 2903, 4406, 4604, 4594
`hop2n`	Number of second-order neighbours	6

(a) Text of intersection 2911, 2024


```
{
 "sid": 2911,
 "yr": 2024,
 "lat": -37.80914,
 "lon": 144.9692,
 "deg": 5,
 "nbrs": [
  {"r": 1, "id": 2919, "w": 0.008, ...},
  {"r": 2, "id": 2902, "w": 0.008, ...},
  {"r": 3, "id": 4405, "w": 0.005, ...},
  {"r": 4, "id": 2918, "w": 0.004, ...},
  {"r": 5, "id": 2912, "w": 0.004, ...}
 ],
 "hop2": [2920, 4603, 2903, 4406, 4604, 4594],
 "hop2n": 6
}
```


(b) Its neighbours (line width = weight)

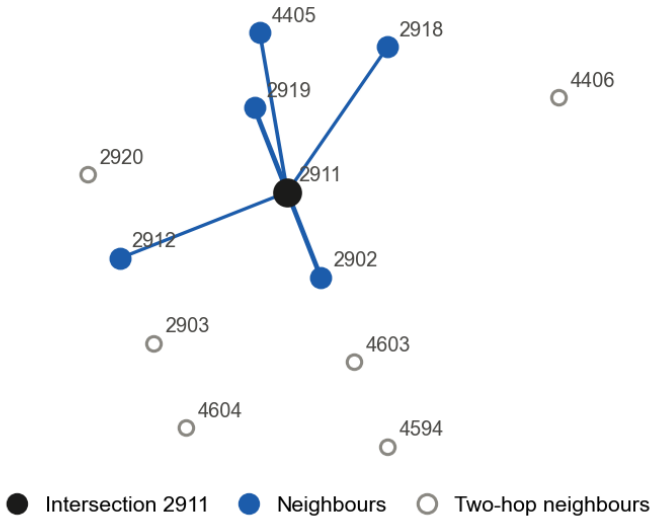


Figure 4.10: Network features of intersection 2911 in 2024. (a) The structured text input to the large language model embedding (neighbour coordinates omitted). (b) The intersection's 5 neighbours and 6 second-order neighbours, with line width showing the link weight.

4.4.3 Large language model embedding

The text is encoded with Qwen3-Embedding-0.6B, a large language model embedding released by Alibaba's Qwen team in 2025 (Zhang et al., 2025). It is based on the Qwen3 large language model, a 28-layer decoder-only Transformer with a hidden size of 1024 and 16 attention heads per layer. It maps texts with similar content to similar vectors, so intersections that are close together and have similar link structures receive similar features.

For encoding, the text is first split into tokens, about 288 on average, and even the longest text stays within the model's length limit and is not truncated. The hidden states of all tokens in the second-to-last layer are then averaged to give a 1024-dimensional vector.

4.4.4 Dimensionality reduction

The 1024-dimensional vectors are reduced with PCA to 64 dimensions, which retain 94.4% of the variance. The reduction is fitted on all 1,859 site-years. Because the text contains only the network structure and no volumes, this step brings no traffic information from the test year into the model.

Figure 4.11 maps the first two principal components for 2024. Both vary continuously over space: the first roughly separates the north of the study area from the south-west, and the second highlights the south-east. The network features thus largely reflect each intersection's location and the structure of its local network.

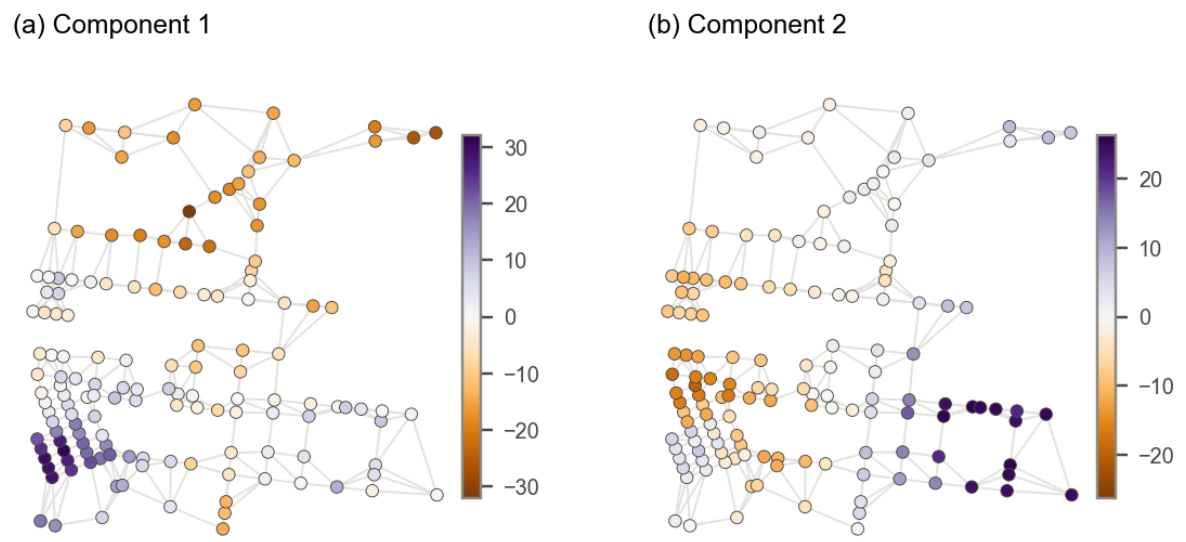


Figure 4.11: Spatial distribution of the first two principal components of the network features (2024, 143 intersections of the original study area). Grey lines are the links of the adjacency graph.

4.5 Temporal features

The temporal features describe the calendar properties of the forecast date and interval, 24 dimensions in all (Table 4.7), and take a value in every 15-minute interval.

4.5.1 Calendar features

There are 10 calendar dimensions. Day of the week, month, day of the year and time of day are cyclical, so each is encoded as a sine and cosine pair. December and January, or 23:45 and 00:00, are then close in value, and the model does not treat them as opposite ends. The day of the year changes with each interval, so the 96 intervals of a day can also be told apart. The year is normalised linearly from 2014–2024 to 0–1 to capture the long-term trend.

4.5.2 Holiday features

There are 14 holiday dimensions, generated from a table of Victorian public holidays compiled by hand in the code. They record whether the day is a public holiday or a restricted trading day, whether the previous or next day is a public holiday, and specific occasions such as the Melbourne Cup, the Friday before the AFL Grand Final, Easter and Christmas. Figure 4.12 shows all 24 features over the week containing the Melbourne Cup (7 November 2023).

Table 4.7: Definitions of the temporal features

Feature	Dimensions	Definition
Day of week	2	sin and cos of 2π·d/7, d being the day of the week (Monday = 0)
Month	2	sin and cos of 2π·m/12, m being the month (January = 0)
Day of year	2	sin and cos of 2π·(j + t/96)/366, j being the day of the year (from 0) and t the interval, so it changes with each interval
Weekend	1	1 on Saturday and Sunday, 0 otherwise
Year	1	(year − 2014) ÷ (2024 − 2014)
Time of day	2	sin and cos of 2π·t/96, t being the interval number (0–95)
Public holiday, restricted trading day	2	The day is a Victorian public holiday, and the day is a restricted trading day (Good Friday, Anzac Day, Christmas Day)
Day before or after a holiday	2	The next day is a public holiday, and the previous day was a public holiday
Specific holidays	9	Melbourne Cup, Friday before the AFL Grand Final, Easter, Christmas, New Year, Australia Day, Labour Day, Anzac Day, King's (Queen's) Birthday
One-off national event	1	One-off holidays such as the National Day of Mourning
Total	24	

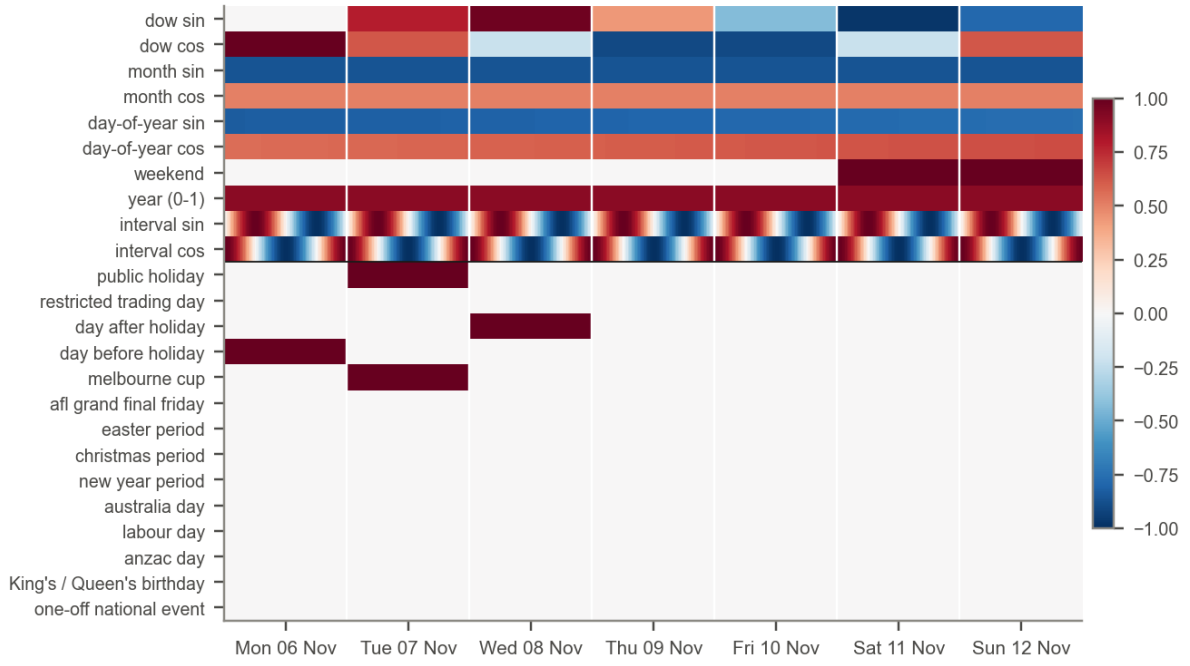


Figure 4.12: Temporal features in the week containing the Melbourne Cup (6–12 November 2023). Each row is one feature and each column one 15-minute interval. Calendar features are above the horizontal line and holiday features below it.

4.5.3 Features not used

The model does not use recent measured volumes, lagged volumes (previous day, same day of the previous week) or measured statistics for the forecast year. These matter greatly in short-term operational prediction but are not available at the planning stage: planning assessment deals with conditions that are not yet built, for which there are neither current-year volumes nor post-development volumes.

4.6 Network architecture

4.6.1 Static branch

The imagery, network and land-use features are projected to 128, 32 and 32 dimensions respectively, each by a linear layer, layer normalisation, SiLU activation and dropout, then concatenated and fused into a 192-dimensional static representation by a further linear layer, layer normalisation and SiLU.

4.6.2 Temporal branch

The 24 temporal features are projected to 32 dimensions in each interval.

4.6.3 Fusion and temporal convolution

The static representation is copied to all 96 intervals, concatenated with each interval's temporal representation and fused into 192 dimensions. It then passes through three residual temporal convolution blocks (Bai et al., 2018), each with two dilated convolution layers (kernel size 3). The dilations of the three blocks are 1, 2 and 4, so the model can use context from both adjacent intervals and a span of several hours. Finally, a 1 × 1 convolution outputs one value per interval. The model with all input branches has 811,793 parameters.

4.7 Prediction target and post-processing

4.7.1 Prediction target

The model does not predict volumes directly but the log ratio relative to the historical profile. Let b be the intersection's mean volume in the same interval on the same day of the week in 2022, and r the model output. The volume is then

$$\hat{y} = (b + 1)\, e^{r} - 1$$

The model thus learns whether volumes are higher or lower than the intersection's typical level, rather than learning from scratch the very different volume levels of different intersections (Section 3.2.4).

4.7.2 Blending with the historical profile

The model output is blended with the historical profile using the weight α:

$$\tilde{y} = \alpha\, \hat{y} + (1 - \alpha)\, b$$

The weight α is searched over 0.60–1.10 in steps of 0.05, and the value with the lowest median GEH statistic (Section 4.11) on the validation set (2023) is chosen, with ties

broken by the lower mean absolute error (MAE). An α above 1 pushes the prediction further away from the historical profile.

4.7.3 Spatial smoothing

Within each interval, each intersection's prediction is averaged with those of its neighbours using the adjacency weights of Section 4.4.1:

$$y^* = (1 - \beta)\,\tilde{y} + \beta \sum_j w_j\ \tilde{y}_j$$

where $\tilde{y}_j$ is the blended prediction of neighbour j (including the intersection itself) and w_j is the row-normalised adjacency weight. The coefficient β is chosen from 0 (no smoothing) and 0.05–0.45 by the same criterion as α.

4.8 Training settings

The loss function has three parts, all computed on the volume scale:

- **Huber loss on 15-minute volumes** (δ = 60 vehicles): quadratic for small errors and linear for large ones (Huber, 1964), so that a few anomalous intervals do not dominate training.
- **GEH term** (weight 0.1): consistent with the GEH statistic commonly used in traffic engineering (Section 4.11), and more sensitive to relative errors in low-volume intervals.
- **Huber loss on hourly volumes** (weight 0.05, δ = 80 vehicles): constrains the total in each hour.

Missing intervals are excluded from the loss. After each epoch, the metrics are computed on the validation set, and the epoch with the lowest median GEH is kept as the final model. Each configuration is trained once with each of 5 random seeds, and the results are reported as the mean and standard deviation of the 5 runs. Of all 30 runs, 18 trained for the full 18 epochs and the rest stopped early (after at least 9 epochs). The settings are listed in Table 4.8.

Table 4.8: Training settings

Item	Setting
Temporal convolution	Hidden size 192, 3 blocks, kernel 3, dilations 1, 2 and 4
Projection sizes	Imagery 128, network 32, land use 32, time 32
Loss	Huber (δ = 60 vehicles) + 0.1 × GEH term + 0.05 × hourly Huber (δ = 80 vehicles)
Optimiser	AdamW, learning rate 5×10^{-5}, weight decay 10^{-4}, gradient clipping 5
Batch size	64 site-days
Epochs	At most 18, stopping after 6 epochs without improvement on the validation set
Model selection	The epoch with the lowest validation median GEH (ties broken by MAE)
Dropout	0.05
Random seeds	42, 73, 128, 2024 and 7
Parameters	753,649–811,793 (depending on the input branches)

4.9 Information available to the model

All inputs of the model are information available at the planning stage, and no traffic information from the test year enters the model in any way:

- The data are split by year, and volumes from the test year (2024) enter no input.
- The historical profile is computed from 2022 alone, one of the training years.
- The dimensionality reduction of the imagery features and the standardisation of all features are fitted on the training years only.
- The dimensionality reduction of the network features is fitted on all years, but the text contains only static network structure and no volumes.
- The blend weight α and the smoothing coefficient β are selected on the validation set (2023) only, and the test set plays no part in any selection.

Table 4.9 lists the source of each input and whether it is available at the planning stage.

Table 4.9: Sources of the model inputs and their availability at the planning stage

Input	Source	At the planning stage
Calendar and public holidays	The forecast date itself	Available
Historical profile	Measured volumes in 2022	Available (existing data)
Imagery features	Satellite imagery of the forecast year	Available (current or scenario imagery at the planning stage)
Land use	CLUE attributes of the forecast year	Available (from the planning proposal)
Network features	Static road network structure	Available
Recent measurements, lagged volumes, same-year statistics	—	Not available, not used

4.10 Model configurations

To assess the separate roles of imagery, the road network and land use, six input configurations were compared (Table 4.10). All configurations use the same data split, network architecture, training settings and post-processing search ranges and differ only in their input branches.

Table 4.10: Inputs of the six model configurations

Configuration	Time features	Historical profile	Imagery	Network	Land use
Time only	✓	✓	–	–	–
Time + land use	✓	✓	–	–	✓
Route 1	✓	✓	✓	–	–
Route 1 + land use	✓	✓	✓	–	✓
Route 2	✓	✓	✓	✓	–
Route 2 + land use	✓	✓	✓	✓	✓

Table 4.11 gives the blend weights and smoothing coefficients selected on the validation set. The four configurations with imagery selected α = 0.60, the lowest value in the search range (Section 4.7.2), for every seed, the two without imagery selected 0.80–1.00, and the smoothing coefficients lie between 0.05 and 0.20.

Table 4.11: Post-processing parameters selected for each configuration

Configuration	Blend weight α (5 seeds)	Smoothing coefficient β (5 seeds)
Time only	0.80–0.95	0.15–0.20
Time + land use	0.85–1.00	0.10–0.20
Route 1	0.60	0.10
Route 1 + land use	0.60	0.05–0.15
Route 2	0.60	0.05–0.15
Route 2 + land use	0.60	0.05–0.15

4.11 Evaluation metrics

The models are evaluated on the test year (2024). Intersection 4889 has no data at all in 2024 (Section 3.2.2), so the evaluation covers the other 80 intersections. Missing intervals (1.43% of all) are excluded from the evaluation. The following metrics are used:

- **MAE**: mean absolute error, in vehicles per 15 minutes (veh/15 min), the main metric of this report.
- **RMSE**: root mean square error, which is more sensitive to large errors.
- **GEH**: a statistic commonly used in calibrating traffic models, GEH $= \sqrt{2(M-C)^2/(M+C)}$, where M and C are the predicted and measured hourly volumes (in this report, 15-minute volumes multiplied by 4). The report gives the median GEH and the shares of intervals with a GEH below 5 and below 10. A GEH below 5 is often used as the criterion for a satisfactory calibration at an individual count site (Department for Transport, 2026).

The mean absolute percentage error (MAPE) is highly unstable in low-volume intervals and is not used in the main text. Full results are given in Appendix C.

4.12 Prediction accuracy

4.12.1 Overall accuracy

Table 4.12 gives the test accuracy of the six configurations. The most accurate is Route 2 + land use, with an MAE of 59.18 veh/15 min, a median GEH of 3.84 and 59.9% of intervals with a GEH below 5. Compared with the time-only baseline, which uses only the temporal features and the historical profile (MAE 62.22), its MAE is 4.9% lower, and the share of intervals with a GEH below 5 is 2.7 percentage points higher.

Table 4.12: Test accuracy of the six configurations (2024, mean ± standard deviation over 5 seeds)

Configuration	MAE (veh/15 min)	RMSE (veh/15 min)	Median GEH	GEH < 5 (%)	GEH < 10 (%)
Time only	62.22 ± 0.52	127.7 ± 1.8	4.14	57.2	81.6
Time + land use	61.36 ± 0.86	130.5 ± 3.5	3.94	59.0	82.4
Route 1	60.55 ± 0.47	122.5 ± 1.5	3.96	58.8	82.0
Route 1 + land use	59.51 ± 0.47	121.4 ± 0.5	3.88	59.6	82.5
Route 2	60.14 ± 0.95	122.3 ± 1.9	3.93	59.1	82.3
Route 2 + land use	59.18 ± 0.50	120.9 ± 0.8	3.84	59.9	82.6

4.12.2 Contribution of each input

The six configurations are compared in pairs with paired t-tests, using the random seeds as the pairing unit (Table 4.13, Figure 4.13):

- **Imagery**: adding the imagery features reduces MAE by 1.67, with all 5 seeds improving (p = 0.002), which makes imagery the most consistent contributor.
- **Land use**: adding land use to Route 1 reduces MAE by 1.04, with all 5 seeds improving (p = 0.028). On the baseline without imagery and on Route 2, MAE also falls (0.87 and 0.97), but the falls are not statistically significant. In addition, adding land use to the baseline without imagery raises RMSE by 2.8, which shows that it reduces the error in most intervals but introduces larger deviations in a few.
- **Road network**: adding the network features to Route 1 changes MAE by −0.41, with only 3 seeds improving (p = 0.384), and the contribution is not statistically significant.
- **All inputs**: the three inputs together reduce MAE by 3.05 (p = 0.001).

Table 4.13: Contribution of each input (MAE change is the mean over 5 seeds, negative values are improvements)

Input added	Comparison	MAE change	p	Seeds improved
Imagery	Route 1 vs Time only	−1.67	0.002	5/5
Land use (no imagery)	Time + land use vs Time only	−0.87	0.218	4/5
Land use (on Route 1)	Route 1 + land use vs Route 1	−1.04	0.028	5/5
Land use (on Route 2)	Route 2 + land use vs Route 2	−0.97	0.124	4/5
Road network	Route 2 vs Route 1	−0.41	0.384	3/5
Road network (with land use)	Route 2 + land use vs Route 1 + land use	−0.34	0.356	3/5
All inputs	Route 2 + land use vs Time only	−3.05	0.001	5/5

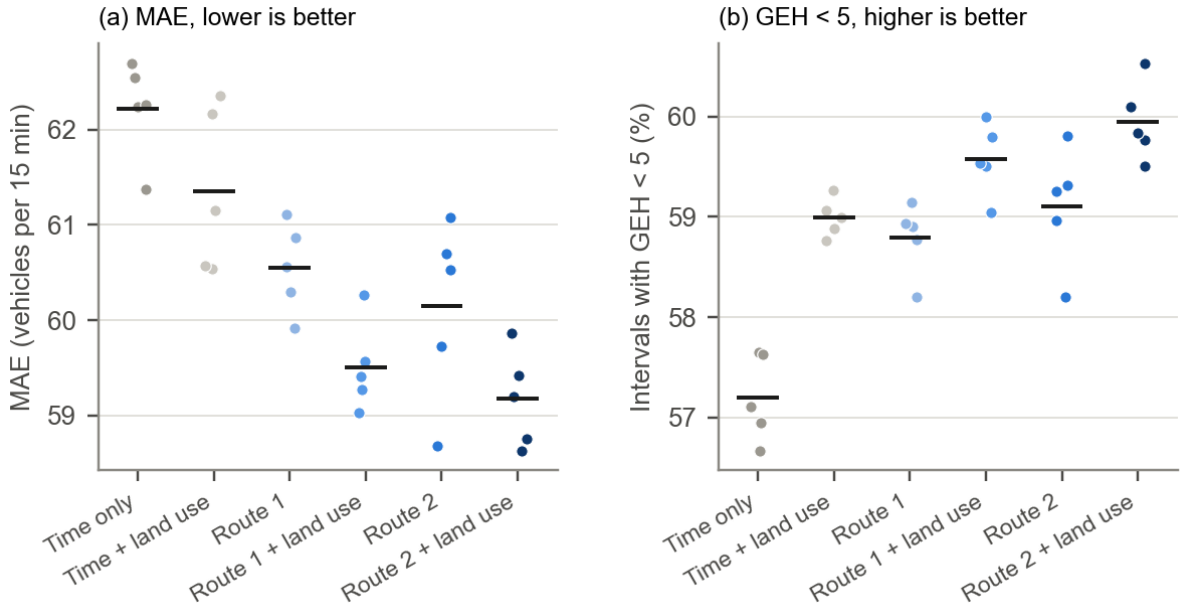


Figure 4.13: Test accuracy of the six configurations over the 5 seeds. (a) MAE. (b) Share of intervals with a GEH below 5. Dots are individual seeds and horizontal lines are the means.

4.12.3 Error by time of day

The error varies over the day, broadly in step with volume (Figure 4.14). In the three analysis windows defined in Section 3.2.3, the MAE of Route 2 + land use is 83.0 in the weekday AM peak, 80.5 in the weekday PM peak and 74.3 veh/15 min at weekend midday, about 14.5%, 13.7% and 14.1% of the mean volume in each window. Compared with the time-only baseline, the improvement is concentrated in the weekday peaks: MAE falls by 9.4 in the AM peak, 5.0 in the PM peak and 2.9 at weekend midday. Overnight (00:00–06:00), the absolute error is smallest, but because volumes are low, the relative error is largest (about 26.7%).

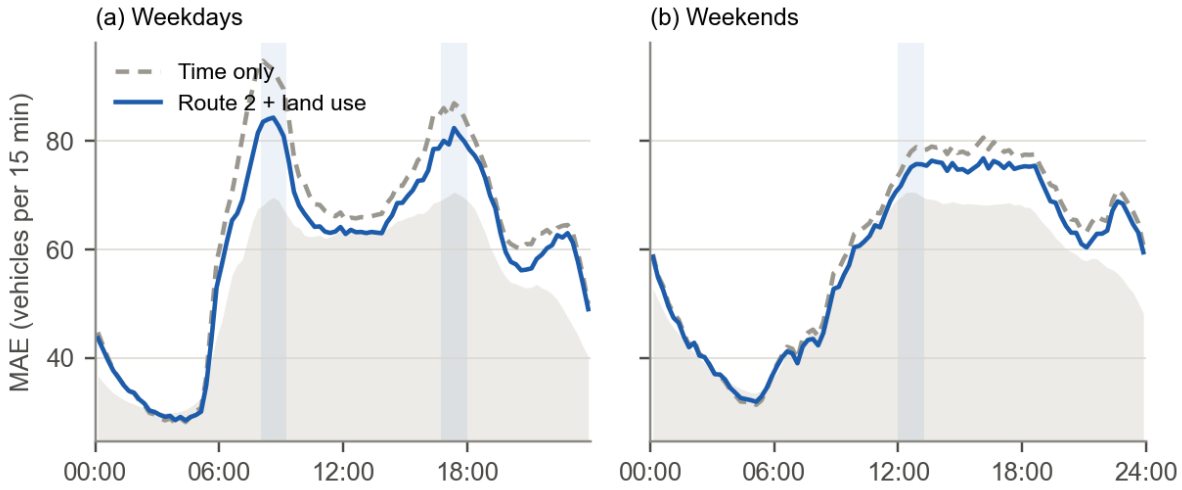


Figure 4.14: Mean absolute error in each interval. (a) Weekdays. (b) Weekends. The grey background shows the daily profile of mean volume (shape only), and blue shading marks the analysis windows of Section 3.2.3.

4.12.4 Error by road class and volume level

The relative error of each group (MAE ÷ mean volume) is computed from the group's mean volume, with the results shown in Figure 4.15:

- **Road class**: the relative error is lowest at trunk intersections (11.3%) and is 15.7%, 19.9% and 20.5% at primary, secondary and tertiary intersections. Unclassified and service intersections carry very low volumes, and their relative errors reach 43.8% and 62.6%.
- **Main and local roads**: compared with the time-only baseline, the MAE of the main-road group falls by 4.4 on average, while the local-road group barely changes (a fall of 0.1). The improvements from imagery, the road network and land use appear mainly on main roads.
- **Volume level**: the high-volume group has the lowest relative error (12.6%), and the medium- and low-volume groups have 22.3% and 21.7%.

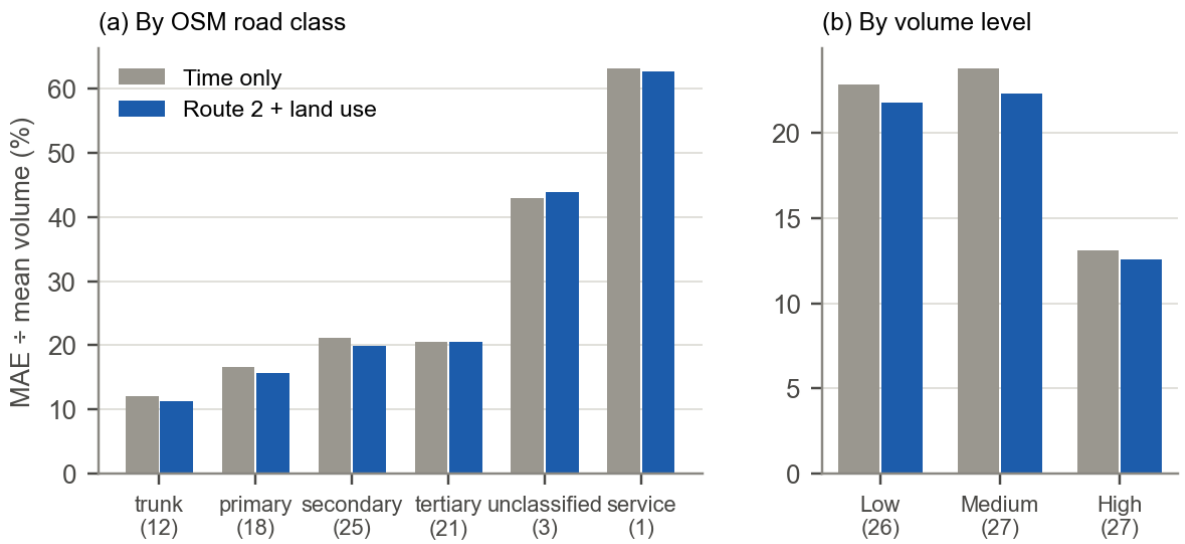


Figure 4.15: Relative error (MAE ÷ mean volume) of different groups of intersections. (a) By OpenStreetMap road class. (b) By the volume groups of Section 3.2.4. The numbers of intersections are in brackets.

4.12.5 Spatial distribution of error

Figure 4.16 shows the change in error at each intersection (Route 2 + land use minus the time-only baseline). Of the 80 intersections, 66 have a lower MAE, and the median change is −2.0 veh/15 min. In relative terms, the median intersection has an error of 14.9%, and 56 intersections are below 20%.

The intersections with the largest relative errors are, in order, 2990, 4588, 4586, 4409, 4529 and 4446, and the reasons differ (see Section 3.2.2):

- Intersections 2990 (MCG access) and 4588 have many days recorded as zero in the test year, which differ greatly from the model's predictions of normal traffic.
- At intersection 4586, the zeros occurred in 2022, so the historical profile used as the reference is almost zero (a mean of 2.5 veh/15 min), whereas the measured mean in 2024 is 23.9 veh/15 min, and the model badly underestimates.
- At intersection 4409, the records for March–April 2024 are unusually high (a mean of 1,847 veh/15 min and a maximum of 3,244 vehicles in a single interval) and stay high until early May, while in the other months (January–February and June–December) the mean is only about 47% of the 2022 level.
- At intersections 4446 and 4529, the mean volume in 2024 changed by +100% and −25% relative to 2022.

These errors stem mainly from data anomalies or large changes in volume, which cannot be foreseen from the inputs available at the planning stage.

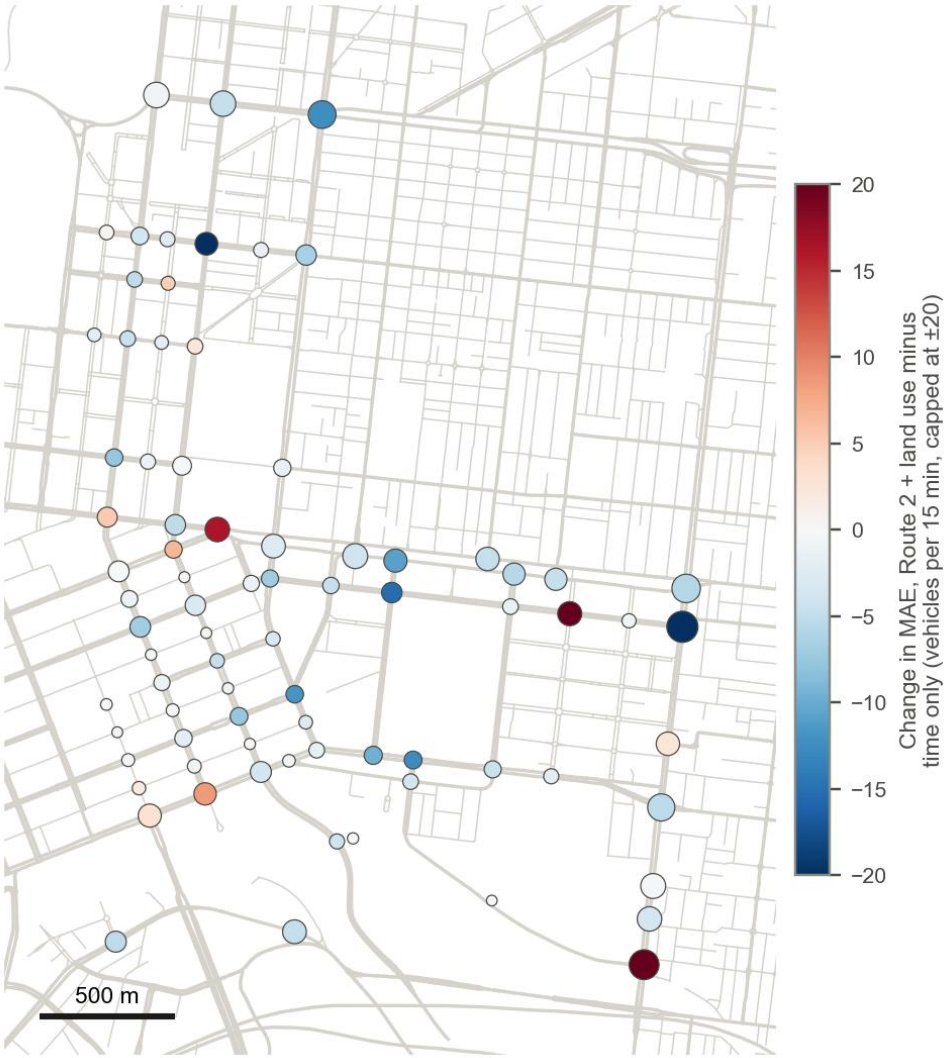


Figure 4.16: Change in MAE at each intersection (Route 2 + land use minus the time-only baseline). Blue shows improvement and red deterioration, and dot size shows mean volume.

4.12.6 Example intersections

Figure 4.17 shows the measured and predicted volumes at four intersections over one week in May 2024. The first three have the median relative error among the trunk, secondary and tertiary intersections, and the fourth has a large error:

- **(a) 3391 Punt / Rowena** (trunk): both the daily pattern and the level are predicted well, with a relative error of 9% over the year.
- **(b) 4561 Russell / Flinders** (secondary): the pattern matches, but daytime volumes are overestimated by about 100 veh/15 min on most days, with a relative error of 15% over the year.
- **(c) 4575 Spring / Collins** (tertiary): the pattern matches, but short-term fluctuations in the peaks are not predicted, with a relative error of 17% over the year.
- **(d) 4409 Rathdowne / Elgin** (secondary): measured volumes in this week are below the usual level, and both configurations clearly overestimate. This

intersection also has unusually high values in March–April 2024 (Section 4.12.5), and its relative error over the year is 69%.

At these intersections, Route 2 + land use and the time-only baseline predict similar patterns and differ mainly in level.

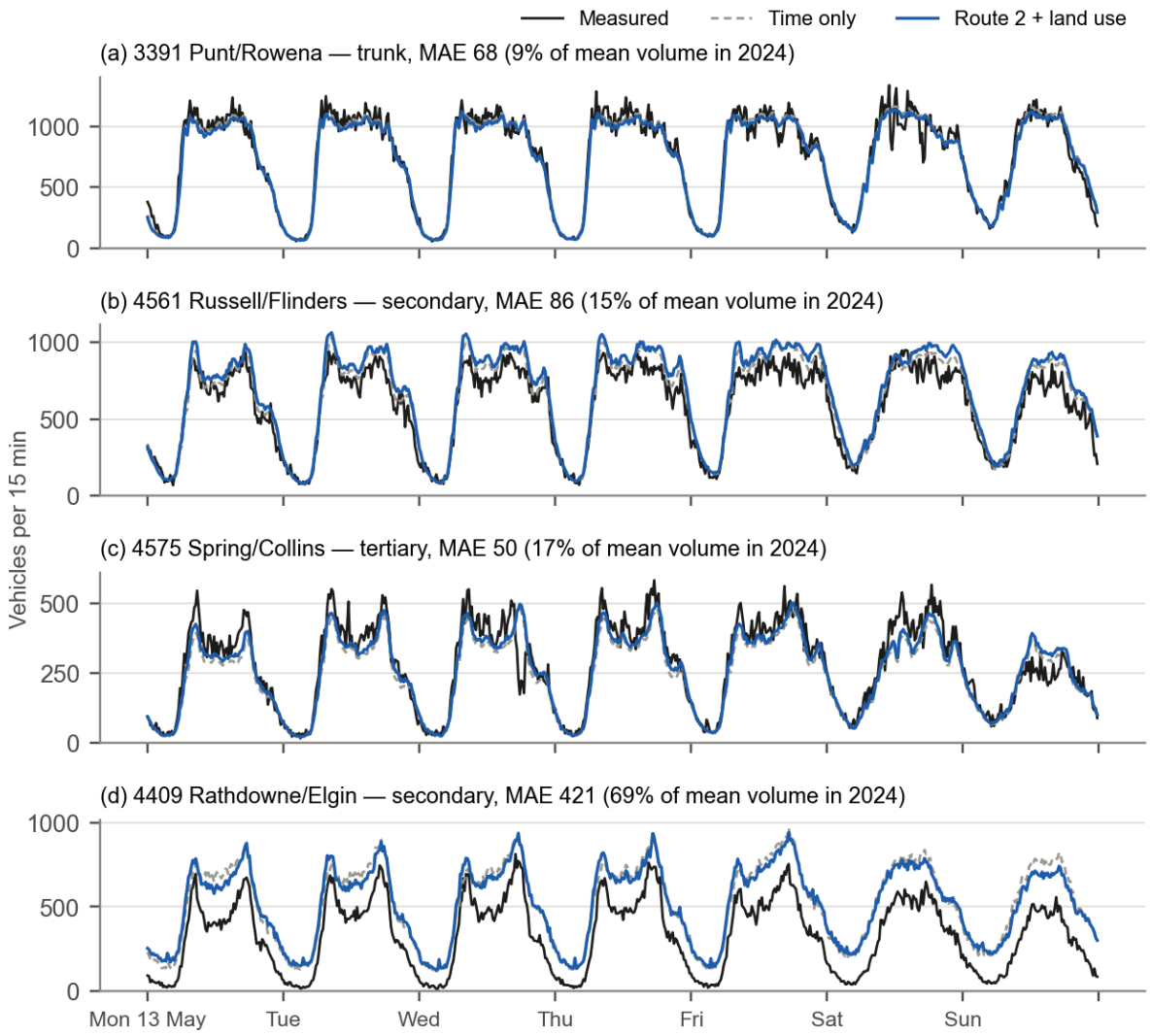


Figure 4.17: Measured and predicted volumes at four intersections, 13–19 May 2024. The MAE and relative error in the titles are for the whole of 2024.

4.13 Extension to 371 signal sites

To test how the method performs over a wider area, the four configurations with imagery were trained on the 371 signal sites of Section 3.1.2 with the same data split, network architecture and training settings, with five seeds each. The adjacency links connect each site to its eight nearest signal sites within 1200 m. No time-only baseline was trained for this set, so the contribution of the imagery cannot be measured here, and only the four configurations with imagery are compared. The results are given in Table 4.14.

The MAE of the four configurations lies between 58.76 and 59.25, a spread of no more than 0.49. Route 2 + land use is again the best, with an MAE 0.35 lower than that of

Route 1 (p = 0.048). The effect of land use differs between the routes. Adding land use to Route 1 brings no improvement, changing MAE by +0.14 (p = 0.358), with lower MAE in only 2 of the 5 seeds. Adding land use to Route 2 gives a small reduction in MAE of 0.48 (p = 0.044), with lower MAE in 4 of the 5 seeds. All four configurations selected a blend weight of 0.60 and a smoothing coefficient of 0.05.

The 371 signal sites include 79 of the 81 study intersections, 78 of which have data in 2024 (intersection 4889 has none). At these 78 intersections, the MAE of Route 2 + land use is 58.02, against 57.15 for the model trained on the 81 intersections alone, 0.87 veh/15 min higher. The method therefore extends to a wider area with accuracy close to that of the study-area model.

Table 4.14: Test accuracy of the four configurations on the 371 signal sites (2024, mean ± standard deviation over 5 seeds)

Configuration	MAE	RMSE	Median GEH	GEH < 5 (%)	MAE (78 shared intersections)
Route 1	59.11 ± 0.33	114.6	4.13	57.2	58.33 ± 0.45
Route 1 + land use	59.25 ± 0.28	115.2	4.11	57.4	58.52 ± 0.57
Route 2	59.24 ± 0.44	115.0	4.13	57.2	58.17 ± 0.35
Route 2 + land use	58.76 ± 0.26	114.2	4.11	57.4	58.02 ± 0.28

4.14 Key findings

1. The imagery features are built from SAM 3 segmentation and DINOv3 region features. On average, roofs cover 25.8% of each image, roads 11.6% and vegetation 5.3%.
2. The most accurate configuration is Route 2 + land use, with a test-year MAE of 59.18 veh/15 min, a median GEH of 3.84 and 59.9% of intervals with a GEH below 5. Its MAE is 4.9% lower than that of the time-only baseline, which uses only the temporal features and the historical profile.
3. The imagery features make the most consistent contribution (MAE −1.67, p = 0.002). Land use improves accuracy on Route 1 by a statistically significant margin (−1.04, p = 0.028) and is consistent in direction but not statistically significant in the other combinations. The contribution of the network features is not statistically significant.
4. The improvement is concentrated in the weekday peaks and on main roads: MAE falls by 9.4 in the AM peak and by 4.4 on average in the main-road group, while the local-road group barely changes.
5. The intersections with the largest errors are affected mainly by data anomalies (zero records and anomalous counts) or large changes in volume, which cannot be foreseen from the inputs available at the planning stage.
6. On the 371 signal sites, the MAE of the four configurations differs by no more than 0.49, and at the 78 shared intersections with 2024 data the MAE of Route 2 + land use (58.02) is close to that of the 81-intersection model (57.15), so the method extends to a wider area. Land use brings no improvement to Route 1 and gives Route 2 a small reduction in MAE of 0.48.

5 Land-use sensitivity analysis

Urban development affects traffic volumes by changing the scale and mix of land use, and planning assessment needs to know how developments of different uses and sizes may change traffic volumes. The land-use sensitivity analysis builds on the prediction models trained in Chapter 4. With the model parameters held fixed, it changes only the land-use attributes of the test year (2024) and compares the predictions before and after the change. It thereby quantifies how changes in the scale and mix of office, residential apartment and retail uses affect predicted traffic volumes, and how the effects differ between times of day and road classes. The results show that office is the most sensitive use when uses are scaled by the same multiple, and that raising the office share while keeping the total number of floors unchanged gives a statistically significant increase in predicted traffic volumes. The response occurs mainly in the periods before and after the peaks and on secondary and local roads.

5.1 Approach

The analysis uses the two configurations of Chapter 4 that include land-use features, Route 1 + land use and Route 2 + land use (the most accurate configuration), each with 5 seeds, giving 10 trained models. The model parameters are held fixed, and the models are not retrained. Each model first predicts the whole of 2024 with the original attribute table, as the base, and then predicts again with the attribute table replaced by a changed version. Each of the three uses has five changes (Section 5.2), so with the base each model makes 16 predictions, 160 in all.

Only the test year 2024 is changed, and only at those of the 81 study intersections where the floor-weighted share of the use is above zero, because a use that is absent cannot be scaled up or down. The imagery, network and temporal features are unchanged, so the response reflects only the model's reaction to changes in the land-use attributes.

The response is defined as (mean predicted volume after the change − mean predicted volume of the base) ÷ α, in vehicles per 15 minutes (veh/15 min). Here α is the blend weight with the historical profile from Section 4.7.2, and all seeds of both configurations selected α = 0.60. The division by α is made because the final prediction is α × model output + (1 − α) × historical profile. The historical profile does not change with the attributes and cancels out when the predictions before and after the change are subtracted, and spatial smoothing (Section 4.7.3) is a linear operation that does not alter this relationship. The difference between predictions therefore equals α times the difference between model outputs, and dividing by α gives the change in the model output itself. In the final reported predictions, the change is 0.60 times the response given here.

The response is averaged over all site-days of a group of intersections and over the specified intervals. Intersections in the group that were not changed are included in the average. They are affected only through spatial smoothing with neighbouring intersections, and their changes are very small (for the office swap, no more than 0.011 veh/15 min in the weekday AM peak). To compare roads with different volume levels, the response is also expressed as a percentage of the group's measured mean volume in 2024, excluding missing intervals.

The response is the model's reaction to a change in its inputs and reflects the association between land use and volume that the model learned from the training data (2014–2019 and 2022). It is not an estimate of a causal effect and does not include changes that a development may bring to the road network, signal timing or travel modes. The results are suited to the screening and comparison of development proposals and cannot replace a traffic impact assessment.

5.2 Experimental design

5.2.1 Uses changed

The analysis selects three uses, office, residential apartment and retail, which correspond to the Census of Land Use and Employment (CLUE) categories Office, Residential Apartment and Retail - Shop (Section 3.4). Table 5.1 lists the number of intersections changed for each use in 2024 and their floor-weighted shares, and Figure 5.1 shows where these intersections are.

- **Office**: 70 intersections have office floors, with the high shares concentrated in the Hoddle Grid, and the median floor-weighted share is 18.4%.
- **Residential apartment**: 76 intersections, the most of the three uses, with a median floor-weighted share of 20.7%, similar to office.
- **Retail**: 57 intersections, with very small floor-weighted shares (median 1.6%, maximum 11.7%). Scaled by the same multiple, retail floors change far less in absolute terms than office and residential floors.

Table 5.1: Intersections changed for the three uses (2024, 81 study intersections)

Use (CLUE category)	Intersections changed	Median floor share (%)	Range of floor share (%)
Office	70	18.4	1.0–46.0
Residential Apartment	76	20.7	3.9–57.3
Retail - Shop	57	1.6	0.2–11.7



Figure 5.1: Floor-weighted shares of the three uses in 2024. (a) Office. (b) Residential apartment. (c) Retail. Filled circles are intersections with the use (the intersections changed), with circle size showing the floor-weighted share, and open circles are intersections without the use, which are not changed. The numbers of intersections changed are in brackets.

5.2.2 Two types of change

Each use has two types of change (Table 5.2):

- **Add-on**: the floors of the use are multiplied by a factor (×0.5, ×1.5, ×2, ×3), which corresponds to adding floors of the use to the existing buildings (or, for ×0.5, removing them). The total number of floors changes accordingly, and the floor-weighted shares of all uses are renormalised. The four building-volume measures (total floors, inner total floors, mean floors and distance-weighted total floors) change in proportion to the change in the total number of floors.
- **Swap**: the floor-weighted share of the use is multiplied by 2, up to at most 100%, the shares of the other 12 uses are reduced in proportion, and the four building-volume measures are unchanged. This corresponds to keeping the total number of floors and converting other uses to this use.

Neither type of change alters the other attributes, such as the number of buildings, the number of tall buildings and the shares by number of buildings. Two types of change are used because an add-on changes the use share and the total number of floors at the same time, so the source of the response cannot be separated. A swap fixes the total number of floors and changes only the land-use mix, which isolates the effect of the use itself.

Table 5.2: Treatment of each attribute in the two types of change

Attribute	Add-on (×0.5, ×1.5, ×2, ×3)	Swap (×2)
Floor-weighted share of the use	Multiplied by the dose, then renormalised	Multiplied by 2, up to at most 100%
Floor-weighted shares of the other 12 uses	Renormalised with the new total (reduced or increased in proportion)	Reduced in proportion so that the shares still sum to 1
Four building-volume measures	Changed in proportion to the added floors	Unchanged
Other 26 attributes	Unchanged	Unchanged

Table 5.3 and Figure 5.2 illustrate the changes at the example intersection of Section 3.3.3 (intersection 2911, Exhibition / Little Lonsdale). An office add-on ×2 adds 341 office floors at this intersection, raising the total number of floors from 1,594 to 1,935 and the office floor share from 21.4% to 35.2%. An office swap ×2 keeps the total number of floors unchanged and raises the office share to 42.8%, with the other uses reduced in proportion. Across all 70 intersections, the office add-on ×2 raises the office floor share by 10.9 percentage points on average, and the office swap ×2 by 20.0 percentage points.

Table 5.3: Attributes of the example intersection under the two office changes

Attribute	Original (2024)	Office add-on ×2	Office swap ×2
Office floor share (%)	21.4	35.2	42.8
Residential apartment floor share (%)	37.0	30.5	26.9
Retail floor share (%)	0.94	0.78	0.68
Floor share of the other 10 uses (%)	40.7	33.5	29.6
Total floors	1,594	1,935	1,594
Inner total floors	480	583	480
Mean floors	8.26	10.03	8.26
Distance-weighted floors	11.29	13.70	11.29



Figure 5.2: Floors of each use within the 500 m square of the example intersection (2911). Top: original data for 2024. Middle: office add-on ×2, with the total number of floors increased. Bottom: office swap ×2, with the total number of floors unchanged. Bar length is the total number of floors, and colours show the part of each use.

5.2.3 Analysis periods and groups of intersections

Sections 5.3 and 5.4 report all-day responses, that is, the mean over all days and all 96 intervals of 2024, a year in which the all-day mean measured volume was 370.6 vehicles

per intersection per 15 minutes. Sections 5.5 and 5.6 also report responses in the three analysis windows defined in Section 3.2.3:

- Weekday AM peak 08:00–09:15, with a mean measured volume of 571 veh/15 min in 2024.
- Weekday PM peak 16:45–18:00, 587 veh/15 min.
- Weekend midday 12:00–13:15, 526 veh/15 min.

Intersections are grouped by the OpenStreetMap road classes of Section 3.5.2 and combined into main roads (trunk, primary and secondary, 55 intersections) and local roads (tertiary and lower, 26 intersections).

5.2.4 Statistical tests

The tests use the random seed as the unit (n = 5). The model of each seed predicts once before and once after the change, and the difference gives that seed's response, so each change has five response values:

- **Whether a response differs from zero**: one-sample t-test.
- **Comparisons between two uses, or between main and local roads**: paired t-test with the seeds as pairs.
- **Dose–response slope**: for each seed, a linear regression on the four add-on doses gives a slope, and the five slopes are then tested with a one-sample t-test.

The significance level is 0.05. Here, statistical significance means that a conclusion is stable across training seeds and does not depend on the randomness of a particular training run.

5.3 Dose–response

Figure 5.3 and Table 5.4 give the all-day responses of the three uses at the four add-on doses. The slope is the increase in response per unit increase in dose, that is, per addition of floors equal to the existing floors of the use.

- **Office**: the response changes monotonically with the dose. In Route 1 + land use, the response is −1.01 at ×0.5 and +3.31 veh/15 min at ×3, and all four doses are statistically significant. The slope is 1.73 ± 0.42 veh/15 min ($p < 0.001$), about 0.5% of the all-day mean volume, and the linear fit has an R^2 of 0.989. In Route 2 + land use, the response also increases monotonically with the dose, with a slope of 1.33 veh/15 min, and the slopes of all 5 seeds are positive.
- **Residential apartment**: in Route 1 + land use, the slope is 0.95 ($p = 0.060$), positive in all 5 seeds but not statistically significant. In Route 2 + land use, the slope is close to zero (0.20, $p = 0.681$).
- **Retail**: the slopes of the two models are 0.32 and 0.20, and neither is statistically significant. The floor-weighted share of retail is itself very small (Section 5.2.1), so the same multiple changes it little, and Section 5.4 adds a test of retail based on the change in share.

The responses of all uses are close to linear in the dose (R^2 of at least 0.96). At the same multiple, office has the largest slope of the three uses.

*Table 5.4: Dose–response of the three uses (all-day mean, veh/15 min, mean over 5 seeds, * marks p < 0.05)*

Use	Model	×0.5	×1.5	×2	×3	Slope (per multiple)	R^2	Slope *p*
Office	Route 1 + land use	−1.01*	+1.02*	+1.92*	+3.31*	1.73 ± 0.42	0.989	< 0.001
Residential	Route 1 + land use	−0.43	+0.47	+0.96	+1.93	0.95 ± 0.81	0.999	0.060
Retail	Route 1 + land use	−0.14	+0.17	+0.34	+0.66	0.32 ± 0.41	0.999	0.159
Office	Route 2 + land use	−0.86	+0.84	+1.52	+2.46	1.33 ± 1.20	0.975	0.069
Residential	Route 2 + land use	−0.06	+0.07	+0.17	+0.45	0.20 ± 1.03	0.963	0.681
Retail	Route 2 + land use	−0.09	+0.10	+0.20	+0.40	0.20 ± 0.22	1.000	0.113

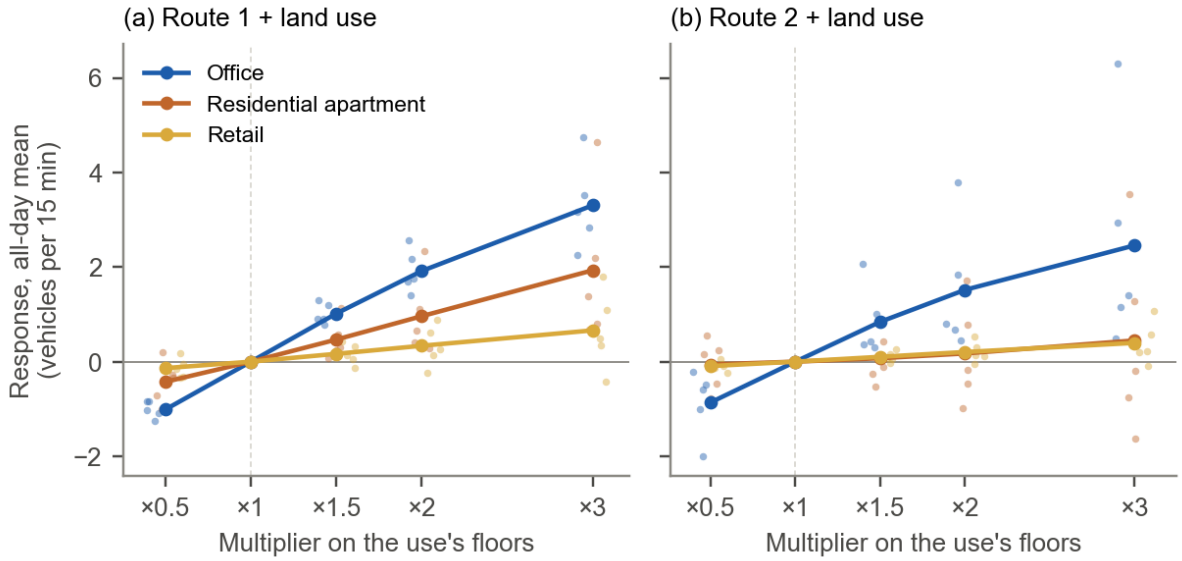


Figure 5.3: Dose–response of the three uses. (a) Route 1 + land use. (b) Route 2 + land use. The horizontal axis is the multiple applied to the floors of the use, with ×1 being the original data. Lines are the means of the 5 seeds and small dots are individual seeds.

5.4 Use swap

A swap changes only the land-use mix while keeping the total number of floors fixed. The results are shown in Figure 5.4 and Table 5.5:

- **Office**: swap ×2 raises the all-day response by +2.59 (Route 1 + land use, $p < 0.001$) and +2.99 veh/15 min (Route 2 + land use, $p = 0.005$), positive in all 5 seeds. This is about 0.7%–0.8% of the all-day mean volume, or roughly 249–287 additional vehicles per intersection per day on average across the 81 intersections. In the final predictions, which blend the model output with the historical profile (Section 5.1), the change is 0.60 times as large: +1.55 and +1.80 veh/15 min, about 0.4%–0.5% of the all-day mean volume, or roughly 149–172 vehicles per intersection per day.
- **Residential apartment and retail**: the responses are smaller (residential +0.91 and +0.55, retail +0.35 and +0.23), and none is statistically significant.
- **Comparison between uses**: the office swap response is significantly larger than that of retail ($p = 0.003$ and 0.005). In Route 2 + land use it is also significantly larger than that of residential ($p < 0.001$), while in Route 1 + land use the

difference from residential is 1.68 veh/15 min, close to but not reaching statistical significance ($p = 0.058$).

When the share doubles while the total number of floors stays the same, therefore, only office brings a statistically significant increase in volume.

The comparison between add-on and swap supports this. The responses to an office add-on ×2 are +1.92 and +1.52 veh/15 min, smaller than for swap ×2. The add-on raises the office share by only 10.9 percentage points on average, about half the increase under the swap (20.0 percentage points), and it also adds to the total number of floors, yet its response is still smaller than that of the swap. In these two models, the office response therefore comes mainly from the change in land-use mix rather than from the increase in the total number of floors.

Table 5.5: Swap responses of the three uses (all-day mean, veh/15 min, mean ± standard deviation over 5 seeds, "p against office" from paired t-tests)

Use	Model	Swap response	Share of volume (%)	Positive seeds	p	p against office	Add-on ×2 response
Office	Route 1 + land use	+2.59 ± 0.54	0.7	5/5	< 0.001	—	+1.92
Residential	Route 1 + land use	+0.91 ± 1.18	0.2	4/5	0.159	0.058	+0.96
Retail	Route 1 + land use	+0.35 ± 0.41	0.1	4/5	0.130	0.003	+0.34
Office	Route 2 + land use	+2.99 ± 1.17	0.8	5/5	0.005	—	+1.52
Residential	Route 2 + land use	+0.55 ± 0.78	0.1	4/5	0.192	< 0.001	+0.17
Retail	Route 2 + land use	+0.23 ± 0.22	0.1	4/5	0.079	0.005	+0.20

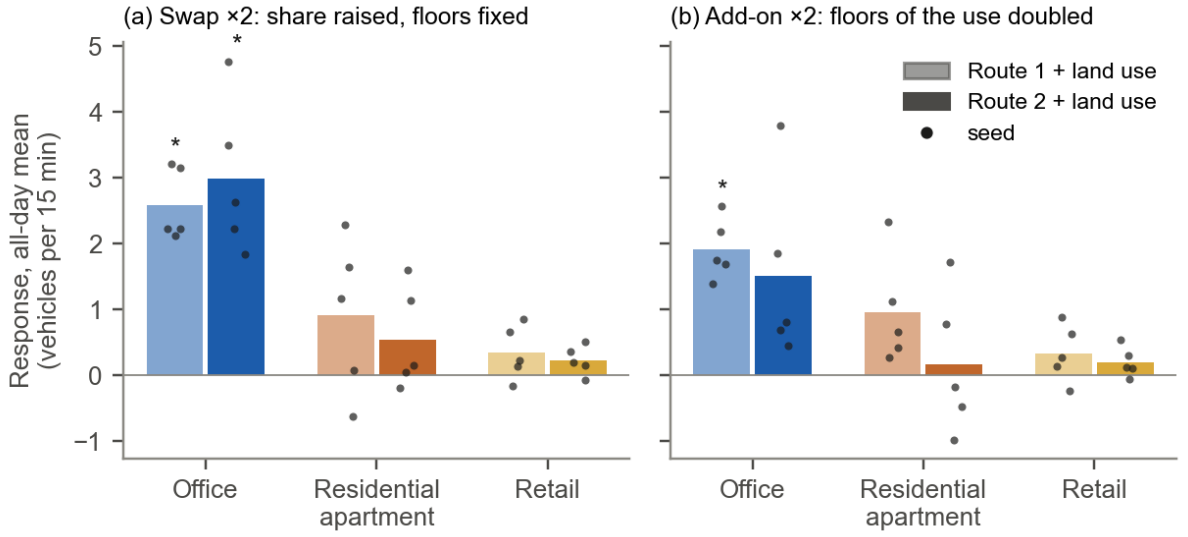


*Figure 5.4: All-day responses of the three uses under the two types of change. (a) Swap ×2, with the share doubled and the total number of floors unchanged. (b) Add-on ×2, with the floors of the use doubled. Light bars are Route 1 + land use and dark bars Route 2 + land use, small dots are individual seeds, and * marks a response significantly different from zero ($p < 0.05$).*

The comparisons above are all based on the same multiple, but the original shares of the three uses differ widely: swap ×2 raises the office floor share by 20.0 percentage points on average, residential by 23.3 and retail by only 3.2. To compare changes of similar size, retail swaps and add-ons at ×5 and ×10 were added, with all other settings as before.

Retail swaps ×5 and ×10 raise the retail share by 12.7 and 27.8 percentage points on average. To allow comparison per percentage point, these responses are averaged only over the intersections changed (Table 5.6, Figure 5.5):

- **The retail response is roughly proportional to the change in share**: per percentage point of share, the all-day response is 0.155–0.157 veh/15 min in Route 1 + land use and 0.098–0.103 veh/15 min in Route 2 + land use, essentially the same at ×2, ×5 and ×10.
- **Compared with office and residential**: per percentage point, retail matches office (0.150) in Route 1 + land use and is about 57%–60% of office (0.173) in Route 2 + land use. Residential is the weakest, at only 0.042 and 0.025.
- **Responses to changes of similar size**: the share increase of a retail swap ×10 is close to that of an office swap ×2, and its all-day response, 4.31 and 2.72 veh/15 min, is of the same order as that of office (3.00 and 3.46). Retail add-ons ×5 and ×10 follow the same pattern (1.84 and 3.78 in Route 1 + land use, 1.11 and 2.34 veh/15 min in Route 2 + land use).
- **Statistical significance**: the all-day responses to retail ×5 and ×10 are still not statistically significant, with 4 of the 5 seeds positive in each case. In the weekday AM peak, the retail responses at every dose are statistically significant and positive in all 5 seeds.

Note that the highest retail floor share in the training data is 17.1%. After retail swaps ×5 and ×10, the retail share exceeds this maximum at 16 and 25 intersections respectively (after ×10, the median retail share of the intersections changed is 16.0%), so these results are extrapolations by the model.

Taken together, "office is the most sensitive use" is a conclusion for changes by the same multiple. The small retail response at ×2 arises mainly because the change is small. Per percentage point of share, retail matches office in Route 1 + land use, while residential is clearly weaker.

*Table 5.6: Swaps of the three uses compared by change in share ("Intersections" is the number of intersections changed, "Share increase" the mean increase in the floor-weighted share of the use in percentage points, responses averaged over the intersections changed only, each cell giving Route 1 + land use / Route 2 + land use in veh/15 min, "Response per point" the all-day response divided by the share increase, * marks p < 0.05)*

Change	Intersections	Share increase	All-day response	Response per point	AM peak response
Office swap ×2	70	20.0	3.00* / 3.46*	0.150 / 0.173	3.35* / 3.51*
Residential swap ×2	76	23.3	0.97 / 0.58	0.042 / 0.025	1.76 / 1.60
Retail swap ×2	57	3.2	0.49 / 0.33	0.156 / 0.103	1.48* / 0.87*
Retail swap ×5	57	12.7	1.99 / 1.24	0.157 / 0.098	4.51* / 2.28*
Retail swap ×10	57	27.8	4.31 / 2.72	0.155 / 0.098	8.27* / 4.03*

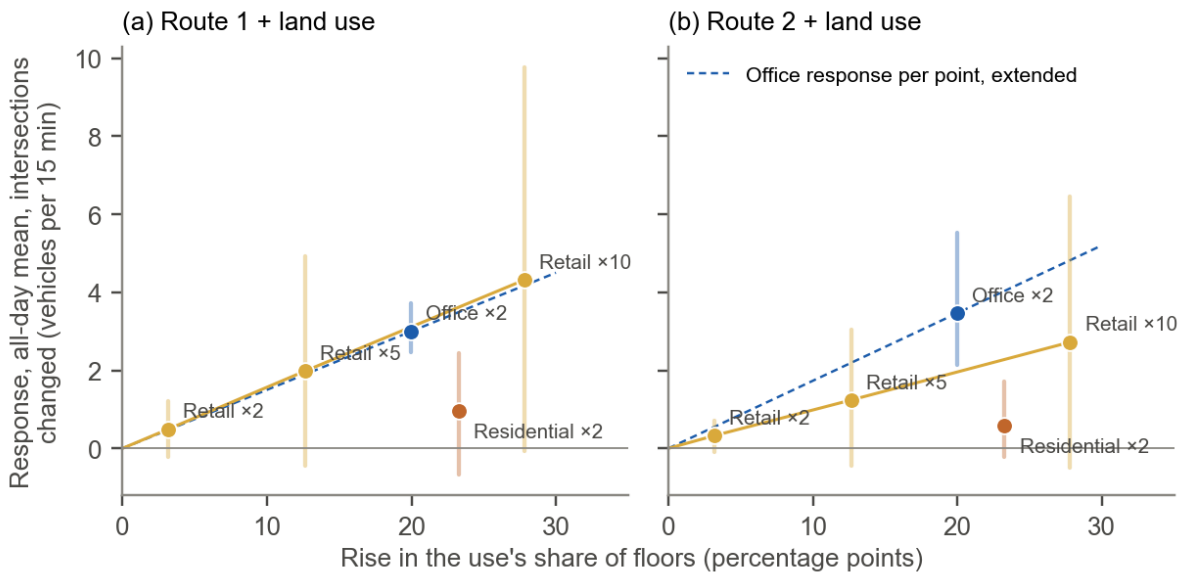


Figure 5.5: Swap response against the increase in share. (a) Route 1 + land use. (b) Route 2 + land use. The horizontal axis is the mean increase in the floor-weighted share of the use in percentage points, and the vertical axis is the all-day mean response of the intersections changed. Dots are the means of the 5 seeds and vertical lines the range across seeds. The yellow line connects retail ×2, ×5 and ×10, and the dashed line extends the office response per percentage point.

5.5 Response by time of day

Table 5.7 gives the responses to swap ×2 of the three uses in the three analysis windows.

- **Office**: office is the use with the largest response in both models and all three windows, including weekend midday. In Route 1 + land use, the responses in the AM peak, the PM peak and at weekend midday are 2.90, 2.02 and 1.86 veh/15 min, all statistically significant. In Route 2 + land use they are 3.03, 1.52 and 1.64 veh/15 min, with the PM peak not statistically significant (p = 0.089). The AM peak response is about 0.51%–0.53% of the volume in that window.
- **Retail**: the AM peak response is statistically significant in both models (1.04 and 0.61 veh/15 min) but only about a third of the office response or less, and it is not statistically significant in the other windows.
- **Residential apartment**: the responses in the two weekday peaks are positive but not statistically significant. At weekend midday the response is negative, statistically significantly so in Route 2 + land use (−0.68, p = 0.034).

*Table 5.7: Responses to swap ×2 of the three uses in the three analysis windows (veh/15 min, share of the measured volume in the window in brackets, * marks p < 0.05)*

Use	Model	Weekday AM peak	Weekday PM peak	Weekend midday
Office	Route 1 + land use	+2.90* (0.51%)	+2.02* (0.34%)	+1.86* (0.35%)
Office	Route 2 + land use	+3.03* (0.53%)	+1.52 (0.26%)	+1.64* (0.31%)
Residential	Route 1 + land use	+1.65 (0.29%)	+0.36 (0.06%)	−0.18 (−0.03%)
Residential	Route 2 + land use	+1.50 (0.26%)	+0.83 (0.14%)	−0.68* (−0.13%)
Retail	Route 1 + land use	+1.04* (0.18%)	+0.55 (0.09%)	+0.37 (0.07%)
Retail	Route 2 + land use	+0.61* (0.11%)	+0.24 (0.04%)	+0.26 (0.05%)

Figure 5.6 shows the response in each interval of the day. The office swap response is largest around 07:15 on weekdays (4.7 veh/15 min in Route 1 + land use and 7.3 veh/15 min in Route 2 + land use), earlier than the volume peak at 08:30, and falls clearly once the AM peak window begins. It is also at a low point in the PM peak window and rises again around 20:15–20:30. In other words, the model predicts that the traffic added by new office floor space occurs mainly before and after the peaks rather than when volumes are highest, so windows centred on the volume peaks give smaller responses than the periods around them. At weekends, the office response is positive throughout the day and is not zero even at night.

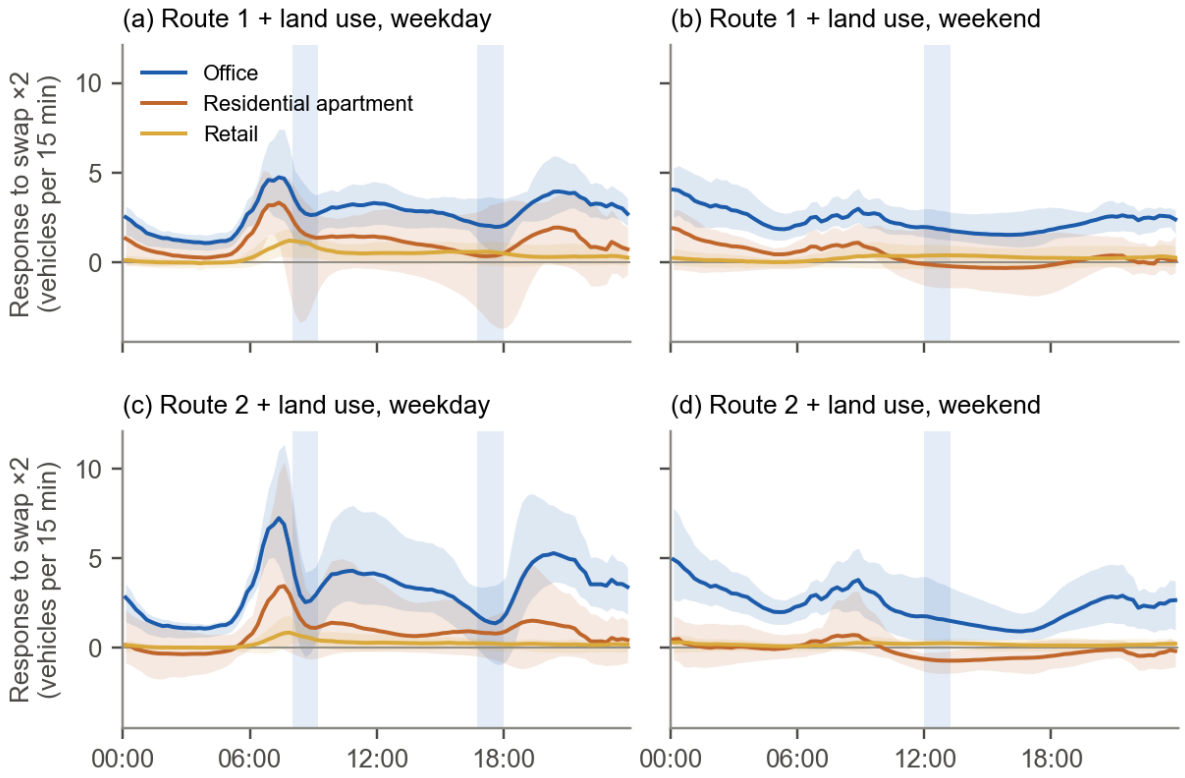


Figure 5.6: Responses to swap ×2 of the three uses in each interval of the day. (a)(b) Weekdays and weekends in Route 1 + land use. (c)(d) Weekdays and weekends in Route 2 + land use. Solid lines are the means of the 5 seeds, shading is the range across seeds, and blue bars mark the analysis windows of Section 3.2.3.

5.6 Response by road and intersection

This section uses the response to an office swap ×2 in the weekday AM peak as its example in comparing road classes and individual intersections, while the comparison of main and local roads in Section 5.6.3 covers all uses, changes and windows.

5.6.1 Road classes

Volume falls step by step with road class, but the response does not grow with volume (Figure 5.7, Table 5.8):

- **trunk**: the highest volume (1,253 veh/15 min in the AM peak) but the smallest response (0.10 and 0.13 veh/15 min), only about 0.01% of its volume.
- **primary**: a response of 2.52 and 2.89 veh/15 min, 0.34%–0.39% of volume.

- **secondary and tertiary**: responses of 3.07–4.81 veh/15 min, 0.7%–1.8% of volume, the highest proportions among the trunk, primary, secondary and tertiary classes.
- **unclassified and service**: only 3 and 2 intersections with very low volumes, so their proportional values fluctuate widely and are not interpreted in detail.

*Table 5.8: Responses of each road class to an office swap ×2 in the weekday AM peak ("With office" is the number of intersections with office floors, which are changed, measured volume is the 2024 AM peak mean, responses in veh/15 min with the share of measured volume in brackets, * marks p < 0.05)*

Road class	Intersections	With office	Measured volume	Route 1 + land use	Route 2 + land use
trunk	12	4	1,253	+0.10 (0.01%)	+0.13 (0.01%)
primary	18	17	743	+2.52* (0.34%)	+2.89* (0.39%)
secondary	25	25	450	+4.65* (1.03%)	+3.17 (0.70%)
tertiary	21	19	274	+3.07* (1.12%)	+4.81* (1.76%)
unclassified	3	3	88	+2.31 (2.64%)	+3.58* (4.09%)
service	2	2	33	+0.29 (0.85%)	+0.66 (1.97%)
Main roads, all	55	46	721	+2.96* (0.41%)	+2.41* (0.33%)
Local roads, all	26	24	242	+2.77* (1.15%)	+4.35* (1.80%)

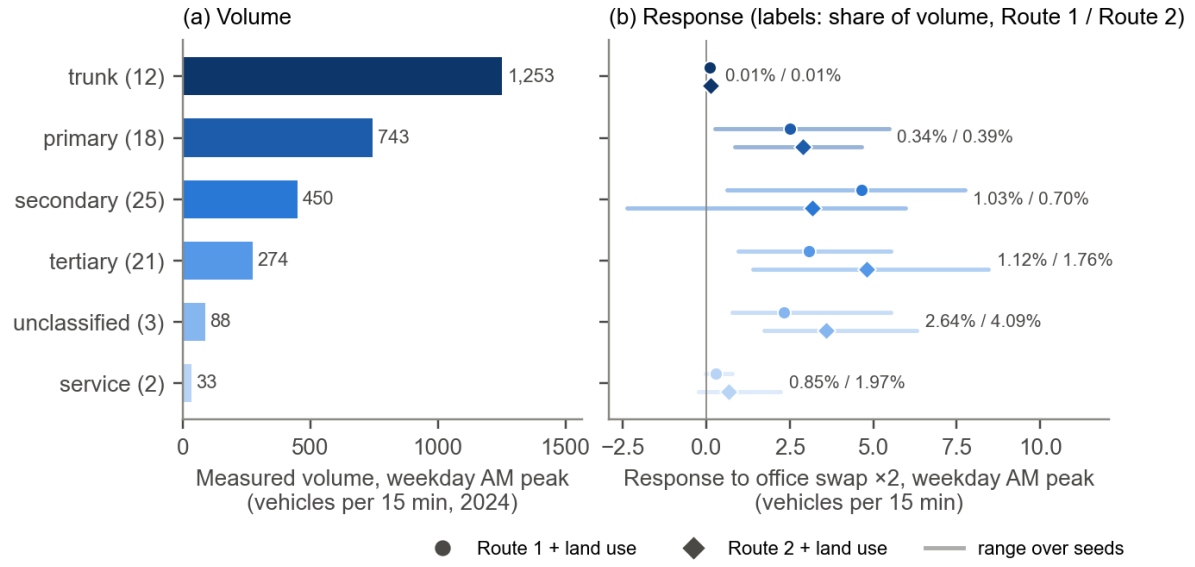


Figure 5.7: Volume and response by road class (weekday AM peak). (a) Measured mean volume in 2024, with the number of intersections in brackets. (b) Response to an office swap ×2, with dots for Route 1 + land use and diamonds for Route 2 + land use, horizontal lines for the range across seeds, and the response as a share of volume labelled on the right.

5.6.2 Responses of individual intersections

Figure 5.8 shows the response at each intersection. The 70 intersections changed differ widely. In Route 2 + land use, the largest response is at intersection 4576 (Collins / Exhibition, tertiary), at 24.6 veh/15 min, 18 intersections exceed 5 veh/15 min and 12 are negative. In Route 1 + land use, the largest response is 22.3 veh/15 min, 16 intersections exceed 5 veh/15 min and 8 are negative. The response at the median intersection is about 0.33%–0.53% of its AM peak volume.

Intersections with larger responses cluster in the parts of the Hoddle Grid with high office shares (compare Figure 5.1a). The response at each intersection is positively correlated with its original office share, with Spearman correlation coefficients of 0.69 and 0.55 ($p < 0.001$): a swap doubles the share, so the more office there is to begin with, the larger the change. The correlation between the intersection responses of the two models is 0.78, so their spatial patterns are largely the same.

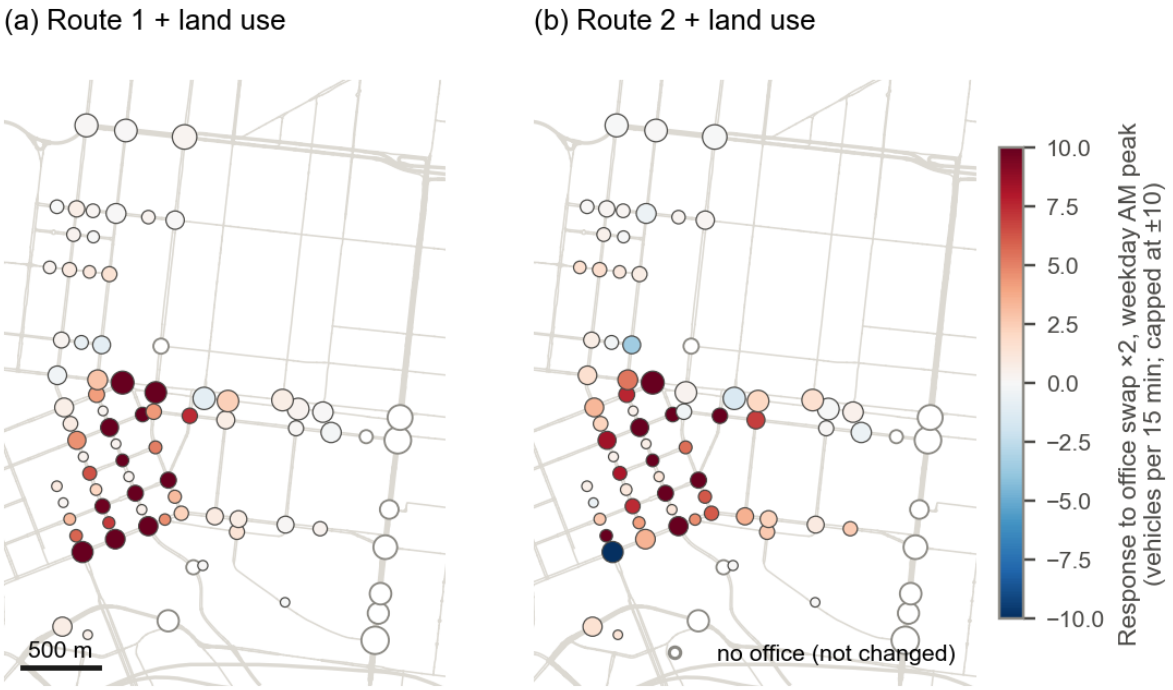


Figure 5.8: Response of each intersection to an office swap ×2 (weekday AM peak). (a) Route 1 + land use. (b) Route 2 + land use. Colour shows the response (red for increases, blue for decreases), dot size shows the measured AM peak volume, and open circles are intersections without office floors, which are not changed.

5.6.3 Main and local roads

The comparison of main and local roads covers 3 uses × 2 models × 3 analysis windows × 5 changes (the four add-on doses and the swap), 90 comparisons in all. Each compares the mean responses of the intersections in the two groups. At ×0.5, land use is reduced and the response is negative, so the comparison is of the size of the fall. The results are shown in Figure 5.9:

- In 44 comparisons the local-road response is significantly larger than the main-road response ($p < 0.05$), in 0 the main-road response is significantly larger, and the other 46 show no significant difference.
- For the office swap in the AM peak, for example, the main-road response is 2.41 and the local-road response 4.35 veh/15 min in Route 2 + land use, while in Route 1 + land use the two are similar (2.96 and 2.77).

Local roads carry far less traffic than main roads (242 veh/15 min in the AM peak, against 721 veh/15 min on main roads). Even where the absolute responses are similar, the change relative to each group's own volume is therefore much larger on local roads:

the response to the office swap in the AM peak is 1.15%–1.80% of the local-road volume, against 0.33%–0.41% of the main-road volume.

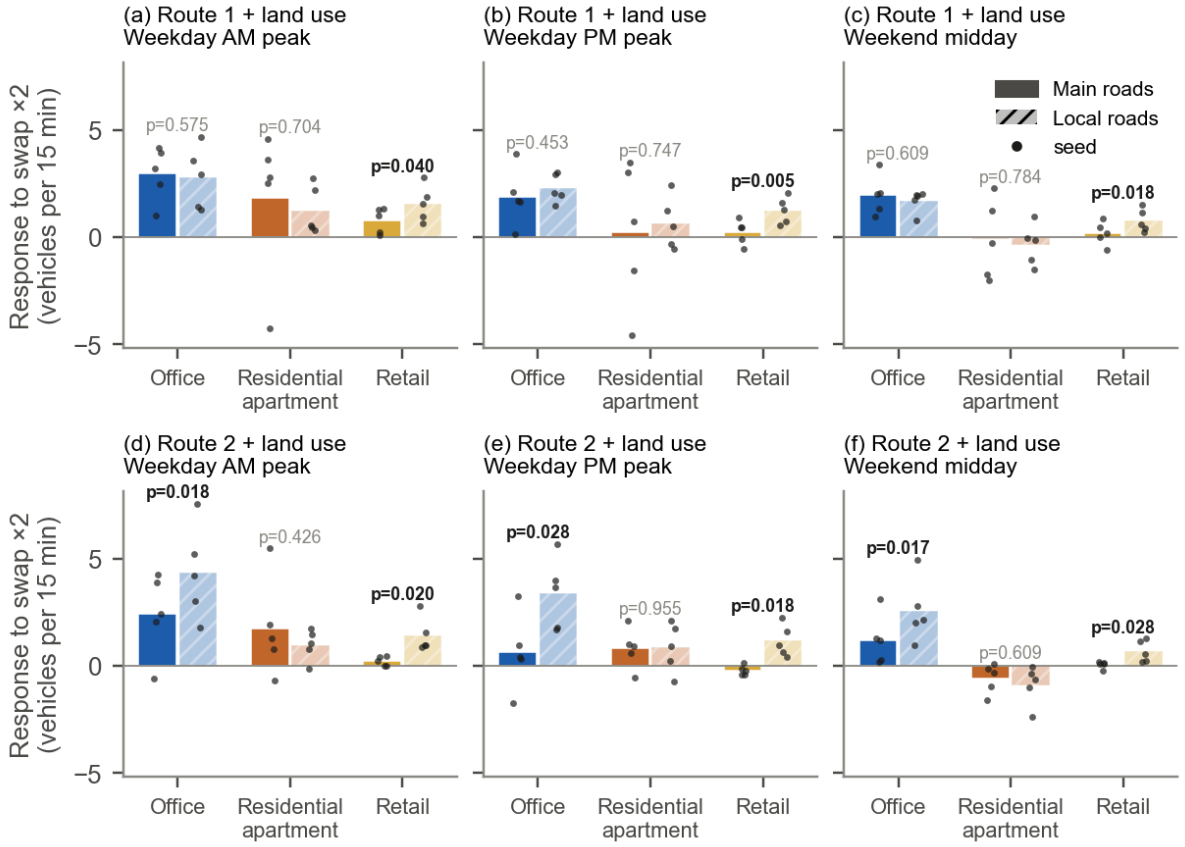


Figure 5.9: Responses of main and local roads to swap ×2 of the three uses. The top row is Route 1 + land use and the bottom row Route 2 + land use. The left, middle and right columns are the weekday AM peak, the weekday PM peak and weekend midday. Solid bars are main roads and hatched bars local roads, and small dots are individual seeds. The p values are from paired t-tests of main against local roads, in bold where p < 0.05.

5.6.4 Interpretation

Section 3.5.3 noted that trunk intersections spend 11 hours of an average weekday above 90% of their peak, against only 3–5.25 hours for primary, secondary and tertiary intersections, which shows that trunk roads operate under heavy load for most of the day. The results above are consistent with this: trunk roads carry the most traffic, yet the model assigns them the least additional traffic, which appears mainly on secondary and local roads. The daily pattern in Section 5.5 points the same way, since the office response falls in the peak periods when volumes are highest and is larger in the periods before and after them.

There are two possible explanations. One is that trunk roads are close to capacity in the peaks, so additional demand cannot show up as higher volumes in those periods and is spread into the periods around them. The other is that traffic on trunk roads is mainly through traffic that is only weakly related to adjacent land use. The model itself has no capacity constraints, and volume data alone cannot distinguish between the two.

In practice, an assessment of the traffic impact of a development should not focus only on main roads: the absolute response of local roads is similar to or larger than that of main roads, and the change relative to their volume is clearly larger.

5.7 Key findings

1. When the land-use attributes of the test year are changed, predicted volumes change in the direction of the change and nearly linearly with the dose. At the same multiple, office is the most sensitive use: the dose–response slope of Route 1 + land use is 1.73 veh/15 min ($p < 0.001$), and in Route 2 + land use the response also increases monotonically with the dose (1.33), with positive slopes in all 5 seeds. The slopes of residential and retail are smaller and not statistically significant.
2. When the share doubles while the total number of floors stays the same, only office brings a statistically significant increase in volume: an office swap ×2 raises the all-day response by +2.59 and +2.99 veh/15 min, about 0.7%–0.8% of all-day volume, while the residential and retail responses are not statistically significant. For office, the add-on ×2 responses are smaller than the swap ×2 responses, which shows that the office response comes mainly from the change in land-use mix.
3. Per percentage point of share, retail matches office in Route 1 + land use (0.155–0.157 against 0.150 veh/15 min) and is about 57%–60% of office in Route 2 + land use, while residential is clearly weaker. The small retail response at ×2 arises mainly because the change is small.
4. Office is the use with the largest response in the weekday AM and PM peaks and at weekend midday. The office response is largest before and after the peaks (around 07:15 on weekdays and in the evening) and falls when volumes are highest.
5. The response is concentrated on secondary and local roads: trunk roads carry the most traffic, but their response to an office swap ×2 in the weekday AM peak is only about 0.01% of their volume, against 0.7%–1.8% on secondary and tertiary roads. Of the 90 comparisons of main and local roads, the local-road response is statistically significantly larger in 44 and the main-road response in 0.
6. The responses of individual intersections vary widely and are positively correlated with the original office share, and the spatial patterns of the two models are largely the same (correlation coefficient 0.78).

6 Development scenario imagery tool

The development scenario imagery tool generates satellite imagery of a development as built on an existing satellite base image, from the development's building footprint, height and use, and supports new building, alteration and demolition scenarios. The tool uses OpenAI's image generation model gpt-image-2 at its core, but the building form is not left to the model. The tool first draws the building's 3D massing on the base image according to its real geometry, then explains what the massing means and what is to be generated in a prompt organised in blocks, and restricts the area the model may rewrite to the massing and the reach of its shadow. The position, footprint and height of the building are therefore fixed by the input geometry, and every pixel outside the editable region is identical to the base image.

The tool has an interactive web interface. Users can choose a development from the development library or draw their own building footprint on the base image, set the height and the scenario, and then generate the imagery. The generated imagery takes the same form as the satellite imagery used by the prediction model, so that planning proposals can be presented visually as imagery, and it can also take the place of actual imagery as input to the prediction model. Section 6.4 uses real developments completed in 2023–2025 to test the prediction accuracy and the change in predictions when a generated building is added to satellite imagery taken before construction, and compares this with updating the land-use attributes.

6.1 Tool overview

The tool's inputs are the building footprint, building height, use and scenario. Footprints can be taken from the development library, which holds 20 developments from the City of Melbourne's development activity data (Section 3.6), or drawn by the user on the base image, and a scene can contain several buildings. The alteration scenario also requires the height of the existing building, and the demolition scenario requires the ground type after demolition, which is one of five: paved ground, a levelled gravel site, a car park, a hoarded construction site and grass. The satellite base image can be taken from any year between 2014 and 2026 and is centred on the development, 400 m across and 1024 × 1024 pixels, about 0.39 m per pixel.

The interface is divided into three columns in the order of operation (Figure 6.1). The left column is for choosing a development or drawing a footprint, the middle column for setting the height, scenario and viewpoint with a live preview of the massing, and the right column for calling the model to generate the imagery and showing the result.

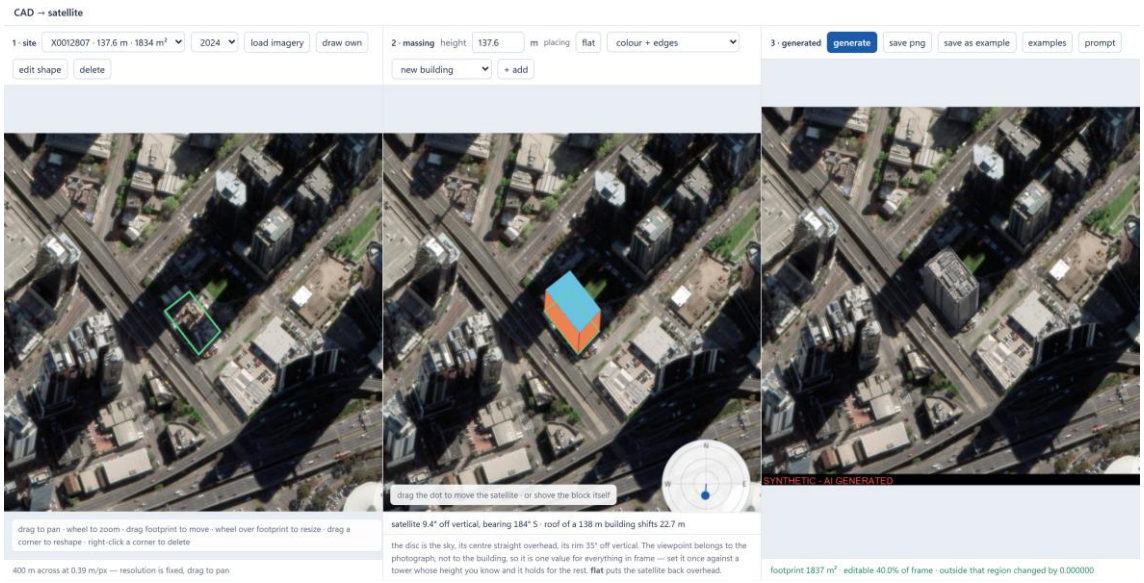


Figure 6.1: Interface of the development scenario imagery tool (development X0012807, 2024 satellite base image, height 137.6 m). The left column is for the site and footprint, the middle column for the massing and viewpoint, and the right column for the generated result and the status bar.

6.2 Image generation method

General-purpose image generation models can produce realistic satellite imagery, but a development described in words alone gives no precise control over the position, footprint and height of the building, nor any assurance that unrelated parts of the image stay unchanged. To make the result follow the real geometry of the development while keeping the rest of the base image unchanged, the tool constrains generation at four stages: the massing, the editable region, the prompt and the model call. Figure 6.2 shows the complete process for a new building.

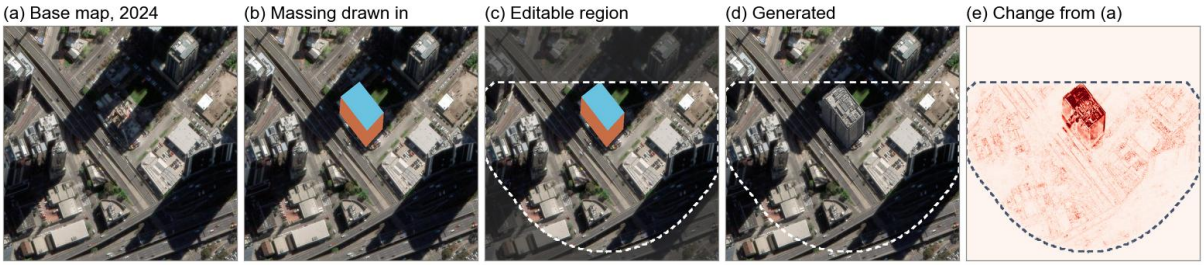


Figure 6.2: Image generation process (development X0012807, height 137.6 m, 2024 satellite base image). (a) Satellite base image. (b) Building massing drawn into the base image by its real geometry, with the roof in blue and the facades in orange. (c) Editable region (inside the dashed line), the area the model may rewrite, made up of the massing and the shadow area to its south and covering 40.0% of the image, with the area outside dimmed. (d) Result after generation and compositing, in which the pixels outside the dashed line are identical to (a). (e) Pixel-by-pixel difference between (d) and (a), with darker colours showing larger differences and the visible changes concentrated at the new building.

6.2.1 3D massing

The tool extrudes the building footprint to the input height to form 3D massing and projects it onto the image according to the viewing angle of the base image, so that the relative position of the massing's roof and base matches that of the existing buildings in the base image.

The massing is drawn into the base image in two strongly contrasting solid colours that mark the roof and the facades, so that the model can tell them apart and reconstruct the building's 3D form. The colours are only markers and do not represent building materials. The three scenarios are marked differently. The new building scenario draws solid massing, the alteration scenario outlines the existing building in wireframe and draws the altered building as solid massing, and the demolition scenario only outlines the building to be demolished in wireframe.

6.2.2 Editable region and compositing

The editable region consists of the massing and the area it may shade. The shadow area is estimated for a sun elevation of at least 45°, at which a shadow is no longer than the building is tall. Melbourne is in the southern hemisphere, where the sun is then in the northern sky and shadows fall only to the south of a building, so the editable region extends only over a semicircle to the south. The alteration and demolition scenarios use the taller of the old and new buildings, so that the existing shadow that needs to be removed also lies within the editable region. The editable region therefore grows with building height: in the saved examples, it covers about 13% of the image for an 80 m building, about 40% for a 137.6 m building and about 73% for a 450 m building.

The model's edit uses a mask to mark the area that may be rewritten, but gpt-image-2 treats the mask as guidance: the model is guided by the mask and is not guaranteed to follow its boundary exactly (OpenAI, 2026b). The tool therefore keeps only the part of the returned image within the editable region, matches colours within the region, feathers the edge inside the region, and takes everything outside the region from the original base image. Composited in this way, all 18 saved generated examples have zero pixel difference outside the editable region.

The editable region is only the area the model is allowed to rewrite and does not mean that everything inside it changes. In the example of Figure 6.2, the editable region covers 40.0% of the image, whereas the pixels that differ markedly from the base image (by more than 25 grey levels on a scale of 255) cover only 3.1% of the image and are concentrated at the new building (Figure 6.2e). Elsewhere in the region there are only slight differences in tone.

6.2.3 Prompt design

For the different development scenarios, the tool has four prompts, for a single new building, several new buildings, alteration and demolition, each 402–551 English words long and given in full in Appendix D. The four prompts share the same six-block structure. The SCENE block explains the input image and what the massing or wireframe means, TASK states the change to be made, GEOMETRY sets the position, footprint and height that must be kept, LIGHT explains how shadows are to be handled, SURFACE sets the texture of the buildings and the ground, and HOLD requires all other pixels to stay unchanged and no annotations to be added. Values such as building height, number of

storeys, footprint area, use and roof offset are filled in automatically by the program from the inputs (Table 6.1), so that the geometry described in the prompt matches the drawn massing exactly.

The prompt for a single new building requires the solid massing to be replaced by a real building. Its GEOMETRY block concentrates on separating roof from facades. It states that the roof face is the only roof and that every other face is a wall seen from the side, distinguished by position rather than by size, because the facades of a tall building may appear larger in the image than its roof. The same block gives the direction and distance of the roof's offset from the ground footprint and explains that the offset arises because the satellite views the scene off vertical. The LIGHT block requires the direction and length of shadows to be read from the existing buildings in the image and the new building to cast a shadow of the same direction and proportionate length. Where that shadow falls on a road, a roof or a footpath, it should darken what is there rather than be painted as a flat shape. The SURFACE block requires roof plant, services, lift overruns and parapets on the roof and glazing lines and material on the facades, consistent with the grain, sharpness and tone of the surrounding imagery, and the marker colours must not appear in the result.

The prompt for several new buildings builds on the single-building prompt and lists each building's height and footprint area. It requires the buildings to stay separate rather than merge and to keep the gaps between them, lets the block drawn in front take precedence where blocks overlap, and makes each roof's offset vary with that building's own height. The LIGHT block requires each building's shadow to be in proportion to its own height, including the shadows that the new buildings cast on one another. The SURFACE block allows the buildings to differ in age, material and colour, as neighbouring real buildings do.

The SCENE block of the alteration prompt explains that the wireframe outlines the existing building and that the solid massing on the same footprint is the altered building, and gives the height of each. The GEOMETRY block requires that no trace of the original height remain, including the old roof, facade above the new roof level, floating roof plant and the old outline, and the marker lines must also be removed. The LIGHT block gives the ratio of the heights before and after the alteration. It requires the original building's shadow to be found and the part that should remain at the new height to be kept, the roads, footpaths, roofs, vegetation and vehicles under the rest to be restored to their unshaded brightness, and the shortened shadow then to be cast. The SURFACE block requires the ground and neighbouring buildings previously hidden by the tower to be completed from their surroundings.

The demolition prompt outlines the building to be demolished in wireframe, including its ground outline, its roof outline and the edges connecting them, and gives the building's height and roof offset. The GEOMETRY block requires the site to become vacant land, with no new building placed on it and no neighbouring building extended into it, and adjoining buildings that share a wall with it keep their own walls. The LIGHT block requires the building's shadow to be removed and the features beneath it to be restored to their unshaded brightness. The SURFACE block sets the vacant site to the ground type the user selected and completes the previously hidden ground and neighbouring buildings from their surroundings.

Table 6.1: Parameters filled into the prompts by the program

Parameter	Meaning	Prompts that use it
Marker colours	Marker colours of the roof and facades, which vary with the massing style	All
Height	Building height (m)	Single new building, alteration, demolition
Storeys, footprint area, use	Storeys converted from height at 3.2 m per storey, footprint area computed from the building footprint, use residential or mixed use	Single new building, alteration
Direction and distance of the roof offset, viewing angle	Computed from the viewpoint and building height	All (several new buildings: direction and angle only)
Number of buildings, list of buildings	Height and footprint area of each building	Several new buildings
Existing height, height ratio	Height before alteration and its ratio to the height after	Alteration
Ground type	Paved ground, levelled gravel site, car park, hoarded construction site or grass	Demolition

6.2.4 Image generation model

The tool uses OpenAI's image generation model gpt-image-2, snapshot gpt-image-2-2026-04-21. OpenAI describes it as a model for fast, high-quality image generation and editing that supports flexible image sizes and high-fidelity image inputs. The model takes text and images as input and produces images as output. It can generate an image from text or edit an input image, and it supports local inpainting of a region specified by a mask (OpenAI, 2026a).

The tool generates imagery through the model's image editing interface. Each request supplies three inputs, the satellite base image with the massing drawn in, the mask marking the editable region and the prompt with its parameters filled in, and returns an image of 1024 × 1024 pixels, the same size as the base image, with the rendering quality set to medium. Requests are submitted one at a time and retried automatically on failure. In the project's usage records, a single generation took about 20–80 seconds, varying with prompt length and service load, in line with OpenAI's note that complex prompts can take up to about 2 minutes (OpenAI, 2026b).

6.3 Application cases

6.3.1 New building

Figure 6.3 uses development X0012807 from the development library. Its site was still under construction in the 2024 satellite base image, and the development activity data record its height as 137.6 m. With the height set to 137.6 m, 250 m and 450 m, the roof offset of the massing grows with height, and the generated buildings all stand on the same footprint. The building height and the visible extent of the facades change with the set height, while the rest of the base image stays unchanged. The position, footprint and height of the building are fixed by the input geometry, and the model turns the massing into a building with a realistic texture and fits it into its surroundings.

The development has been built in the 2026 satellite imagery (Figure 6.3e). The generated building and the actual building stand on the same building footprint. The two

images were captured from different angles, with the roof offset to the north in the 2024 base image and to the north-east in the 2026 imagery, so different sides of the facade are visible in the two, and details such as roof plant and facade materials also differ from the actual building.

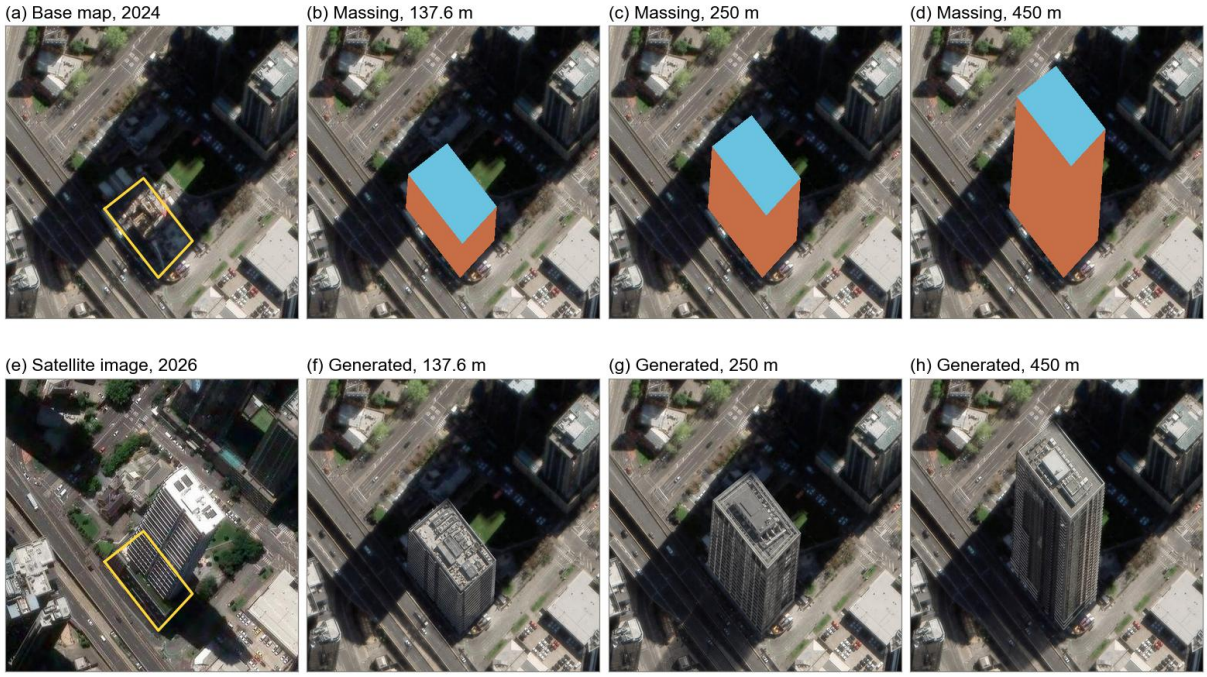


Figure 6.3: New building case (development X0012807). (a) 2024 satellite base image, with the yellow line marking the building footprint of the development, whose site was then under construction. (b)–(d) Massing with the height set to 137.6 m (the height recorded in the development activity data), 250 m and 450 m. (e) 2026 satellite imagery, in which the development has been built, with the yellow line marking the same footprint and a viewing angle different from 2024. (f)–(h) The corresponding generated results.

6.3.2 Several buildings

In Figure 6.4, four building footprints were drawn by hand on the 2026 satellite base image and all four buildings were generated at once. Each building keeps its own footprint and spacing, and the buildings do not merge. In a second generation, the height of the lower-left building was raised to 250 m with all other settings unchanged. After regeneration, that building's height and facades change accordingly, while the other buildings remain on their own footprints.

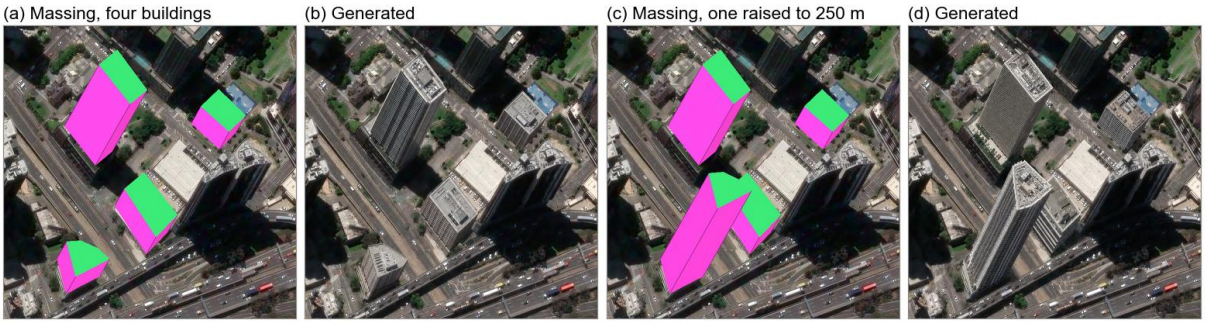


Figure 6.4: Several buildings case (2026 satellite base image, four footprints drawn by hand). (a) Massing of the four buildings. (b) Generated result. (c) Massing with the lower-left building raised to 250 m. (d) Regenerated result.

6.3.3 Alteration and demolition

Figure 6.5 takes development X0012807 as built in the 2026 satellite base image. The alteration scenario lowers the tower to 20 m on its original footprint. In the generated result, the part of the tower above that height is removed together with its long shadow to the south, and the roads, car parks and green space that were previously hidden or in shadow are restored. The demolition scenario outlines the building in wireframe, and the figure shows the results of four generations, in which the site appears as paved ground, grass, a levelled gravel site and a car park. In all four results the building and its shadow are removed, while the surrounding buildings, roads and trees are unchanged.

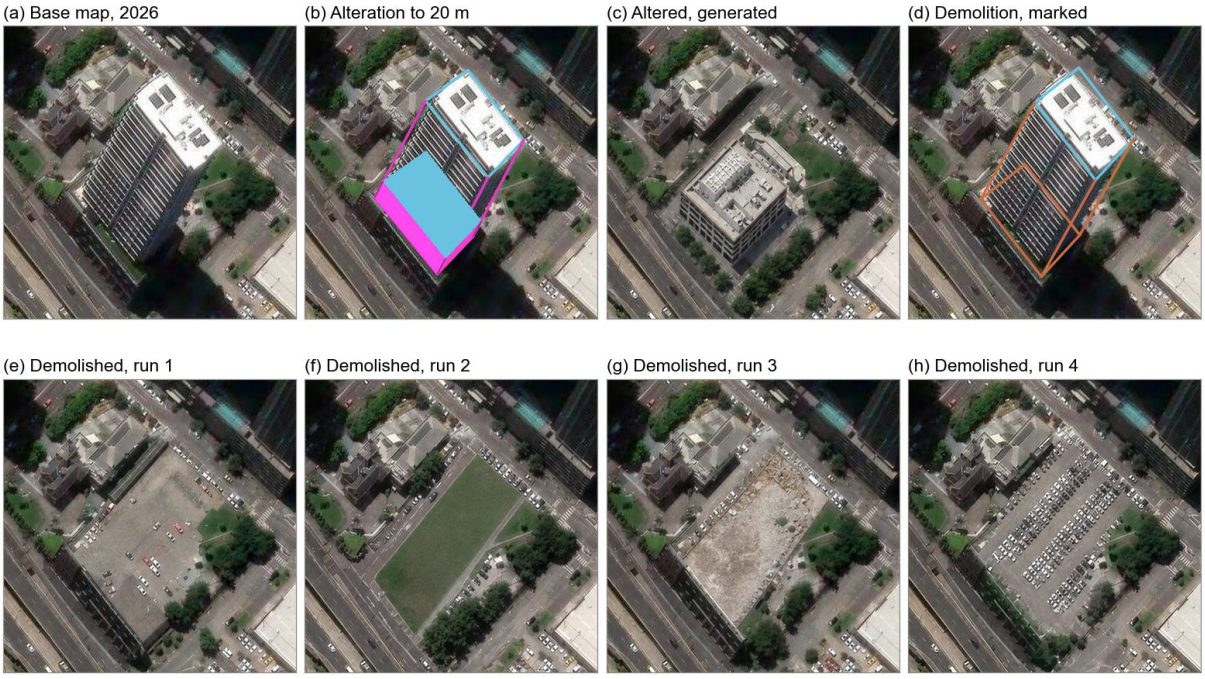


Figure 6.5: Alteration and demolition cases (2026 satellite base image, development X0012807 as built). (a) The existing building in the base image. (b) Alteration to 20 m, with the existing building in wireframe and the altered building as solid massing. (c) Generated result of the alteration. (d) Demolition, with the building to be demolished outlined in wireframe. (e)–(h) Results of four demolition generations, in which the site appears as paved ground, grass, a levelled gravel site and a car park.

6.4 Testing generated imagery for traffic volume prediction

The imagery generated by the tool takes the same form as the satellite imagery used by the prediction model. In planning, a proposed building can be added to satellite imagery taken before a development is built and used in place of actual imagery as input to the prediction model (Chapter 4), so that the predicted volumes with and without the development can be compared. This use requires that the generated imagery does not reduce prediction accuracy and that adding a building changes the predictions in a reasonable way. To test this, 17 real developments completed in 2023–2025 were selected, and the results were compared with adding the same developments to the land-use attributes (Chapter 5).

6.4.1 Test method

The test centres on Route 2 + land use, the most accurate configuration in Chapter 4, and uses its trained models for the 5 seeds without retraining. For the developments completed in 2023–2024, Route 1 + land use serves as a comparison. The test cases are the intersections around each development that lie within 250 m of the centre of its building footprint and whose 500 m image contains the whole footprint. Satellite imagery taken before the development was built serves as the base image, and a building generated from the development's footprint and height is added to it (Section 6.4.2). In the completion year, the imagery and the land-use attributes each take two values, which combine into four cases (Table 6.2):

- **No development**: the base image and the original land-use attributes.
- **Image only**: the image with the generated building and the original land-use attributes.
- **Land use only**: the base image and the land-use attributes with the development added.
- **Full scenario**: the image with the generated building and the land-use attributes with the development added.

The land-use attributes with the development added are produced by the method of Section 4.3. The development is written into the Census of Land Use and Employment (CLUE) as one building record, with its number of floors converted from its height at 3.5 m per floor. This conversion serves only the land-use attributes and is separate from the 3.2 m per storey with which the tool's prompts describe a building (Table 6.1). It is recorded as office if it has commercial floor space and as residential apartment otherwise, and the attributes of each intersection's 500 m square are then recomputed. The inputs of all other intersections and years, the network features and the temporal features are unchanged, so the predictions of the four cases differ only through the imagery and the land-use attributes.

Table 6.2: The four test cases

Case	Imagery in the completion year	Land-use attributes in the completion year
No development	Base image (satellite imagery before construction)	Original attributes
Image only	Base image with the generated building	Original attributes
Land use only	Base image (satellite imagery before construction)	Development added
Full scenario	Base image with the generated building	Development added

The generated imagery is turned into imagery features by the segmentation, region feature and aggregation process of Section 4.2.

The developments completed in 2023 fall in the model's validation year. The blend weight and the spatial smoothing coefficient of Section 4.7 are selected on the validation year, and changing that year's inputs would change the selection, so every test fixes both parameters at the values each model selected in training. The developments completed in 2025 lie one year beyond the test year (2024). Their measured volumes come from the same data source, the original land-use attributes carry forward those of 2024 (the CLUE data end in 2024), and the year in the temporal features keeps the scaling of Section 4.5.1, with 2014–2024 mapped to 0–1, so that 2025 takes the value 1.1.

The test uses the following measures:

- **Prediction accuracy**: the median GEH statistic (Section 4.11) at the affected intersections in the completion year, the share of intervals with a GEH below 5, and the mean absolute error (MAE). GEH is computed over all intervals, and the MAE is computed for each intersection case and then averaged.
- **With-and-without comparison**: the difference in the mean predicted volume between each case and the no-development case, divided by the blend weight α (Section 4.7.2), in vehicles per 15 minutes (veh/15 min). This is the response defined in Chapter 5, the change in the model output itself, and it is also expressed as a percentage of the mean measured volume.
- **Closeness to measured volumes**: the difference in MAE between each case and the no-development case, where a negative value means that the predictions are closer to the measured volumes, together with the share of cases that come closer.

Two units of analysis are used. With the random seed as the unit (n = 5), each seed is first averaged over all cases and then tested with a one-sample t-test, as in Chapter 5, which shows whether a result is stable across training seeds. With the case as the unit, each case is averaged over the 5 seeds and then tested with a sign test and a Wilcoxon signed-rank test, which shows whether a result holds across developments and intersections. The image of each case was generated only once and is shared by the 5 seeds, so the seed-level test does not include the randomness of generation itself, and both tests are therefore reported. The significance level is 0.05.

6.4.2 Test cases

The test developments are taken from the developments completed in 2023–2025 in the City of Melbourne's development activity data (Section 3.6), as listed in Table 6.3. They are 15.8–119.9 m tall, and their uses include residential, student accommodation, hotel, serviced apartments, office and retail. The cases fall into two groups by completion year (Table 6.4):

- **Developments completed in 2023–2024**: 10 developments. Those completed in 2023 use the 2019 and 2022 imagery as base images, and those completed in 2024 use the 2022 and 2023 imagery, giving 35 intersections and 75 intersection cases. Each generated building was produced with the same massing and editable-region method as the tool on an image centred on the development, with the massing drawn for a viewpoint 15° off vertical, and was then moved by its ground position into the base image of each intersection.

- **Developments completed in 2025**: 7 developments. They use the 2022 and 2023 imagery, two to three years before completion, as base images, so that the base image does not show the site under construction. They involve 11 intersections, each with two base images, giving 22 intersection cases. The generated buildings were produced by the tool's new building scenario directly on each intersection's base image (500 m across, 1024 × 1024 pixels), with a viewpoint 3° off vertical and the roof offset to the north, and several developments around one intersection were added together.

Figure 6.6 shows one case from each group, and Appendix E shows every case.

Table 6.3: Developments used in the test (predominant use from the development activity data, office and retail listed only above 1,000 m^2 of floor space, "Intersections" being the number of intersections whose image includes the generated building)

Development	Address	Completed	Height (m)	Predominant use	Intersections
X0004000	13–23 Spring St	2023	119.9	Residential	6
X0008603	276–284 Russell St	2023	89.6	Student accommodation	5
X0009218	200 Victoria Pde, East Melbourne	2023	45.5	Office, Retail	3
X000146	20–30 Bourke St	2023	41.6	Hotel	5
X001536	273–279 Little Collins St	2023	32.6	Serviced apartments	4
X0008015	23–29 Victoria St	2024	80.0	Student accommodation	4
X0008019	124–130 Russell St	2024	54.8	Hotel, Retail	6
X001172	502–506 Albert St, East Melbourne	2024	41.6	Residential	5
X0005621	90 Jolimont St, East Melbourne	2024	30.0	Office	1
X0001630	2–12 Elgin St, Carlton	2024	25.6	Residential	2
X0008610	309–325 Bourke St	2025	42.7	Hotel, Retail	2
X0011602	226–232 Flinders Lane	2025	37.5	Office	3
X0014503	121–131 Cardigan St, Carlton	2025	35.2	Residential	1
X0011004	299 Bourke St	2025	34.2	Office, Retail	2
X0011204	372–376 Albert St, East Melbourne	2025	34.1	Residential	3
X0013802	364–370 Albert St, East Melbourne	2025	28.8	Residential, Office	3
X0014301	19–21 Bourke St	2025	15.8	Retail	3

Table 6.4: The two groups of test cases ("intersection cases" being combinations of an intersection and a base image)

Completed	Base image	Developments	Intersections	Intersection cases
2023	2019	4	18	18
2023	2022	5	21	21
2024	2022	5	18	18
2024	2023	5	18	18
2023–2024 total	—	10	35	75
2025	2022	7	11	11
2025	2023	7	11	11
2025 total	—	7	11	22
Total	—	17	37	97

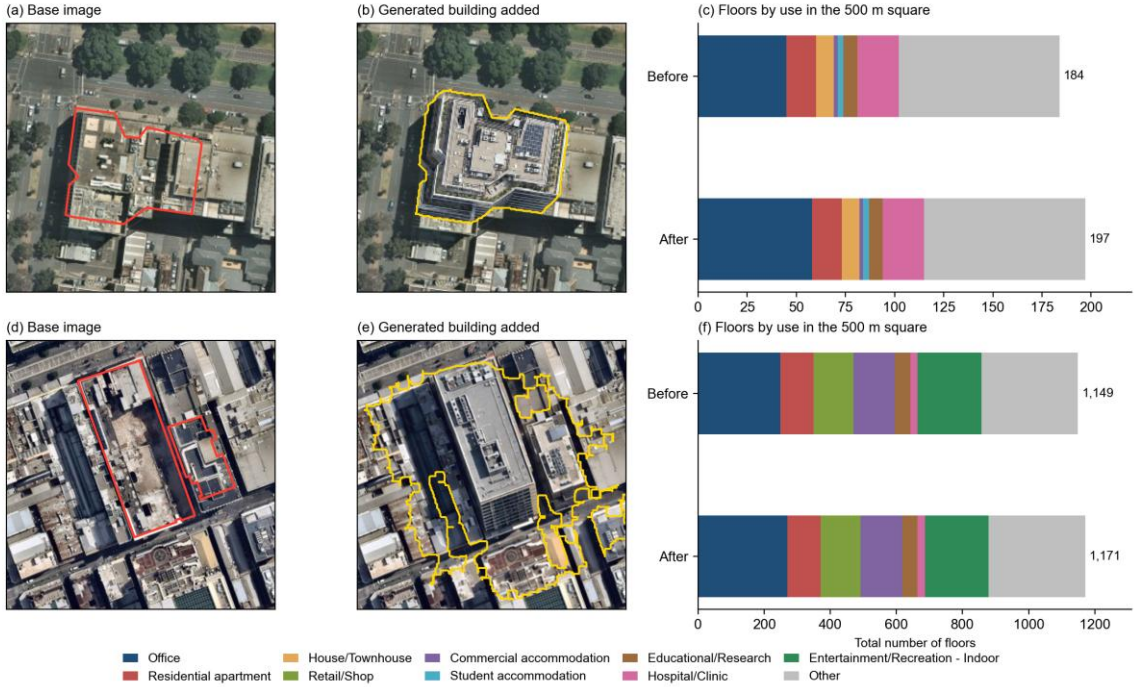


Figure 6.6: Examples of test cases. The top row is a case from the developments completed in 2023–2024: intersection 4455, development X0009218 (45.5 m tall), on the 2019 imagery. The bottom row is a case from the developments completed in 2025: intersection 4596, developments X0008610 and X0011004 (42.7 m and 34.2 m tall), on the 2022 imagery. (a)(d) Base image, with the red line marking the building footprint of the development. (b)(e) Generated building added to the base image, with the yellow line marking the area that changed markedly. (c)(f) Floors by use in the intersection's 500 m square before and after the development is added, recomputed after the development is written into the CLUE census.

6.4.3 Prediction accuracy

Table 6.5 gives the prediction accuracy at the affected intersections in the completion year for the four cases. For Route 2 + land use, the median GEH is 3.50–4.06, all below 5, and 57.8%–63.6% of intervals have a GEH below 5, comparable with the accuracy in the test year of Chapter 4 (median GEH 3.84). The developments completed in 2025 lie one year beyond the test year, and their median GEH is slightly higher than that of the

developments completed in 2023–2024. Compared with the no-development case, the median GEH of the other three cases changes by no more than 0.05 and the MAE by no more than 0.15 veh/15 min, so adding a generated building or updating the land-use attributes changes prediction accuracy very little.

Table 6.5: Prediction accuracy at the affected intersections in the completion year for the four cases (Route 2 + land use, mean of 5 seeds, GEH over all intervals, MAE averaged over intersection cases)

Group	Case	Median GEH	GEH < 5 (%)	MAE (veh/15 min)
Completed 2023–2024	No development	3.50	63.5	41.10
Completed 2023–2024	Image only	3.50	63.6	40.95
Completed 2023–2024	Land use only	3.51	63.5	41.15
Completed 2023–2024	Full scenario	3.50	63.6	41.00
Completed 2025	No development	4.01	58.1	63.83
Completed 2025	Image only	4.06	57.8	63.95
Completed 2025	Land use only	4.00	58.1	63.81
Completed 2025	Full scenario	4.06	57.8	63.93

6.4.4 With-and-without comparison

Table 6.6 and Figure 6.7 give the response of each case relative to the no-development case:

- **Image only**: for the developments completed in 2023–2024, the all-day response of Route 2 + land use is +0.14 veh/15 min (p = 0.314), and for those completed in 2025 it is +0.14 (p = 0.391). As a comparison, Route 1 + land use gives +0.69 (p = 0.017) for the developments completed in 2023–2024.
- **Land use only**: the developments completed in 2023–2024 and in 2025 give +0.35 (p = 0.100, with 68% of cases positive) and +0.04 veh/15 min (p = 0.807). Route 1 + land use gives +0.45 (p = 0.006).
- **Full scenario**: the two groups give +0.49 (p = 0.090) and +0.19 veh/15 min (p = 0.513), about 0.18% and 0.07% of the mean measured volume. Route 1 + land use gives +1.12 (p = 0.007).
- **Additional effect of the image**: with the development already in the land-use attributes, adding the generated building gives an additional response of +0.14 (p = 0.316) and +0.16 veh/15 min (p = 0.331).

Under Route 2 + land use, the all-day responses to adding the generated building, to updating the land-use attributes and to both together are all positive but very small, and none is statistically significant. The responses of Route 1 + land use are larger, and for the developments completed in 2023–2024 all three cases are positive and statistically significant. Figure 6.7 shows that the response varies through the day and differs considerably between the 5 seeds. For the developments completed in 2023–2024, the positive response occurs mainly before the morning peak and in the evening. For those completed in 2025, the response to the image alone occurs mainly during the day.

Table 6.6: All-day response of each case relative to the no-development case (veh/15 min, divided by α, mean of 5 seeds, seed-level p values in brackets, "Cases positive" being the case-level share under Route 2 + land use, Route 1 + land use not computed for the developments completed in 2025)

Group	Case	Route 2 + land use	Cases positive	Route 1 + land use
Completed 2023–2024	Image only	+0.14 (0.314)	51%	+0.69 (0.017)
Completed 2023–2024	Land use only	+0.35 (0.100)	68%	+0.45 (0.006)
Completed 2023–2024	Full scenario	+0.49 (0.090)	55%	+1.12 (0.007)
Completed 2023–2024	Full scenario − land use only	+0.14 (0.316)	52%	+0.67 (0.018)
Completed 2025	Image only	+0.14 (0.391)	59%	—
Completed 2025	Land use only	+0.04 (0.807)	55%	—
Completed 2025	Full scenario	+0.19 (0.513)	59%	—
Completed 2025	Full scenario − land use only	+0.16 (0.331)	59%	—

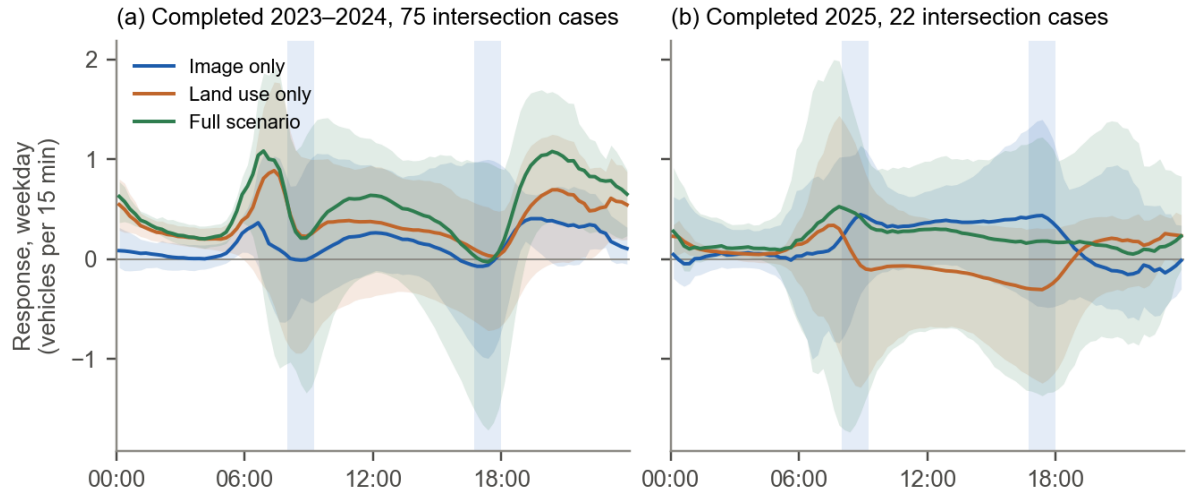


Figure 6.7: Response of each case relative to the no-development case through the day (weekdays, Route 2 + land use). (a) Developments completed in 2023–2024. (b) Developments completed in 2025. Solid lines are the mean of 5 seeds, shading shows the range across seeds, and the blue bands are the analysis windows of Section 3.2.3.

6.4.5 Closeness to measured volumes

Table 6.7 compares the prediction error of each case with that of the no-development case:

- **Image only**: for the developments completed in 2023–2024, the MAE changes by −0.15 veh/15 min on average (p = 0.016), but only 52% of cases come closer to the measured volumes, and the mean change comes mainly from a few cases (case-level signed-rank test p = 0.558). For the developments completed in 2025, the MAE changes by +0.11 (p = 0.205), and 41% of cases come closer.
- **Land use only**: the MAE of the two groups changes by +0.05 (p = 0.029) and −0.03 (p = 0.365), and 57% and 55% of cases come closer.

- **Full scenario**: the MAE of the two groups changes by −0.09 (p = 0.038) and +0.10 (p = 0.183), and 53% and 36% of cases come closer.

Against mean errors of about 41.1 and 63.8 veh/15 min at the affected intersections, the changes from all cases are very small. A single development has a limited effect on the predicted volumes at the surrounding intersections, and neither adding a generated building nor updating the land-use attributes brings the predictions markedly closer to the measured volumes.

Table 6.7: Change in prediction error of each case relative to the no-development case (Route 2 + land use, difference in MAE, veh/15 min, negative values being closer to the measured volumes, seed-level p values in brackets, "Cases closer" and the signed-rank test at case level)

Group	Case	Difference in MAE	Cases closer	Signed-rank p
Completed 2023–2024	Image only	−0.15 (0.016)	52%	0.558
Completed 2023–2024	Land use only	+0.05 (0.029)	57%	0.759
Completed 2023–2024	Full scenario	−0.09 (0.038)	53%	0.463
Completed 2025	Image only	+0.11 (0.205)	41%	0.098
Completed 2025	Land use only	−0.03 (0.365)	55%	0.290
Completed 2025	Full scenario	+0.10 (0.183)	36%	0.156

6.5 Key findings

1. The tool supports new building, alteration and demolition scenarios, can use developments from the development library or footprints drawn by the user, and allows several buildings in one scene.
2. The position, footprint and height of a building are fixed by the input geometry. When the same development is set to different heights, the generated building follows the massing rather than being left to the model.
3. Edits occur only within the massing and its shadow area, and all 18 saved generated examples have zero pixel difference outside the editable region.
4. The tool uses gpt-image-2's image editing interface as its generation core and constrains generation with 3D massing, the editable region and prompts organised in blocks, and a single generation takes about 20–80 seconds.
5. When a generated building is added to satellite imagery taken before construction, the median GEH at the affected intersections is 3.50–4.06, all below 5. The median GEH changes by no more than 0.05 and the MAE by no more than 0.15 veh/15 min from the no-development case, so the generated imagery leaves prediction accuracy essentially unchanged.
6. In the with-and-without comparison, the all-day responses of Route 2 + land use to the image only, the land use only and the full scenario are all positive. The full scenario gives +0.49 and +0.19 veh/15 min (developments completed in 2023–2024 and in 2025), about 0.18% and 0.07% of the mean measured volume, and none of the responses is statistically significant.
7. Against the measured volumes, the MAE of each case changes by no more than 0.15 veh/15 min and 36%–57% of cases come closer, so a single development has a limited effect on the predicted volumes at the surrounding intersections.

7 Conclusion

For the planning assessment of urban development, this project developed a traffic volume prediction method that describes the built environment with satellite imagery. Using large pretrained vision foundation models and a large language model embedding (LLM embedding), the project turned satellite imagery and road network structure into features, combined them with the land-use census and calendar information, and predicted traffic volumes in all 96 fifteen-minute intervals of the day using only information available at the planning stage. Building on this, the project quantified how changes in the scale and mix of office, residential apartment and retail land use affect predicted traffic volumes, and built a development scenario imagery tool with an interactive web interface.

The results show that satellite imagery features consistently improve prediction accuracy and that the method extends to 371 signal sites across the City of Melbourne. The predicted response to land-use change is clearest for office and arises mainly on secondary and local roads. The scenario imagery tool generates satellite imagery of new building, alteration and demolition scenarios from the input building geometry, with edits confined strictly to the building and its shadow. At the same time, the model can predict only for intersections seen in training, and the sensitivity results reflect statistical associations that the model learned from data rather than causal effects. These limitations define where the results apply and point to the directions of future work.

7.1 Summary

The project used central Melbourne as its study area, with the traffic volumes recorded by the Sydney Coordinated Adaptive Traffic System (SCATS) at 81 signalised intersections from 2014 to 2024 (excluding 2020 and 2021) as the observed data. These were aligned by location and year with Nearmap satellite imagery, the Census of Land Use and Employment (CLUE) and the OpenStreetMap road network, with 2024 as the test year. The following summarises the project's achievements and main findings against the three objectives set out in Section 1.3.

7.1.1 Planning-oriented traffic volume prediction

The project built a traffic volume prediction model that relies only on information available at the planning stage. SAM 3 and DINOv3 turn satellite imagery into region features, Qwen3-Embedding encodes each intersection's position relative to its neighbouring signalised intersections, and both are fed, together with land-use attributes and calendar features, into a temporal convolutional network. In the test year, the configuration that uses imagery, the road network and land use together (Route 2 + land use) is the most accurate, with a mean absolute error (MAE) of 59.18 vehicles per 15 minutes (veh/15 min), a median GEH of 3.84 and 59.9% of intervals with a GEH below 5. Its MAE is 4.9% lower than that of the time-only baseline, which uses only the temporal features and the historical profile.

Among the inputs, satellite imagery makes the most consistent contribution, reducing MAE by 1.67 veh/15 min (p = 0.002), with improvements in all 5 random seeds. Land

use improves accuracy by a statistically significant margin in the configuration that adds imagery only (Route 1), reducing MAE by 1.04 (p = 0.028), and is consistent in direction but not statistically significant in the other combinations. The contribution of the network features is not statistically significant. The gains in accuracy are concentrated in the weekday AM peak and on main roads: MAE falls by 9.4 in the AM peak and by 4.4 on average in the main-road group, while the local-road group barely changes. On the 371 signal sites of the City of Melbourne, the MAE of the four configurations with imagery differs by no more than 0.49, and at the 78 shared intersections with test data the MAE of Route 2 + land use (58.02) is close to that of the 81-intersection model (57.15). Land use brings no improvement to Route 1 (MAE change +0.14) and gives Route 2 a small reduction in MAE of 0.48, with lower MAE in 4 of the 5 seeds. These results show that the prediction method, which describes the built environment with satellite imagery, is effective within the study area and extends to a wider area. (Chapter 4)

7.1.2 Effect of land-use change on traffic volume

With the trained models of the two configurations that include land use (5 seeds each) held fixed, the project changed only the land-use attributes of the test year and compared the predictions before and after the change. The predicted response follows the direction of the change and varies roughly linearly with its size. When uses are changed by the same multiple, office is the most sensitive. In Route 1 + land use, the dose–response slope of office is 1.73 veh/15 min (p < 0.001), and in Route 2 + land use the response also increases monotonically with the dose (slope 1.33), with positive slopes in all 5 random seeds. When the office share doubles while the total number of floors stays the same, the all-day responses of the two configurations are +2.59 and +2.99 veh/15 min in the model output, about 0.7%–0.8% of all-day volume, and +1.55 and +1.80 veh/15 min in the final predictions. For office, the add-on ×2 responses are smaller than the swap ×2 responses, which shows that the office response comes mainly from the change in land-use mix. The swap responses of residential apartment and retail are not statistically significant. Per percentage point of share, the retail response matches that of office in Route 1 + land use (0.155–0.157 against 0.150 veh/15 min) and is about 57%–60% of office in Route 2 + land use, while residential is clearly weaker.

By time of day, office is the use with the largest response in the weekday AM and PM peaks and at weekend midday. Its response is largest before the peak (around 07:15 on weekdays) and in the evening, and falls when volumes are highest. By road class, trunk roads carry the most traffic, but in the weekday AM peak their response to an office swap ×2 is only about 0.01% of their volume, against 0.7%–1.8% on secondary and tertiary roads. Of the 90 comparisons of main and local roads, the local-road response is statistically significantly larger in 44 and the main-road response in 0. (Chapter 5)

7.1.3 Development scenario imagery generation

The project built a development scenario imagery tool with an interactive web interface. The tool supports new building, alteration and demolition scenarios, can use the 20 developments in the development library or building footprints drawn by the user, and allows several buildings in one scene. It uses gpt-image-2 as its generation core and constrains generation with 3D massing, an editable region and prompts organised in blocks. The position, footprint and height of a building are fixed by the input geometry, and when the same development is set to different heights, the generated building

follows the massing. Edits occur only within the massing and its shadow area, and all 18 saved generated examples have zero pixel difference outside the editable region. A single generation takes about 20–80 seconds. Tested on 17 real developments completed in 2023–2025, generated imagery used in place of actual imagery as input to the prediction model leaves accuracy essentially unchanged (median GEH 3.50–4.06, MAE changing by no more than 0.15 veh/15 min), and the predicted responses to the full scenario (+0.49 and +0.19 veh/15 min) are positive but not statistically significant. (Chapter 6)

7.2 Limitations

On the data side, the land-use features depend on the CLUE census, which covers only the City of Melbourne, and this limits both the study area and the extended validation. Applying the method to other regions requires land-use data with building-level detail. SCATS records traffic volumes only at signalised intersections, so volumes on road links and at unsignalised intersections cannot be observed directly, and some intersections have long runs of zero records that are probably caused by detector faults. Only one satellite image survey is used each year, captured around February, so a development completed during a year appears only in the following year's imagery.

On the method side, the model uses each intersection's 2022 historical profile as the reference for prediction, so it can predict only for intersections seen in training and must be retrained for new locations. The network features come from an adjacency graph built on straight-line distances, which does not include the actual connectivity, direction of travel or capacity of the roads and is the same in every year, so it can hardly represent changes in the network. The model also has no capacity constraints or route choice and cannot represent how traffic generated by a development redistributes across the network. In addition, data anomalies and large changes in volume cannot be foreseen from the inputs available at the planning stage and are the main source of error at the intersections with the largest errors.

On the evaluation side, the model is evaluated on a single test year (2024), apart from a check on 2025 at the intersections around the test developments of Section 6.4, and its performance in other years and traffic conditions remains to be tested. The data are split by year, and all test intersections appear in the training data, so the evaluation reflects the ability to predict known locations in future years. The median spacing between neighbouring intersections is about 126 m, so the image areas commonly overlap and the intersections are not independent samples. Each configuration was trained with only 5 random seeds, which limits the power of the statistical tests, and differences that are consistent in direction but not statistically significant (such as the contribution of the network features) remain inconclusive.

On the application side, the sensitivity results are the model's response to changes in its attributes. They reflect the association between land use and volume learned from the training data of the study area (2014–2019 and 2022), are not causal effects, and do not include changes that a development may bring to the road network, signal timing or travel modes. When land-use changes go beyond the range of the training data, the model's responses are extrapolations: for example, when the retail share is scaled up to 5 and 10 times, 16 and 25 intersections respectively exceed the maximum in the training data (17.1%). Road classes are taken from the OpenStreetMap classification, and roads of the same class still differ in traffic function and capacity. Building appearance in the

scenario imagery is synthesised by an image model and serves only to show the location and bulk of a proposal, not its architectural design. The project's results are suited to the screening and comparison of proposals at the planning stage and cannot replace a formal traffic impact assessment, traffic simulation or field surveys, and applying them to other regions also requires retraining and validation.

7.3 Recommendations and outlook

Based on the findings and limitations above, the following recommendations are made for applying the results.

1. At the planning stage, the project's model can be used to screen and compare development proposals. Impacts are best measured as the difference between the same model's predictions before and after a proposal. Components common to both predictions then cancel out in the difference, which makes the result more robust than the predicted absolute volumes.
2. When a development proposal is described by land-use attributes, the building scale and share should be given for each use, with particular attention to changes in the mix of uses such as office. The sensitivity analysis shows that the predicted response comes mainly from changes in the land-use mix, and that raising the office share while keeping the total number of floors unchanged produces a larger response than an add-on alone.
3. Traffic impacts should be interpreted with road class in mind. The same land-use change produces markedly larger relative changes on secondary and local roads than on trunk roads, so the assessment of a development's impact should not focus only on trunk roads.
4. The development scenario imagery tool is suited to communicating proposals and to preliminary comparison, presenting the changes before and after a development visually as satellite imagery. Its results are illustrative only and are not a basis for architectural design. Generated imagery can take the place of actual imagery as input to the prediction model without loss of accuracy, but the predicted change from a single development is small and not statistically significant (Section 6.4), so it is not on its own a basis for traffic assessment.
5. Planning-oriented traffic prediction depends on annually updated satellite imagery and building-level land-use data, which should be acquired continuously with consistent statistical definitions to support updating and extending the model.

Future work could focus on two directions. The first is to train a single prediction model across several areas of Melbourne and test its ability to predict at unseen locations, so that it can be used at new locations that lack long-term observations. The second is to build a directed road network automatically from satellite imagery and OpenStreetMap, so that the network features can reflect the upstream and downstream relationships of traffic in the network.

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Appendix A Data dictionary

The prediction model draws on SCATS traffic volumes, the signalised intersections, Nearmap satellite imagery and CLUE building data (Chapter 3), and the OpenStreetMap road attributes of the intersections are used to group the results by road class. The following lists, dataset by dataset, the fields the project uses, with their meaning and processing, and the site-years with a high share of days recorded as zero or with no record. Appendix B gives the rules by which the CLUE building data are aggregated into land-use attributes.

A.1 SCATS traffic volumes

Each record of the raw data gives the number of vehicles that passed one detector in each interval of a day (Department of Transport and Planning, 2026a). The fields and their processing are given in Table A.1.

Table A.1: Fields of the raw SCATS volume data

Field	Meaning	Processing
`NB_SCATS_SITE`	SCATS site number	Used as the intersection identifier, `SITE_NO` in the other tables
`QT_INTERVAL_COUNT`	Date of the record	Date part taken
`NB_DETECTOR`	Detector number	Counts of all detectors at an intersection summed by interval
`V00`–`V95`	Vehicles passing the detector in each of the 96 15-minute intervals, `V00` being 00:00–00:15 and `V95` 23:45–24:00	Negative and empty values treated as missing
`NM_REGION`, `CT_RECORDS`, `QT_VOLUME_24HOUR`, `CT_ALARM_24HOUR`	Other fields	Not used

The processed volume table has one record per intersection per day, with the fields `SITE_NO`, `date` and `V00`–`V95`. Here `V00`–`V95` are the sums of the vehicles counted by all detectors at the intersection in each interval, in vehicles per 15 minutes, with missing intervals left empty. For the 81 study intersections, the table holds 322,051 records from 1 January 2014 to 31 December 2024, 263,083 of them in the years used by the model.

Following the rules of Section 3.2.2, a site-day with all 96 intervals missing is a day with no record, and a site-day with records that sum to zero over the 96 intervals is a day recorded as zero. Table A.2 lists the site-years, within the years used by the model, in which either kind of day makes up more than 25% of the recorded days.

Table A.2: Site-years with a high share of days recorded as zero or with no record

Site	Name	Year	Use	Days recorded as zero (%)	Days with no record (%)	Longest run of zero days
2990	MCG Access / Brunton	2014	Training	75.6	0.0	18
2990	MCG Access / Brunton	2015	Training	67.0	0.0	17
2990	MCG Access / Brunton	2016	Training	67.8	0.0	15
2990	MCG Access / Brunton	2017	Training	71.4	0.0	21
2990	MCG Access / Brunton	2018	Training	58.7	0.0	12
2990	MCG Access / Brunton	2019	Training	60.4	0.0	12
2990	MCG Access / Brunton	2022	Training	71.9	0.0	23
2990	MCG Access / Brunton	2023	Validation	78.8	0.0	28
2990	MCG Access / Brunton	2024	Test	72.6	0.0	20
4586	Exhibition / Little Collins	2022	Training	84.3	0.0	204
4588	Swanston / Little Collins	2023	Validation	63.5	0.0	153
4588	Swanston / Little Collins	2024	Test	73.4	0.0	205
4603	Exhibition / Little Bourke	2018	Training	0.9	32.9	–
4603	Exhibition / Little Bourke	2022	Training	40.2	0.0	127
4889	Sturt / Kavanagh	2024	Test	0.0	100.0	–

There are 13 site-years with a high share of days recorded as zero, 9 of them at the MCG access road intersection, number 2990. Its zero days are spread across every month of the year, with the longest run lasting 12–28 days, which is consistent with the intersection carrying traffic mainly on event days. In the other 4 site-years the zeros run on without a break, the longest runs lasting 127–205 days. The site-years with a high share of days with no record are intersection 4603 in 2018 (33%) and intersection 4889 in 2024 (100%). Of these, intersection 4889 in 2024 has no record at all, so the test evaluation covers the other 80 intersections (Section 4.11).

A.2 Signalised intersections

The number, name, type and location of the study intersections come from the list of SCATS signal sites, and the road class, number of lanes, speed limit and road names were matched from OpenStreetMap (Section 3.5.2). All 81 study intersections are intersection signals, 72 in the City of Melbourne and 9 in the City of Yarra. The fields of the intersection table are given in Table A.3.

Table A.3: Fields of the intersection table

Field	Meaning
SITE_NO	SCATS site number
SITE_NAME	Intersection name, usually the names of the crossing roads separated by "/"
TYPE	Signal type, INT an intersection signal and POS a pedestrian crossing signal
MUNICIPALITY	Local government area, MBN the City of Melbourne and YRA the City of Yarra
LATITUDE, LONGITUDE	Latitude and longitude of the intersection (WGS 84)
osm_class	Highest OpenStreetMap road class among the roads passing within 40 m of the intersection
road_group	Road-class group, main for main roads (trunk, primary, secondary) and local for local roads (tertiary and lower)
osm_lanes	Number of lanes of that highest-class road recorded in OpenStreetMap, empty if not recorded
osm_maxspeed	Speed limit of that road (km/h), empty if not recorded
osm_names	Names of the roads passing within 40 m of the intersection

A.3 Nearmap satellite imagery

For each intersection and year, the survey closest to 1 February is used (Section 3.3.2), and the selected surveys are recorded in the survey table, whose fields are given in Table A.4. For the study intersections in the years used by the model, the table has 729 records, with capture dates from 31 days before 1 February to 16 days after.

Table A.4: Fields of the survey table

Field	Meaning
SITE_NO	Intersection number
year	Year
capture_date	Capture date of the selected survey
survey_id	Nearmap survey identifier, used to download the survey's tiles
days_from_target	Days between the capture date and 1 February of the year, negative before 1 February
n_surveys_that_year	Number of surveys covering the intersection in that year
LATITUDE, LONGITUDE	Centre of the image patch, that is, the intersection

A.4 CLUE building data

The CLUE building data list every building in the City of Melbourne each year (City of Melbourne, 2026a), and the project uses the 154,284 records of 2014–2024. The fields used and the handling of missing values are given in Table A.5, where the missing records of fields other than location are counted among the records that have a location. Other fields, such as building name, address and accessibility, do not enter the model.

Table A.5: Fields of the CLUE building data used

Field	Meaning	Missing records	When missing
`census_year`	Census year	0	–
`longitude`, `latitude`	Building location	2,587	Excluded from aggregation
`number_of_floors_above_ground`	Floors above ground	50	Counted as one floor
`predominant_space_use`	Predominant use	0	–
`construction_year`	Construction year	70,671	Excluded from building age and from buildings built in the last 5 years
`bicycle_spaces`	Bicycle spaces	82,467	Counted as zero

Appendix B Definitions of the land-use attributes

The land-use attributes are aggregated building by building from the CLUE building data over the same area as the satellite image patch, the 500 m square centred on the intersection. Each intersection has 43 attributes per year, of which 17 describe the building stock and 26 the mix of 13 use classes. The following gives the aggregation rules, the definition and unit of each attribute, and how the original CLUE uses correspond to the 13 classes. Table 4.4 summarises them.

B.1 Aggregation rules

Buildings are located by the longitude and latitude in the CLUE records, and a building counts towards an intersection when its location falls within the square. The square extends 250 m east, west, north and south of the intersection, and the inner square 125 m. Distances are computed with a planar approximation, taking 1° of latitude as 110,540 m and 1° of longitude as 111,320 m times the cosine of the mean latitude of the 143 intersections of the original study area. The attributes of each year are aggregated only from the records of the same census year.

Buildings with a missing number of floors count as one floor, missing bicycle spaces count as zero, and buildings with a missing construction year are left out of the building age and of the number built in the last 5 years (Table A.5). The number built in the last 5 years counts buildings whose construction year is no earlier than the census year minus 5. In the distance-weighted attributes, buildings closer than 20 m to the intersection are treated as being 20 m away. When the square has no building of 10 or of 30 or more floors, the corresponding distance to the nearest such building is set to 500 m.

When the square has no building records, all 43 attributes are 0. When none of the buildings in the square has a construction year, the median building age is 0. Among the study intersections, only intersections 1038 and 3392 are affected. Both lie on Punt Road, and their squares hold at most 1 building record in any year. There are 4 site-years with no building records and 14 site-years in which no building has a construction year.

The attributes are standardised with the mean and standard deviation of the training years before entering the model (Section 4.3.5).

B.2 Building stock attributes

Table B.1 gives the definitions and units of the 17 building stock attributes and their medians over the 729 site-years of the 81 study intersections in the 9 years used.

Table B.1: Definitions of the building stock attributes

Attribute	Name	Unit	Definition	Median
`buildings`	Buildings	buildings	Number of buildings in the square	140
`floors_mean`	Mean floors	floors	Mean number of floors above ground of the buildings in the square	4.0
`floors_max`	Maximum floors	floors	Largest number of floors above ground in the square	19
`floors_sum`	Total floors	floors	Sum of the floors above ground of the buildings in the square	665
`floors_p90`	90th percentile of floors	floors	90th percentile of the number of floors above ground in the square	8.0
`tall_over_10`	Buildings of 10+ floors	buildings	Number of buildings with 10 or more floors above ground	7
`tall_over_30`	Buildings of 30+ floors	buildings	Number of buildings with 30 or more floors above ground	0
`bicycle_spaces`	Bicycle spaces	spaces	Sum of the bicycle parking spaces recorded for the buildings in the square	242
`built_last_5y`	Built in the last 5 years	buildings	Number of buildings with a construction year no earlier than the census year minus 5	2
`median_age`	Median building age	years	Census year minus the median construction year, over buildings with a construction year	97.0
`buildings_inner`	Inner buildings	buildings	Number of buildings in the central 250 m square	36
`floors_sum_inner`	Inner total floors	floors	Sum of the floors of the buildings in the central 250 m square	168
`floors_max_inner`	Inner maximum floors	floors	Largest number of floors in the central 250 m square	14
`floors_sum_weighted`	Distance-weighted floors	floors/m	Sum over buildings of floors divided by the distance to the intersection	4.32
`buildings_weighted`	Distance-weighted buildings	buildings/m	Sum over buildings of the inverse distance to the intersection	0.872
`nearest_over_10_m`	Distance to nearest 10+ floor building	m	Distance to the nearest building with 10 or more floors	87
`nearest_over_30_m`	Distance to nearest 30+ floor building	m	Distance to the nearest building with 30 or more floors	500

B.3 Land-use mix attributes

CLUE records 28 predominant uses, and 12 of them each form a class of their own: the 10 most common uses in the CLUE records of the whole municipality over 2014–2024, and Student Accommodation and Hospital/Clinic, the 12th and 13th most common. The other 16 uses, including Equipment Installation, the 11th most common, are pooled as "other", giving 13 classes (Table B.2). Each class has two shares, by number of buildings and by number of floors, in columns named `share_` and `floorshare_` followed by the suffix in the table. When the square has buildings, the shares of the 13 classes sum to 1.

The building counts and floor shares in Tables B.2 and B.3 are computed over the union of the squares of the study intersections in 2024, counting each building once, 3,594 buildings in all.

Table B.2: Use classes and their composition in the study area (2024)

Column suffix	CLUE use	Buildings	Floor share (%)
house_townhouse	House/Townhouse	1,813	24.7
residential_apartment	Residential Apartment	399	23.7
office	Office	263	16.7
retail__shop	Retail - Shop	144	2.7
entertainment_recreation__indoor	Entertainment/Recreation - Indoor	308	5.8
educational_research	Educational/Research	69	3.2
storage	Storage	29	0.5
commercial_accommodation	Commercial Accommodation	77	6.4
hospital_clinic	Hospital/Clinic	48	1.7
student_accommodation	Student Accommodation	44	2.7
workshop_studio	Workshop/Studio	10	0.2
unoccupied__unused	Unoccupied - Unused	206	6.5
other	The other 16 uses (Table B.3)	184	5.1

The 16 uses pooled as “other” are listed in Table B.3. Together they account for 5.1% of the buildings in the study area.

Table B.3: CLUE uses pooled as “other” (2024)

CLUE use	Buildings
Performances, Conferences, Ceremonies	41
Parking - Commercial Covered	26
Unoccupied - Under Renovation	23
Equipment Installation	21
Parking - Private Covered	16
Unoccupied - Under Construction	16
Public Display Area	12
Unoccupied - Under Demolition/Condemned	11
Community Use	6
Retail - Showroom	4
Transport	4
Institutional Accommodation	3
Wholesale	1
Manufacturing	0
Retail - Cars	0
Retail - Stall	0

Appendix C Full experimental results

The accuracy reported in Chapter 4 is the mean of 5 random seeds for each configuration. The following gives the test accuracy, number of epochs and post-processing parameters of each of the 30 training runs, together with the results of each run in the extended validation and the test error at each intersection. All metrics are computed on the test year (2024), excluding missing intervals (1.43% of all), and are defined in Section 4.11.

The two metrics not used in the main text are defined as follows. The mean absolute percentage error (MAPE) is the mean over intervals of the absolute error divided by the measured volume, with a denominator of 1 where the measured volume is below 1 vehicle, so intervals with a measured volume of zero or close to zero raise MAPE sharply. The mean GEH is the arithmetic mean of the GEH over all intervals.

C.1 Study intersections

Tables C.1 and C.2 list the results of each run by configuration, with the seeds in the order 42, 73, 128, 2024 and 7. MAE and RMSE are in vehicles per 15 minutes. The best epoch is the epoch with the lowest median GEH on the validation set, which gives the model finally used (Section 4.8).

Table C.1: Test accuracy of each run of the six configurations (2024, 80 intersections)

Configuration	Seed	MAE	RMSE	MAPE (%)	Mean GEH	Median GEH	GEH < 5 (%)	GEH < 10 (%)
Time only	42	62.24	128.7	59.1	6.54	4.08	57.6	81.6
	73	61.37	126.4	59.6	6.50	4.15	57.1	81.6
	128	62.70	130.3	59.1	6.58	4.17	56.9	81.7
	2024	62.26	127.2	61.6	6.55	4.09	57.6	81.7
	7	62.55	126.1	59.6	6.59	4.20	56.7	81.3
	Mean	62.22	127.7	59.8	6.55	4.14	57.2	81.6
Time + land use	42	62.17	134.3	60.4	6.49	3.94	59.0	82.4
	73	62.35	134.1	61.1	6.49	3.93	59.1	82.2
	128	60.54	129.3	58.4	6.36	3.96	58.9	82.5
	2024	61.16	127.7	60.3	6.43	3.91	59.3	82.3
	7	60.57	127.1	58.9	6.37	3.97	58.8	82.4
	Mean	61.36	130.5	59.8	6.43	3.94	59.0	82.4
Route 1	42	59.92	120.6	60.8	6.36	3.94	58.9	82.0
	73	60.30	122.7	58.2	6.33	3.96	58.8	82.0
	128	60.56	122.7	59.6	6.34	3.92	59.1	82.1
	2024	60.87	124.6	59.7	6.37	3.94	58.9	82.3
	7	61.11	122.2	59.4	6.42	4.03	58.2	81.8
	Mean	60.55	122.5	59.6	6.36	3.96	58.8	82.0
Route 1 + land use	42	59.57	120.9	60.2	6.29	3.88	59.5	82.5
	73	59.03	121.1	58.9	6.23	3.84	60.0	82.9
	128	60.26	122.1	60.1	6.37	3.93	59.0	82.0
	2024	59.41	121.3	60.2	6.27	3.88	59.5	82.6
	7	59.27	121.5	59.2	6.25	3.86	59.8	82.7
	Mean	59.51	121.4	59.7	6.28	3.88	59.6	82.5
Route 2	42	60.70	121.4	60.6	6.41	4.02	58.2	81.6
	73	58.69	119.8	58.8	6.21	3.85	59.8	82.7
	128	59.73	122.0	59.5	6.30	3.91	59.3	82.4
	2024	61.08	124.6	60.4	6.39	3.90	59.3	82.2
	7	60.53	123.5	60.3	6.38	3.95	59.0	82.3
	Mean	60.14	122.3	59.9	6.34	3.93	59.1	82.3
Route 2 + land use	42	58.63	119.6	57.8	6.20	3.82	60.1	82.7
	73	59.42	121.6	58.6	6.24	3.85	59.8	82.5
	128	59.20	121.4	58.5	6.23	3.77	60.5	82.7
	2024	59.87	121.5	58.7	6.31	3.88	59.5	82.3
	7	58.76	120.5	58.6	6.22	3.85	59.8	82.8
	Mean	59.18	120.9	58.4	6.24	3.84	59.9	82.6

Table C.2: Epochs and post-processing parameters of each run of the six configurations

Configuration	Seed	Epochs trained	Best epoch	Blend weight α	Smoothing coefficient β	Parameters
Time only	42	10	4	0.95	0.15	753,649
	73	9	3	0.80	0.20	
	128	18	18	0.85	0.20	
	2024	11	5	0.80	0.20	
	7	9	3	0.80	0.15	
Time + land use	42	18	17	0.90	0.20	761,265
	73	18	12	1.00	0.10	
	128	13	7	1.00	0.20	
	2024	11	5	0.85	0.15	
	7	18	13	1.00	0.15	
Route 1	42	14	8	0.60	0.10	795,889
	73	18	13	0.60	0.10	
	128	18	17	0.60	0.10	
	2024	18	12	0.60	0.10	
	7	18	15	0.60	0.10	
Route 1 + land use	42	14	8	0.60	0.15	803,505
	73	18	16	0.60	0.05	
	128	11	5	0.60	0.15	
	2024	18	18	0.60	0.15	
	7	18	16	0.60	0.10	
Route 2	42	10	4	0.60	0.15	804,177
	73	18	18	0.60	0.05	
	128	12	6	0.60	0.15	
	2024	18	17	0.60	0.10	
	7	12	6	0.60	0.15	
Route 2 + land use	42	18	18	0.60	0.10	811,793
	73	18	18	0.60	0.05	
	128	18	18	0.60	0.10	
	2024	18	13	0.60	0.15	
	7	18	18	0.60	0.15	

C.2 Extended validation

The extended validation (Section 4.13) trained four configurations with imagery on the 371 signal sites. All configurations selected a blend weight of 0.60 and a smoothing coefficient of 0.05 for every seed and trained for 18 epochs. The test accuracy of each run is given in Table C.3, whose last column is the MAE at the 78 intersections, of the 79 shared with the study area, that have test data.

Table C.3: Test accuracy of each run of the four configurations on the 371 signal sites (2024)

Configuration	Seed	MAE	RMSE	MAPE (%)	Median GEH	GEH < 5 (%)	GEH < 10 (%)	MAE (shared intersections)
Route 1	42	58.95	114.4	151.9	4.11	57.4	80.6	58.76
	73	59.23	114.7	154.5	4.14	57.0	80.3	57.85
	128	59.53	115.2	154.5	4.13	57.2	80.3	58.83
	2024	58.65	113.8	153.5	4.11	57.4	80.7	57.99
	7	59.21	114.8	154.2	4.15	57.0	80.5	58.20
	Mean	59.11	114.6	153.7	4.13	57.2	80.5	58.33
Route 1 + land use	42	59.55	116.6	154.1	4.10	57.5	80.7	59.28
	73	59.33	115.3	152.5	4.10	57.4	80.5	58.67
	128	59.45	115.2	150.9	4.15	57.0	80.4	58.20
	2024	58.87	114.4	151.7	4.10	57.4	80.7	58.68
	7	59.06	114.7	151.2	4.10	57.4	80.5	57.76
	Mean	59.25	115.2	152.1	4.11	57.4	80.6	58.52
Route 2	42	59.07	115.1	151.4	4.11	57.4	80.5	58.13
	73	59.37	115.5	151.0	4.11	57.4	80.3	58.03
	128	59.80	115.4	152.1	4.20	56.5	80.0	58.73
	2024	58.61	114.3	155.2	4.08	57.7	80.8	57.77
	7	59.37	114.5	152.5	4.14	57.1	80.2	58.19
	Mean	59.24	115.0	152.5	4.13	57.2	80.4	58.17
Route 2 + land use	42	58.53	113.8	148.9	4.13	57.3	80.7	57.82
	73	58.93	114.4	150.1	4.13	57.2	80.6	58.35
	128	59.10	114.7	150.6	4.12	57.3	80.6	58.23
	2024	58.73	114.4	151.4	4.07	57.8	80.8	57.98
	7	58.51	113.6	149.8	4.11	57.4	80.7	57.70
	Mean	58.76	114.2	150.2	4.11	57.4	80.7	58.02

C.3 Test error at each intersection

Table C.4 lists the test MAE of the 80 study intersections as the mean of the 5 seeds. Intersection 4889 has no data at all in 2024 and is not included. Volume is the mean measured volume over the valid intervals of the test year, in the same unit as MAE.

Table C.4: Test MAE at each intersection (2024, mean of 5 seeds, veh/15 min)

Site	Name	Volume	Time only	Time + land use	Route 1	Route 1 + land use	Route 2	Route 2 + land use
1029	Grattan / Drummond	109	29.5	27.2	27.1	27.0	27.8	27.5
1034	Faraday / Drummond	125	27.4	27.1	32.0	30.3	31.7	32.2
1036	Queensberry / Drummond	178	35.3	33.3	35.3	33.6	37.1	34.2
1038	Punt / Gate 6	740	74.8	79.7	74.3	71.4	74.7	71.2
1052	Wellington / Clarendon	271	66.4	64.3	60.5	61.0	60.7	61.9
2024	Spring / Flinders Lane	109	19.2	17.5	17.0	16.6	16.8	16.6

Site	Name	Volume	Time only	Time + land use	Route 1	Route 1 + land use	Route 2	Route 2 + land use
2100	Eastern / Nicholson / Princes	996	83.0	77.4	73.7	73.1	72.9	70.8
2901	Spring / Lonsdale	225	31.7	31.6	32.5	30.1	31.5	30.4
2902	Exhibition / Lonsdale	460	58.4	55.7	55.7	53.3	54.8	55.4
2903	Russell / Lonsdale	475	51.2	49.4	44.5	46.1	44.6	44.2
2911	Exhibition / Little Lonsdale	11	3.5	3.6	3.5	3.5	3.7	3.7
2912	Russell / Little Lonsdale	257	40.0	39.7	38.5	38.8	38.8	38.8
2918	Victoria / Latrobe	742	89.9	93.1	109.8	101.9	105.2	106.3
2919	Exhibition / Latrobe	286	78.0	82.0	90.8	85.5	84.9	84.6
2920	Russell / Latrobe	488	73.5	85.5	74.5	74.6	69.3	73.4
2931	Flinders / Wellington Sth	72	13.6	13.8	13.6	13.4	13.4	13.0
2932	Wellington / Spring / Flinders	201	42.0	41.7	44.1	42.7	42.1	40.4
2933	Victoria / Clarendon	576	63.2	59.4	57.6	57.7	58.1	57.7
2934	Wellington / Jolimont	310	52.4	41.2	41.1	41.3	42.2	39.9
2990	MCG Access / Brunton	0	0.6	0.6	0.4	0.4	0.4	0.4
3385	Hoddle / Victoria	1,059	76.5	75.7	72.7	73.7	76.1	70.7
3386	Hoddle / Albert / Elizabeth	1,306	204.0	193.4	113.6	114.7	111.5	113.6
3389	Hoddle / Freeman	661	226.6	232.2	235.3	235.3	235.1	229.0
3390	Punt / Bridge / Hoddle / Wellington	954	82.8	80.1	79.8	81.2	81.1	77.6
3391	Punt / Rowena	772	68.1	67.9	69.4	69.6	69.5	67.9
3392	Punt / Brunton	1,185	154.4	143.2	197.2	197.9	198.9	205.5
3417	Elgin / Drummond	169	25.4	23.5	23.1	23.1	23.5	23.0
3450	Nicholson / Elgin / Johnston	458	71.5	68.7	69.1	65.5	64.1	65.0
3498	Elgin / Cardigan	155	41.0	41.5	41.9	41.3	41.5	41.4
4218	Victoria / Smith	668	72.6	71.1	66.2	66.0	66.0	67.9
4380	Cemetery / Lygon / Princes	824	67.8	65.0	67.5	67.8	69.2	67.2
4381	Rathdowne / Princes	817	106.7	110.2	103.3	103.0	103.0	102.0
4395	Grattan / Cardigan	107	30.2	28.9	28.7	27.8	28.9	28.3
4396	Grattan / Lygon	205	34.9	32.0	30.8	30.4	32.1	30.6

Site	Name	Volume	Time only	Time + land use	Route 1	Route 1 + land use	Route 2	Route 2 + land use
4397	Rathdowne / Grattan	199	25.9	26.7	27.2	27.5	25.6	28.3
4399	Lygon / Queensberry	289	64.6	59.5	57.3	58.1	57.5	56.9
4400	Rathdowne / Queensberry	371	70.5	69.2	70.8	70.2	69.4	70.4
4404	Victoria / Lygon	432	106.6	107.0	113.7	115.4	113.1	111.8
4405	Victoria / Rathdowne	467	53.6	50.1	50.5	48.9	49.8	48.6
4406	Victoria Pde and Nicholson St	690	61.7	62.7	69.4	58.4	62.1	59.1
4407	Lygon / Faraday	197	34.3	31.7	30.7	29.7	30.3	29.3
4408	Lygon / Elgin	299	34.7	32.1	32.2	31.8	32.3	31.2
4409	Rathdowne / Elgin	611	481.8	501.8	442.1	447.8	431.6	421.4
4410	Canning / Elgin	164	23.0	22.0	22.5	22.4	22.0	21.6
4419	Nicholson / Gertrude	286	31.8	31.0	31.4	31.3	31.1	30.5
4441	Victoria / Brunswick / Gisborne	775	121.4	118.2	126.8	118.2	121.9	117.4
4443	Victoria / Wellington / Powlett	576	81.7	79.2	76.0	76.8	78.9	77.0
4444	Albert / Nicholson / Lonsdale	274	33.7	29.9	27.3	27.3	27.2	26.7
4445	Albert / Gisborne	239	29.2	28.2	25.6	25.3	27.4	24.6
4446	Albert / Lansdowne	461	234.8	232.2	195.7	218.7	198.3	219.5
4447	Albert / Clarendon	202	30.1	29.6	30.0	30.5	29.9	28.8
4448	Albert / Powlett	709	166.7	168.3	209.7	182.5	261.6	207.8
4449	Albert / Simpson	131	20.1	19.8	20.5	20.4	21.0	19.5
4451	Wellington Sth / Jolimont	199	41.4	39.2	39.9	38.3	40.3	37.8
4453	Wellington / Powlett	176	32.8	32.7	31.4	30.8	31.7	30.7
4454	Wellington / Lansdowne	330	52.9	45.3	43.9	44.6	42.4	43.3
4455	Victoria / Lansdowne	643	105.6	101.5	92.8	96.0	93.4	94.7
4528	Batman / Melb Park Nth Access	174	32.9	30.8	28.8	28.8	28.6	28.7
4529	Melbourne Park Nth Access / Fed Sq Access	20	12.9	13.3	13.0	13.2	12.9	12.8
4549	Alexandra / Linlithgow	688	64.3	64.1	59.5	59.6	61.2	59.5
4560	Exhibition / Flinders	497	63.4	62.0	61.1	59.6	59.9	59.7

Site	Name	Volume	Time only	Time + land use	Route 1	Route 1 + land use	Route 2	Route 2 + land use
4561	Russell / Flinders	585	78.0	81.3	90.0	87.4	82.0	86.5
4562	Swanston / Flinders	642	170.2	182.1	204.5	174.0	163.0	173.3
4568	Exhibition / Flinders Lane	18	4.4	4.5	4.4	4.2	4.4	4.2
4569	Russell / Flinders Lane	120	17.6	17.6	17.1	17.0	17.4	17.0
4570	Swanston / Flinders Lane	103	29.2	29.6	35.8	30.8	33.6	31.0
4575	Spring / Collins	293	62.2	57.5	47.7	51.6	49.4	50.4
4576	Collins / Exhibition	304	37.6	31.7	31.3	30.3	29.9	30.0
4577	Russell / Collins	290	34.9	34.3	33.6	33.7	33.6	32.9
4578	Swanston / Collins	82	28.8	28.6	28.6	28.4	28.2	28.4
4586	Exhibition / Little Collins	24	20.5	20.4	20.2	20.5	20.0	19.5
4587	Russell / Little Collins	81	12.6	12.2	12.1	12.1	12.3	12.2
4588	Swanston / Little Collins	7	23.0	23.2	22.8	23.2	23.3	22.4
4593	Spring / Bourke	142	22.0	20.4	19.5	19.1	19.4	19.0
4594	Bourke / Exhibition	145	29.3	26.4	26.8	25.6	27.5	24.7
4595	Russell / Bourke	223	27.4	27.4	27.8	26.8	27.6	26.3
4596	Swanston / Bourke	35	10.1	10.0	10.0	10.0	10.1	10.0
4603	Exhibition / Little Bourke	12	5.5	5.4	5.0	5.2	4.8	4.9
4604	Russell / Little Bourke	21	6.4	6.4	6.3	6.4	6.3	6.4
4920	City / Southgate	496	52.4	46.6	48.3	45.1	47.8	47.4

Appendix D Prompts for the development scenario imagery tool

The four prompts described in Section 6.2.3 are given in full below. The tool sends them to gpt-image-2 in English as written, with the parameters in braces filled in by the program from the inputs (Table 6.1) and rounded to whole numbers.

D.1 Single new building (495 words)

SCENE

This is an overhead satellite photograph of a site in Melbourne. A massing study of an approved building has been painted onto it at the exact position, footprint and scale the building will occupy. The {roof_word} face is its roof; the {wall_word} faces are its sides. Those two colours identify the surfaces and are not the building's materials.

TASK

Replace the painted solid with a photographic building, so the result reads as an ordinary satellite photograph in which that building exists.

GEOMETRY

The ground position, the footprint outline and the roof outline are fixed. Do not move, rotate, resize or simplify them. The building is {height} m tall, about {storeys} storeys, on a footprint of {area} square metres, land use {use}.

The one {roof_word} face is the roof and the only roof. Every {wall_word} face is a vertical wall seen from the side, not roof: put no plant, no services and no roof texture on it. Telling them apart is a matter of position, not size – at this height the walls can cover more of the picture than the roof does, and they are still walls. The roof is the {roof_word} outline, offset about {shift} m towards the {bearing} from the ground outline, because a satellite {lean} degrees off vertical sees the top of a {height} m building displaced that far. The ground outline is where the building meets the ground.

Show facade on the same side and to the same extent the solid does, and leave the roof and the ground footprint exactly where they are drawn.

LIGHT

Every shadow in this frame runs the same direction, and the length of each relative to its own building tells you the sun elevation. Read both off the real buildings in frame rather than assuming them, then give the new building a shadow of matching direction and proportionate length. Where that shadow falls across a road, a roof or a footpath, darken what is there rather than painting a flat shape. Where the new building blocks light that used to reach the ground, remove the shadow that no longer belongs.

SURFACE

Give it the substance a real building has at this resolution: roof plant, services, lift overruns and parapets on the roof; glazing lines, spandrels and material on the facades. Match the grain, sharpness, colour balance and haze of the surrounding photograph. The

edges where it meets the ground, the neighbouring roofs and the street should look photographed, not cut out.

Neither {roof_word} nor {wall_word} may appear in the result. They are markings, not materials: the finished roof is the grey of membrane, metal deck or concrete with its plant and services, and the finished walls are glazing, precast or brick. A {roof_word} roof or a {wall_word} wall in the output means the marking was copied instead of read.

HOLD

Apart from the new building and the shadows it changes, every pixel stays as it is. Add no outline, label, annotation or watermark anywhere.

D.2 Several new buildings (476 words)

SCENE

This is an overhead satellite photograph of a site in Melbourne. Massing studies of {count} approved buildings have been painted onto it at the exact positions, footprints and scales they will occupy. The {roof_word} faces are roofs; the {wall_word} faces are sides. Those two colours identify the surfaces and are not the buildings' materials.

TASK

Replace the painted solids with photographic buildings, so the result reads as an ordinary satellite photograph in which all {count} exist.

GEOMETRY

Ground positions, footprint outlines and roof outlines are fixed. Do not move, rotate, resize, merge or simplify them. They stay separate buildings and the gaps between them are preserved. Where two solids overlap, the one painted in front stands in front. The buildings are:

{buildings}

Each {roof_word} face is a roof and the only roof of its building. Every {wall_word} face is a vertical wall seen from the side, not roof: put no plant, no services and no roof texture on it. At these heights the walls can cover more of the picture than the roofs do, and they are still walls. Each roof outline is offset towards the {bearing} from its own ground outline, further for a taller building, because the satellite is {lean} degrees off vertical. Show facade on the same side and to the same extent each solid does, and leave every roof and ground footprint exactly where it is drawn.

LIGHT

Every shadow in this frame runs the same direction, and the length of each relative to its own building tells you the sun elevation. Read both off the real buildings in frame, then give each new building a shadow of matching direction and a length proportionate to its own height – a taller one casts a longer shadow than a shorter one in the same frame. Include the shadow one new building casts onto another, and remove any existing shadow the new buildings now block.

SURFACE

Give each the substance a real building has at this resolution: roof plant, services, lift overruns and parapets on the roof; glazing lines, spandrels and material on the facades. Match the grain, sharpness, colour balance and haze of the surrounding photograph. The edges where they meet the ground, neighbouring roofs and the street should look photographed, not cut out. They may differ from each other in age, material and colour, as neighbouring buildings do.

Neither {roof_word} nor {wall_word} may appear in the result. They are markings, not materials: the finished roof is the grey of membrane, metal deck or concrete with its plant and services, and the finished walls are glazing, precast or brick. A {roof_word} roof or a {wall_word} wall in the output means the marking was copied instead of read.

HOLD

Apart from the new buildings and the shadows they change, every pixel stays as it is. Add no outline, label, annotation or watermark anywhere.

D.3 Alteration (551 words)

SCENE

This is an overhead satellite photograph of a site in Melbourne. A building stands on the parcel now, and two things have been drawn over it. Outlined in {roof_word} and {wall_word} lines, with the building still visible inside, is the building as it stands today – about {existing} m tall. Painted as a filled solid on the same footprint is what that same building becomes after the alteration: {height} m tall, its {roof_word} face the new roof and its {wall_word} faces the new sides. Those colours identify surfaces and are not the building's materials.

TASK

Alter the building that is there so it matches the painted solid, and make the photograph consistent with that alteration.

GEOMETRY

The altered building's ground position, footprint outline and roof outline are fixed as painted. It is {height} m tall, about {storeys} storeys, on a footprint of {area} square metres, land use {use}.

The one {roof_word} face is the new roof and the only roof. Every {wall_word} face is a vertical wall seen from the side, not roof: put no plant, no services and no roof texture on it. The roof is the {roof_word} outline, offset about {shift} m towards the {bearing} from the ground outline, because a satellite {lean} degrees off vertical sees the top of a {height} m building displaced that far. The ground outline is where the building meets the ground.

Nothing of the building's former height may remain: no part of its old roof, no facade above the new roofline, no rooftop plant left floating, and no trace of its old outline showing through. Remove the marking lines themselves along with the height they mark – they are instructions, not part of the scene.

LIGHT

This is where the change shows most. The building was {existing} m and becomes {height} m, so it was shading roughly {ratio} times as far as it now will. Find the shadow the old building cast, work out how much of it a {height} m building would still cast, and give back the rest: restore road surface, footpath, neighbouring roof, vegetation and parked cars at the brightness those same surfaces have elsewhere in this photograph where nothing shades them. Then cast the shortened shadow in the direction every other shadow in the frame runs.

SURFACE

Give the altered building the substance a real one has at this resolution: roof plant, services and parapets on its new roof; glazing lines and material on its facades. Ground and neighbouring buildings that the taller building used to hide from view must be filled in from what surrounds them – continue the road, the footpath, the parking, the planting and the adjacent roofs as they run on either side, at the grain, sharpness and colour balance of the rest of the photograph.

Neither {roof_word} nor {wall_word} may appear in the result. They are markings, not materials: the finished roof is the grey of membrane, metal deck or concrete with its plant and services, and the finished walls are glazing, precast or brick. A {roof_word} roof or a {wall_word} wall in the output means the marking was copied instead of read.

HOLD

Apart from the altered building, what it used to hide and the shadows that change, every pixel stays as it is. Add no outline, label, annotation or watermark anywhere.

D.4 Demolition (402 words)

SCENE

This is an overhead satellite photograph of a site in Melbourne. One building in it has been outlined: a {wall_word} line around where it meets the ground, a {roof_word} line around its roof, and {wall_word} lines up its visible corners joining the two. The building itself is still visible inside that outline, unchanged. The lines mark which building is meant and how tall it stands – about {height} m, the two outlines being {shift} m apart towards the {bearing} because the satellite is {lean} degrees off vertical.

TASK

Demolish the outlined building. The result should read as an ordinary satellite photograph of the same place taken after demolition, with nothing standing on that parcel and no drawn lines left anywhere.

GEOMETRY

The ground the building occupied becomes open. Do not put another building there, and do not enlarge the neighbouring ones into the space. Buildings that share a wall with the one being demolished keep their own walls, which now face open ground. Remove the marking lines themselves along with the building – they are instructions, not part of the scene.

LIGHT

The building's shadow goes with it. Find the shadow it casts – it runs the same direction as every other shadow in the frame and its length matches {height} m against the neighbours – and restore whatever that shadow was lying on: road surface, footpath, neighbouring roof, vegetation, parked cars, all at the brightness the unshaded parts of those same surfaces have in this photograph. Light reaching ground that was in shade is the clearest sign the building is gone.

SURFACE

Leave the cleared parcel as {ground}, rendered the way that surface looks from above in this part of the city, at the grain, sharpness, colour balance and haze of the rest of the photograph. Ground and neighbouring buildings the demolished building used to hide must be filled in from what surrounds them – continue the road, the footpath, the parking, the planting and the adjacent roofs as they run on either side. Its edges against the street, the footpath and the neighbouring buildings should look photographed, not cut out.

Neither {roof_word} nor {wall_word} may appear in the result – the marking lines go with the building they mark.

HOLD

Apart from the demolished building, what it used to hide and the shadows that change, every pixel stays as it is. Add no outline, label, annotation or watermark anywhere.

Appendix E Test cases for generated imagery

This appendix shows all the test cases of Section 6.4, in six groups by completion year and base image year. Each case is a pair of images. The left image is the base image, with the red line marking the building footprint of the development. The right image is the base image with the generated building added, with the yellow line marking the area that changed markedly from the base image. Both images are cropped to the same extent. Where several developments lie around one intersection, the generated buildings of all of them are added.

E.1 Completed in 2023, on the 2019 imagery

18 intersection cases, involving 4 developments.



Figure E.1: Cases completed in 2023 on the 2019 imagery (1/3). In each pair, the left image is the base image and the right image has the generated building added.

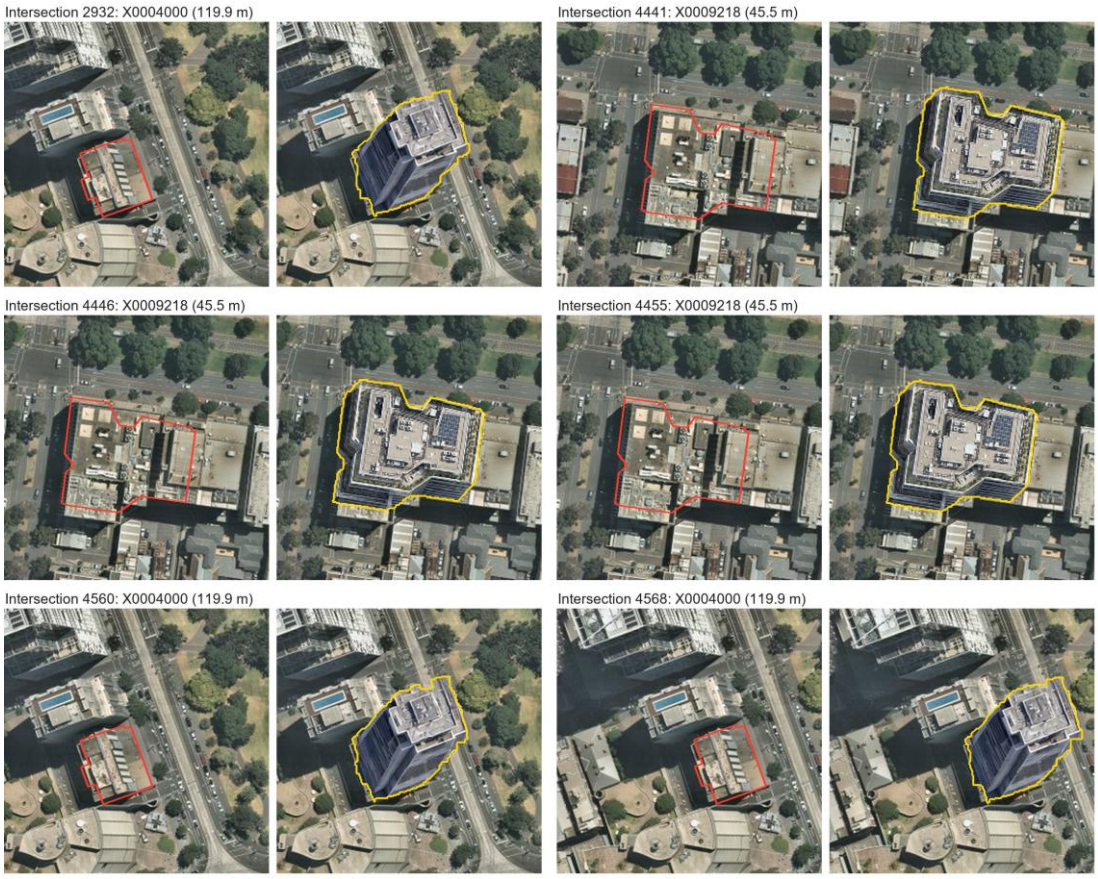


Figure E.2: Cases completed in 2023 on the 2019 imagery (2/3). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.3: Cases completed in 2023 on the 2019 imagery (3/3). In each pair, the left image is the base image and the right image has the generated building added.

E.2 Completed in 2023, on the 2022 imagery

21 intersection cases, involving 5 developments.

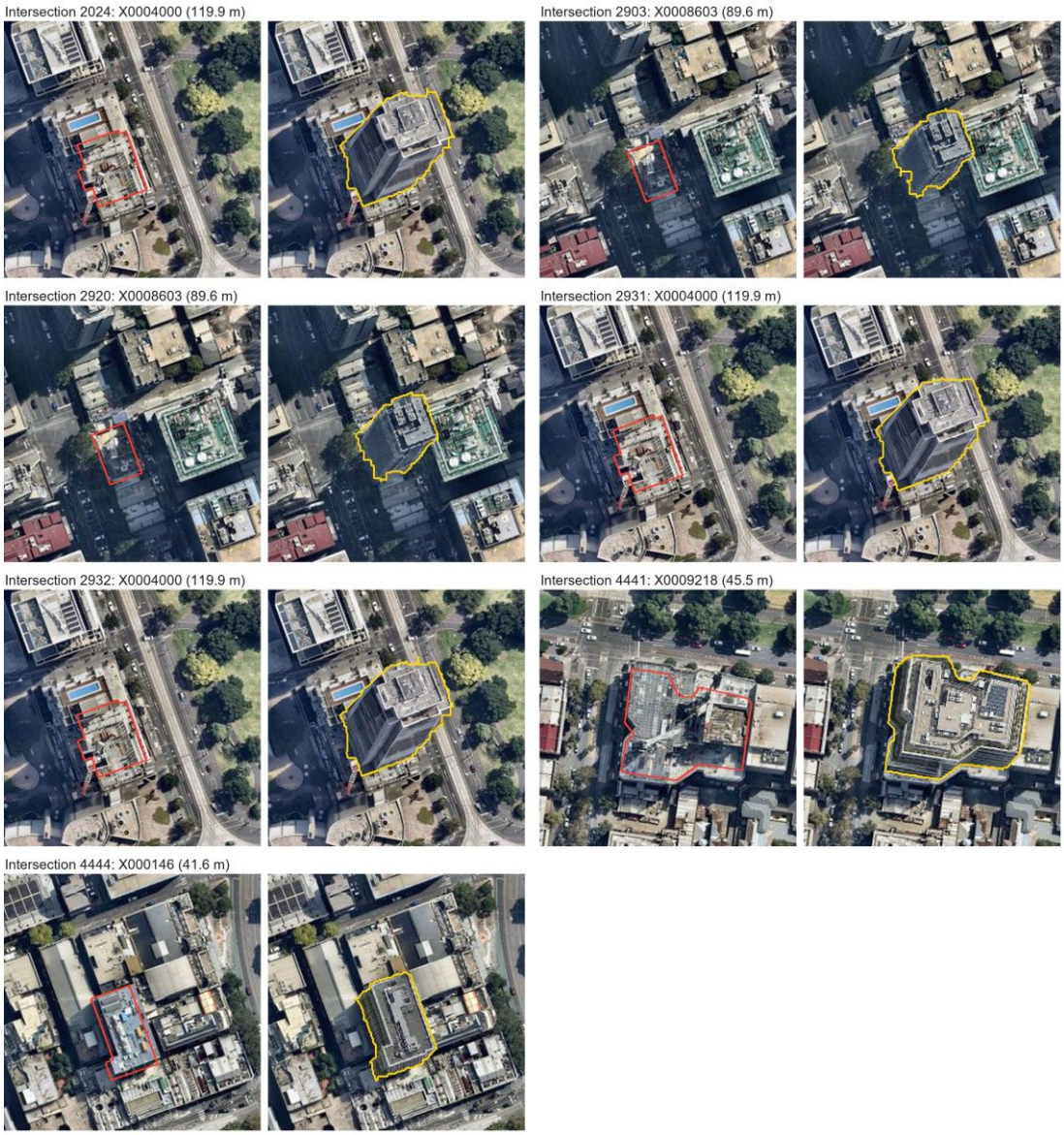


Figure E.4: Cases completed in 2023 on the 2022 imagery (1/3). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.5: Cases completed in 2023 on the 2022 imagery (2/3). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.6: Cases completed in 2023 on the 2022 imagery (3/3). In each pair, the left image is the base image and the right image has the generated building added.

E.3 Completed in 2024, on the 2022 imagery

18 intersection cases, involving 5 developments.



Figure E.7: Cases completed in 2024 on the 2022 imagery (1/3). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.8: Cases completed in 2024 on the 2022 imagery (2/3). In each pair, the left image is the base image and the right image has the generated building added.

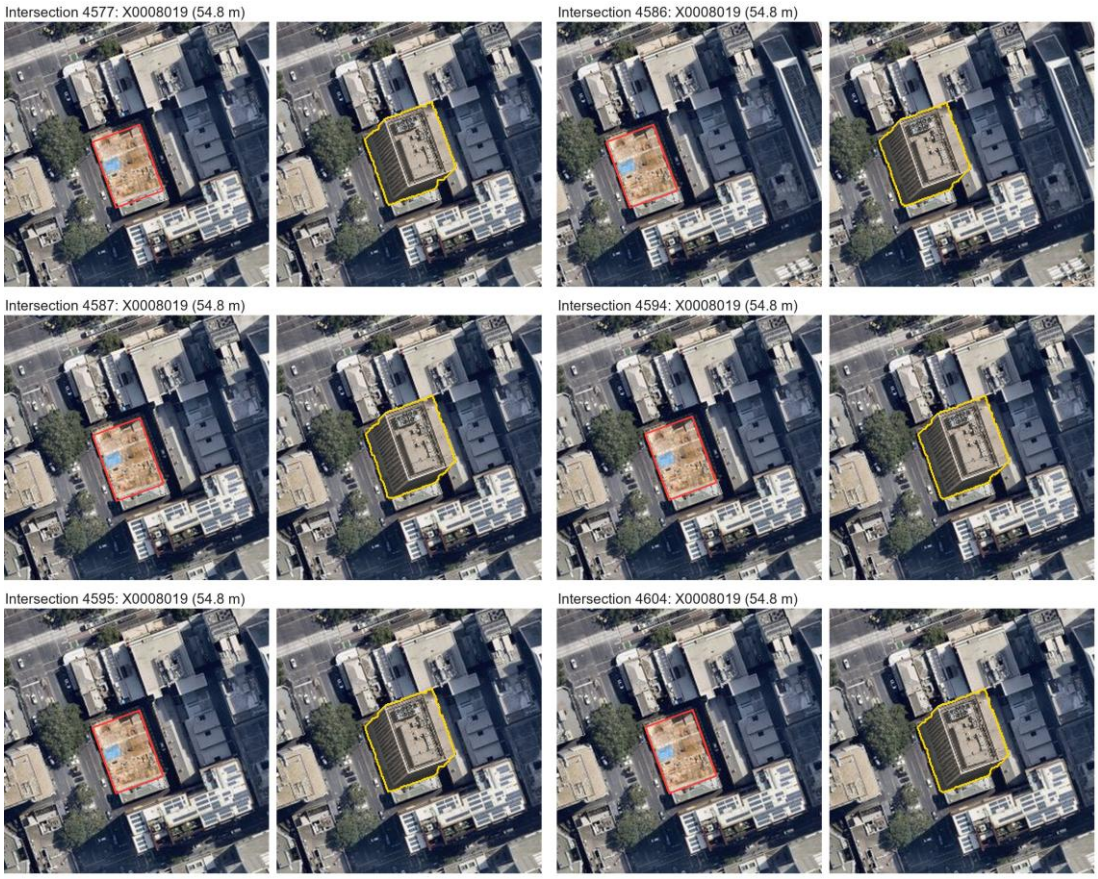


Figure E.9: Cases completed in 2024 on the 2022 imagery (3/3). In each pair, the left image is the base image and the right image has the generated building added.

E.4 Completed in 2024, on the 2023 imagery

18 intersection cases, involving 5 developments.



Figure E.10: Cases completed in 2024 on the 2023 imagery (1/3). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.11: Cases completed in 2024 on the 2023 imagery (2/3). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.12: Cases completed in 2024 on the 2023 imagery (3/3). In each pair, the left image is the base image and the right image has the generated building added.

E.5 Completed in 2025, on the 2022 imagery

11 intersection cases, involving 7 developments.



Figure E.13: Cases completed in 2025 on the 2022 imagery (1/2). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.14: Cases completed in 2025 on the 2022 imagery (2/2). In each pair, the left image is the base image and the right image has the generated building added.

E.6 Completed in 2025, on the 2023 imagery

11 intersection cases, involving 7 developments.



Figure E.15: Cases completed in 2025 on the 2023 imagery (1/2). In each pair, the left image is the base image and the right image has the generated building added.



Figure E.16: Cases completed in 2025 on the 2023 imagery (2/2). In each pair, the left image is the base image and the right image has the generated building added.